DevOps Project

Problem Statement:

Create an end-to-end CI/CD pipeline in AWS platform using Jenkins as the orchestration tool, GitHub as scm, maven as the build tool, deploy in a docker instance and create a docker image, store the docker image in ECR, Kubernetes deployment using ECR image. Build sample java web app using maven.

CI/CD Pipeline Setup Approach:

Requirements:

- Git (Local Version Control System)
- GitHub (Distributed Version Control System)
- Jenkins (Continuous Integration Tool)
- Maven (Build Tool)
- Docker (Containerization)
- Kubernetes (Container Orchestration)

Step 1: CI/CD with GitHub, Jenkins, Maven & Tomcat

1. Install & Configure Jenkins
2. Install & Configure Maven and Git
3. Set Up Apache Tomcat Server
4. Integrate Jenkins with GitHub, Maven, and Tomcat
5. Create CI/CD Jobs in Jenkins
    - CI Job: Build code using Maven
    - CD Job: Deploy to Tomcat
6. Test Deployment on Tomcat

Step 2: CI/CD with GitHub, Jenkins, Maven & Docker

1. Set Up Docker Environment
2. Create Docker Image and Run Container
3. Integrate Docker with Jenkins

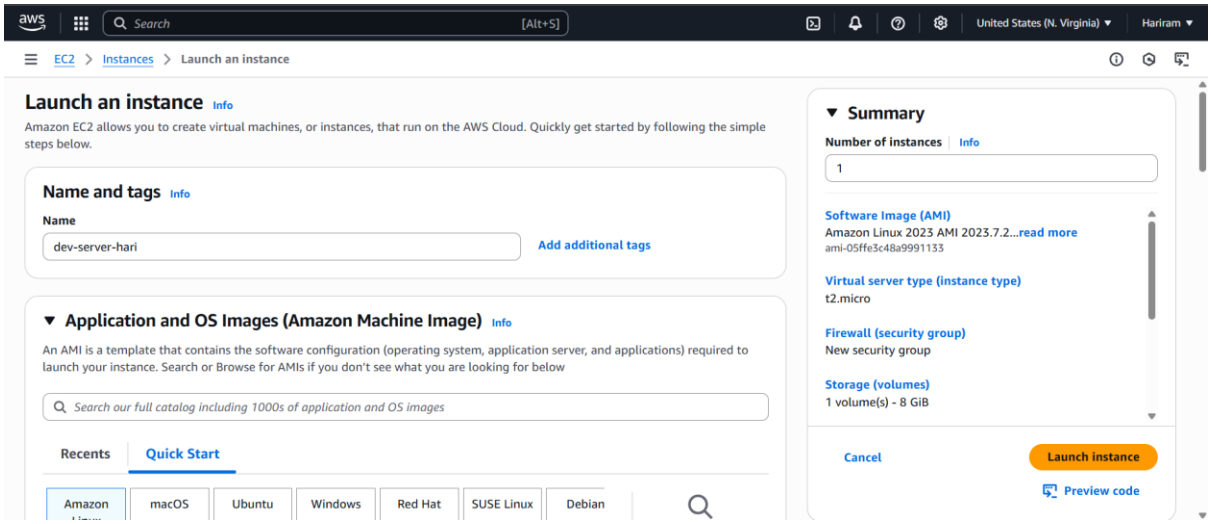
4. Create Jenkins CI/CD Job
    - CI Job: Build code with Maven
    - CD Job: Package into Docker image and deploy to container

Step 3: CI/CD with GitHub, Jenkins, Maven & Kubernetes

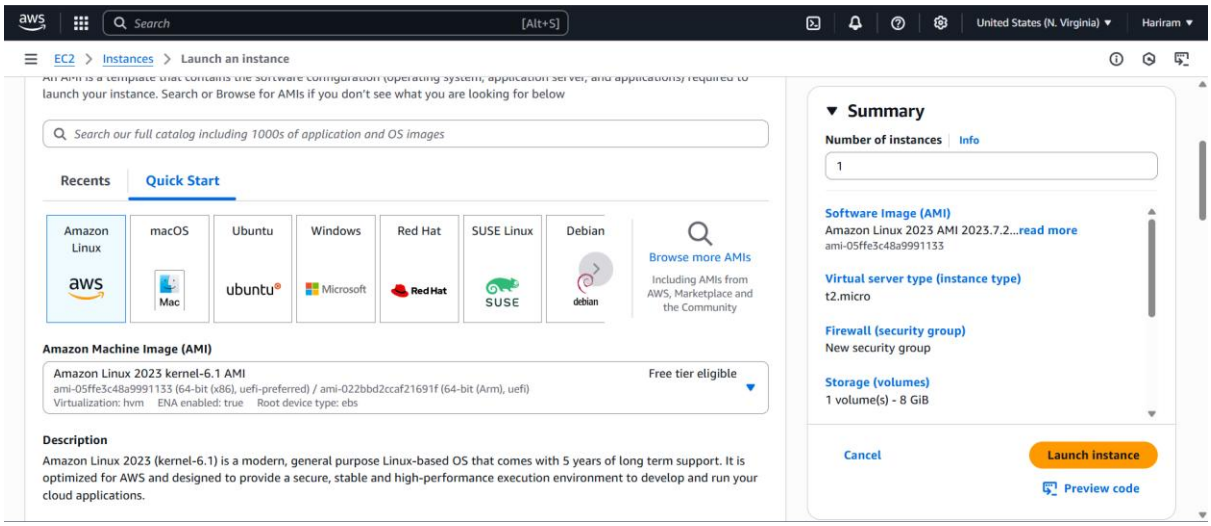
1. **Set Up Kubernetes Cluster (e.g., EKS)**
2. **Write Kubernetes Manifest Files**
    - Deployment
    - Service
    - Pod specifications
3. **Create Jenkins CI/CD Job**
    - CI Job: Build code with Maven
    - CD Job: Deploy to Kubernetes using kubectl

Step 4: Final Deployment & Validation

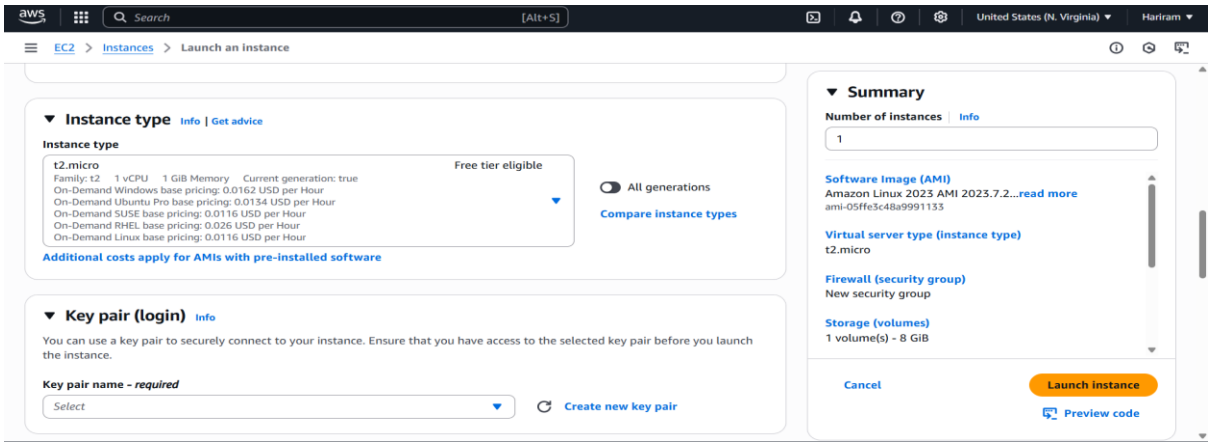
1. Deploy Artifacts to Kubernetes
2. Ensure Docker and Kubernetes Artifacts Contain Required Code
3. Trigger Jenkins Build
4. Verify Pod Creation in Kubernetes
5. Access Application via Kubernetes Service IP
    - Copy service IP
    - Paste in browser to validate output



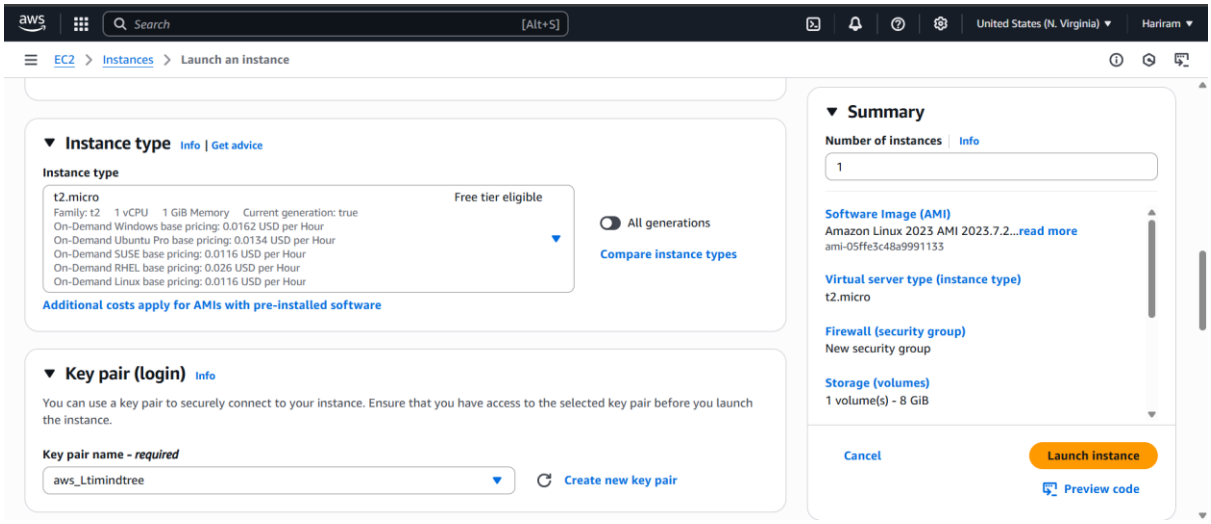
Creating Dev-server instance .



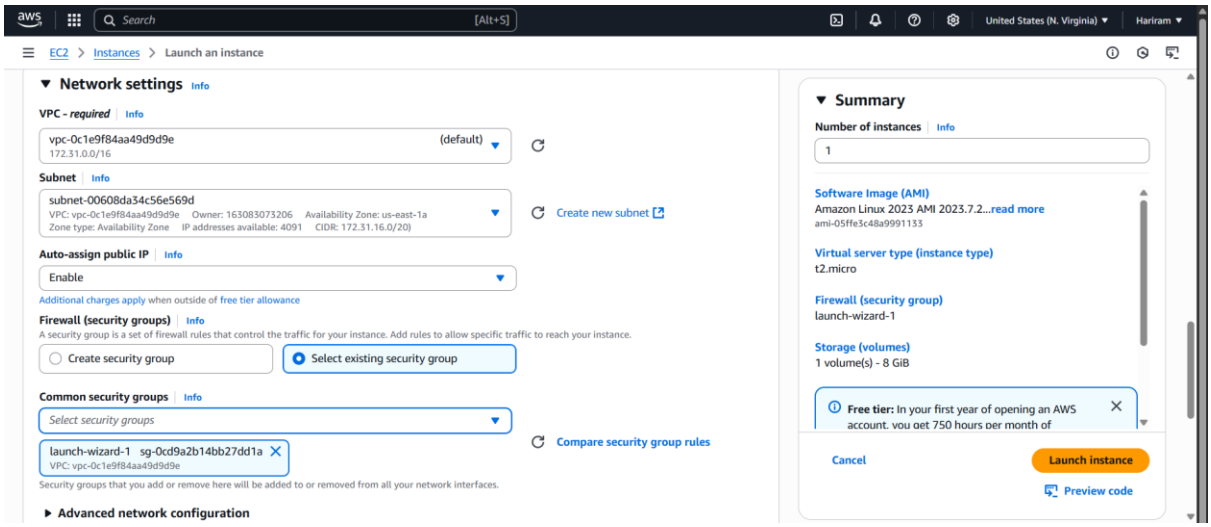
Selected amazon linux.



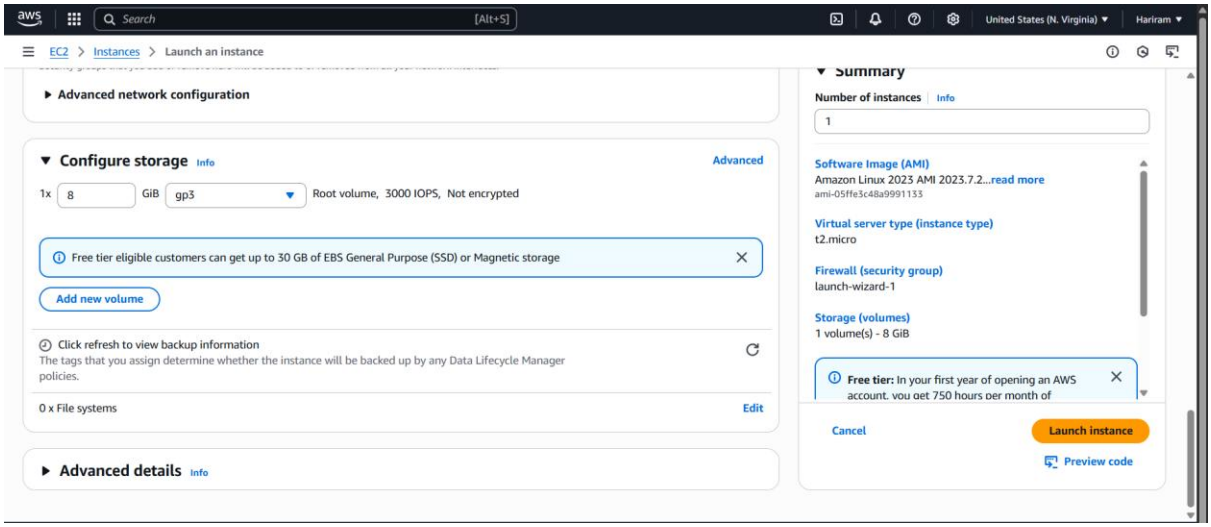
T2.micro is selected.



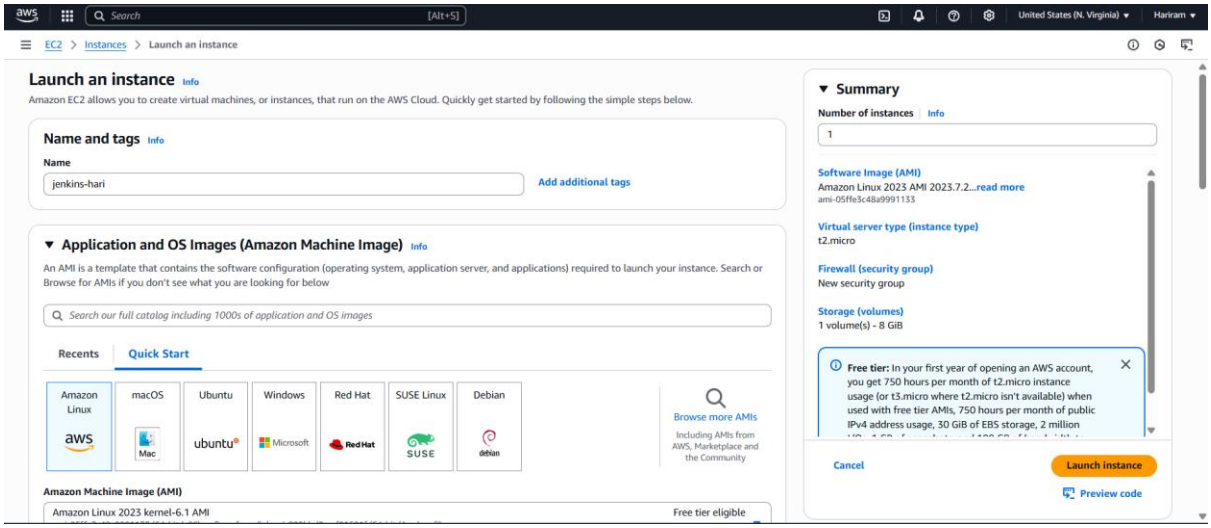
Existing key pair is selected.



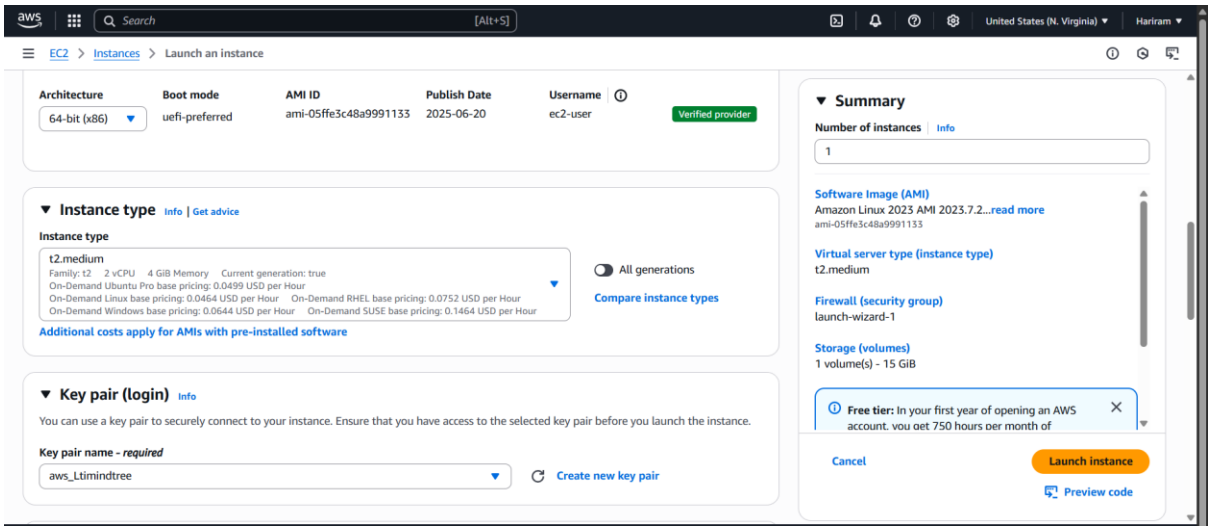
Existing security group is selected with required ports allowed.



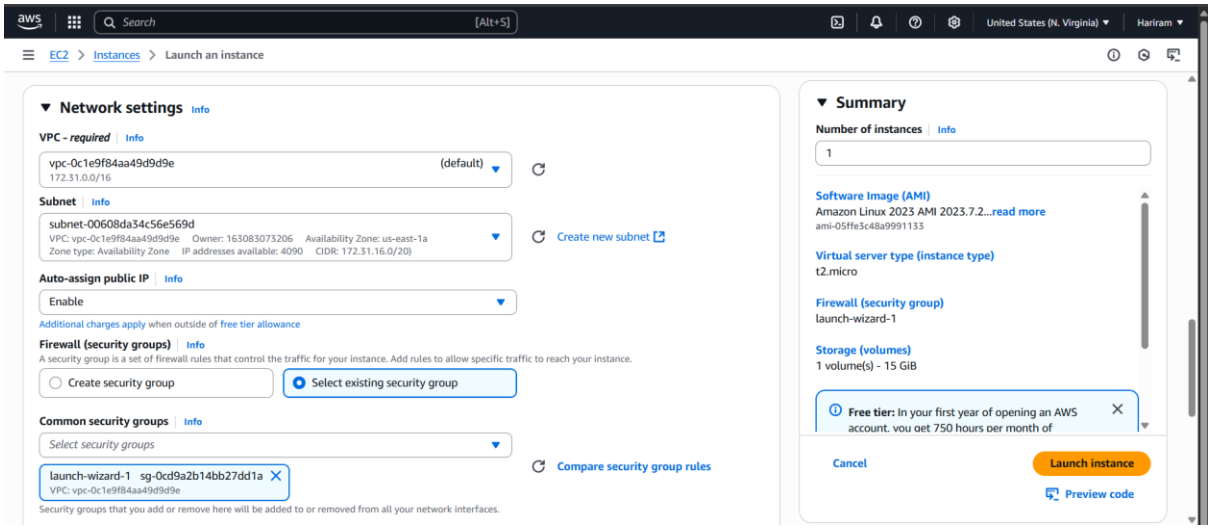
8GB storage and gp3 is selected.



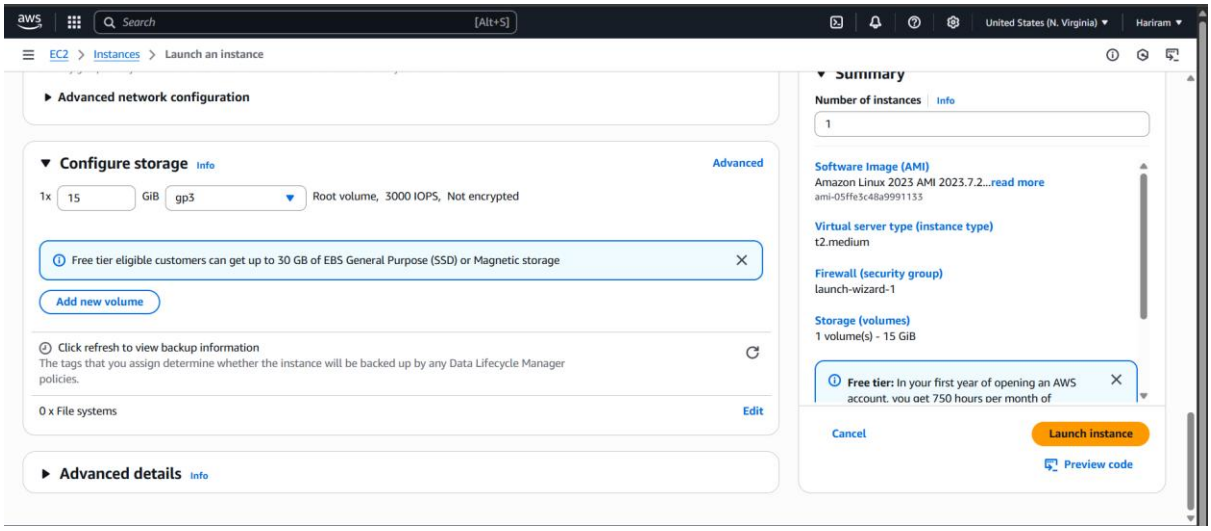
Jenkins server is created with amazon linux.



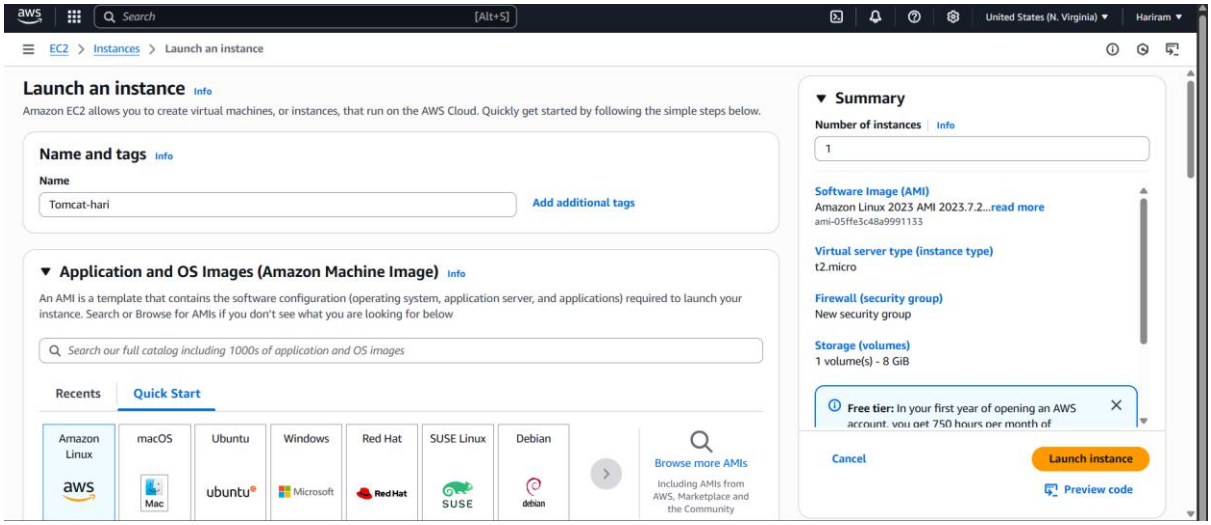
T2.medium is selected as a requirement for Jenkins.



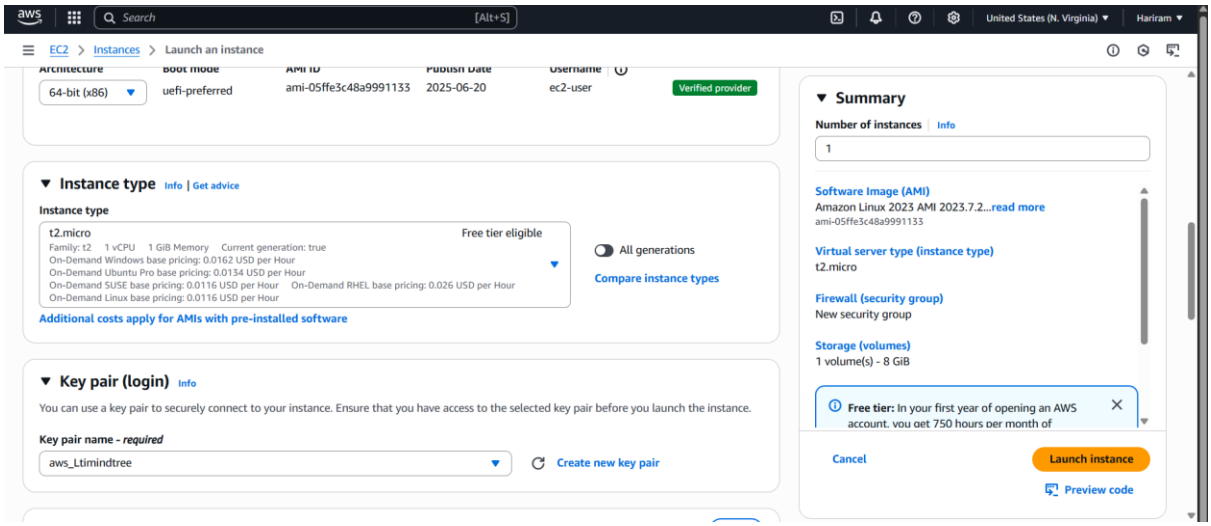
Existing security group is selected.



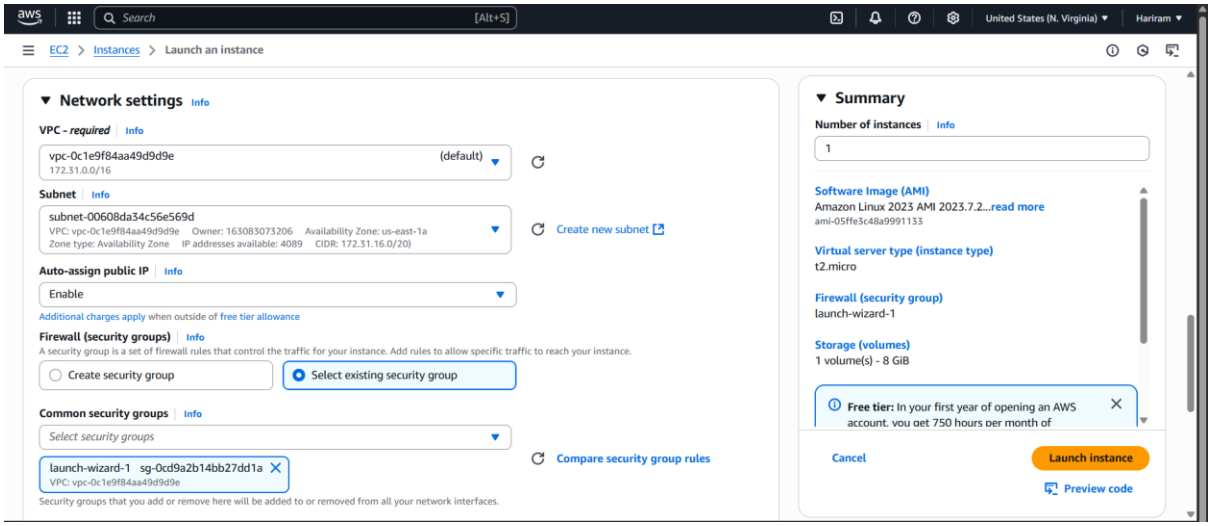
15GB storage and gp3 is selected.



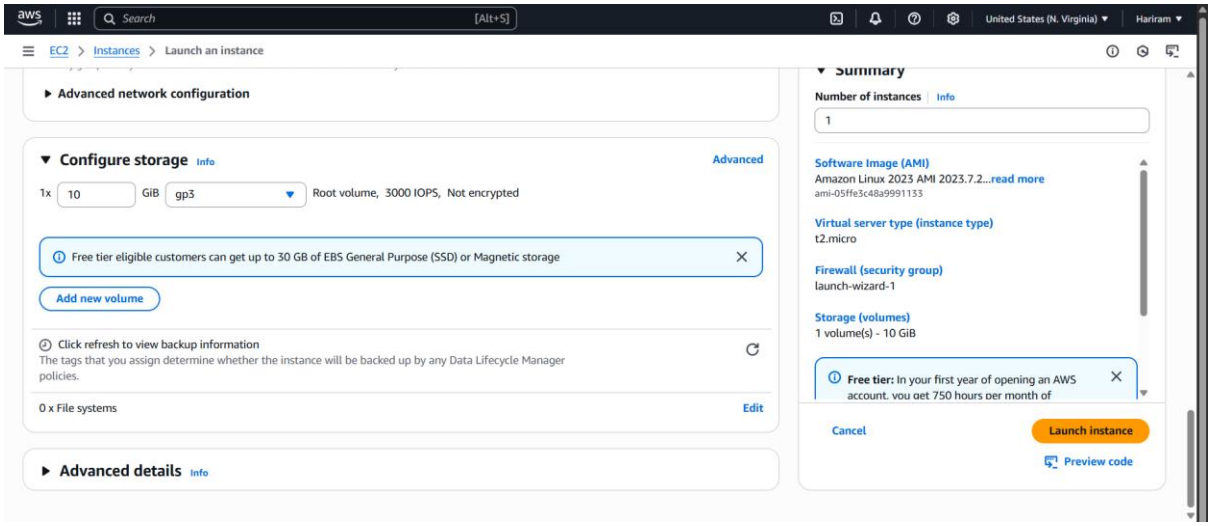
Tomcat server is created with amazon linux.



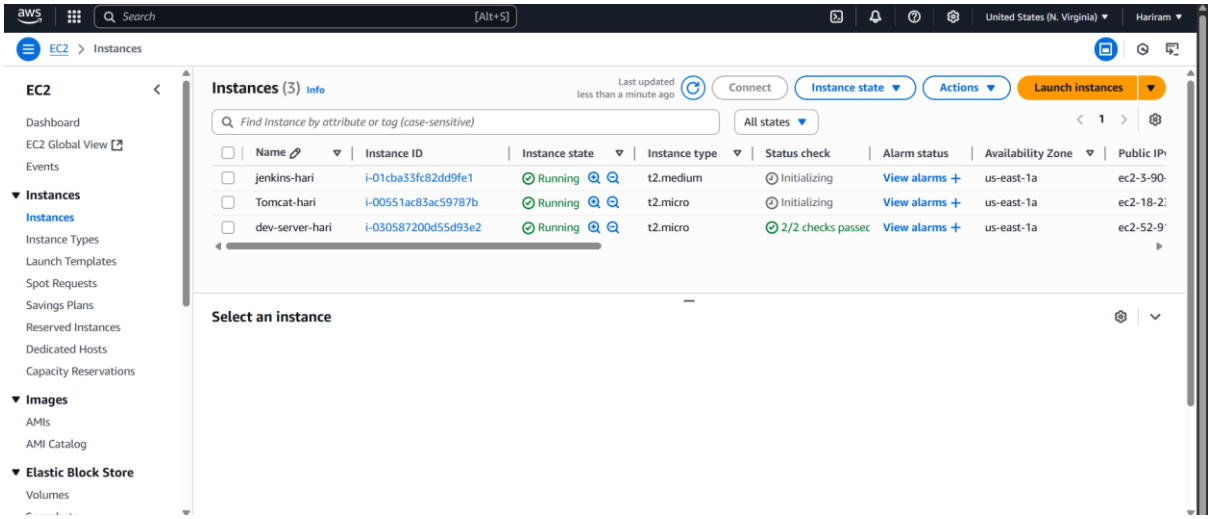
T2.micro is selected with existing key apir.



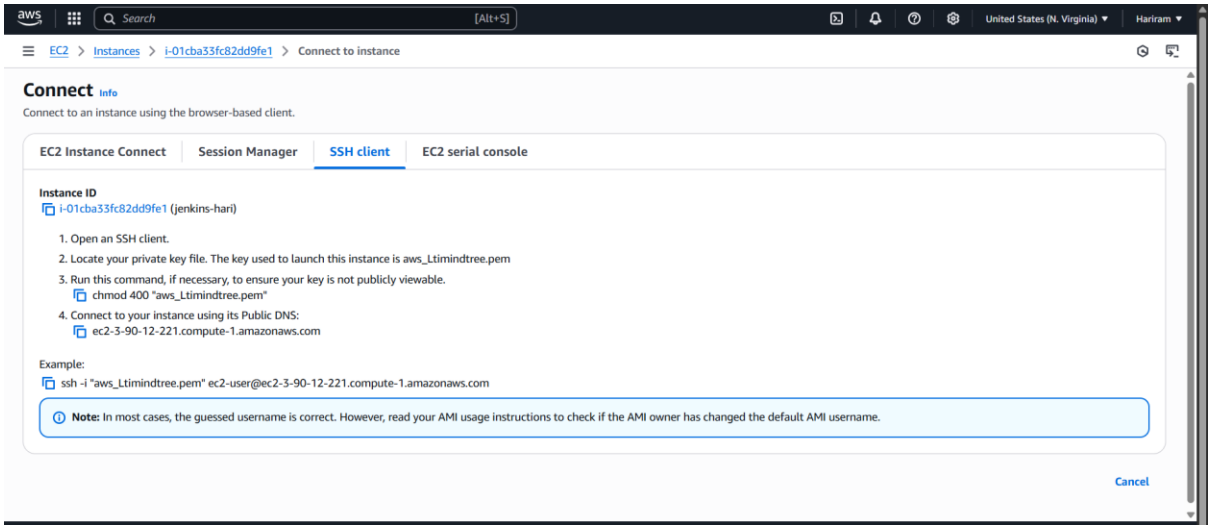
Existing security group is selected.



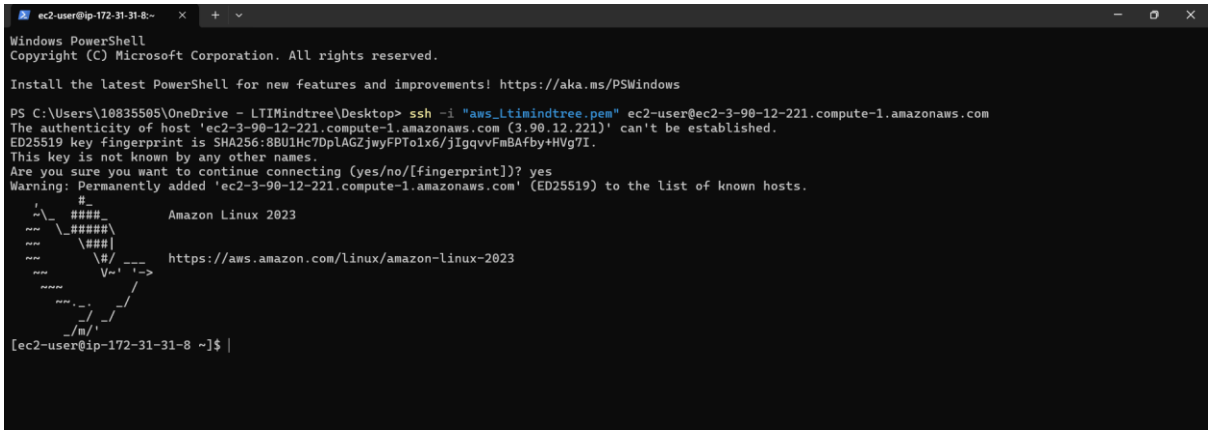
10GB storage and gp3 is selected.



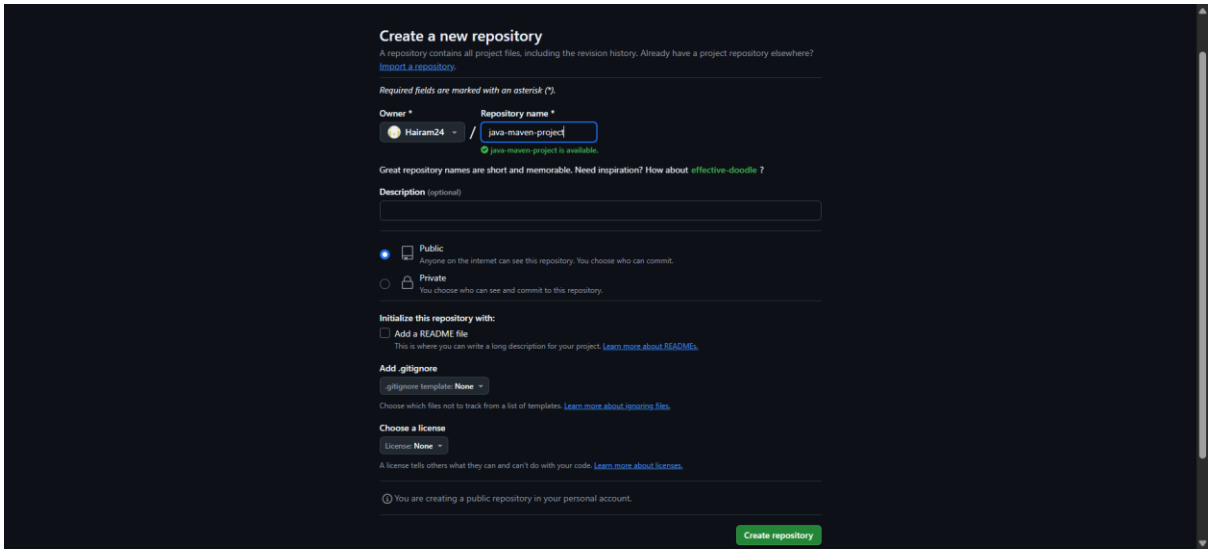
All the three servers are ready to run.



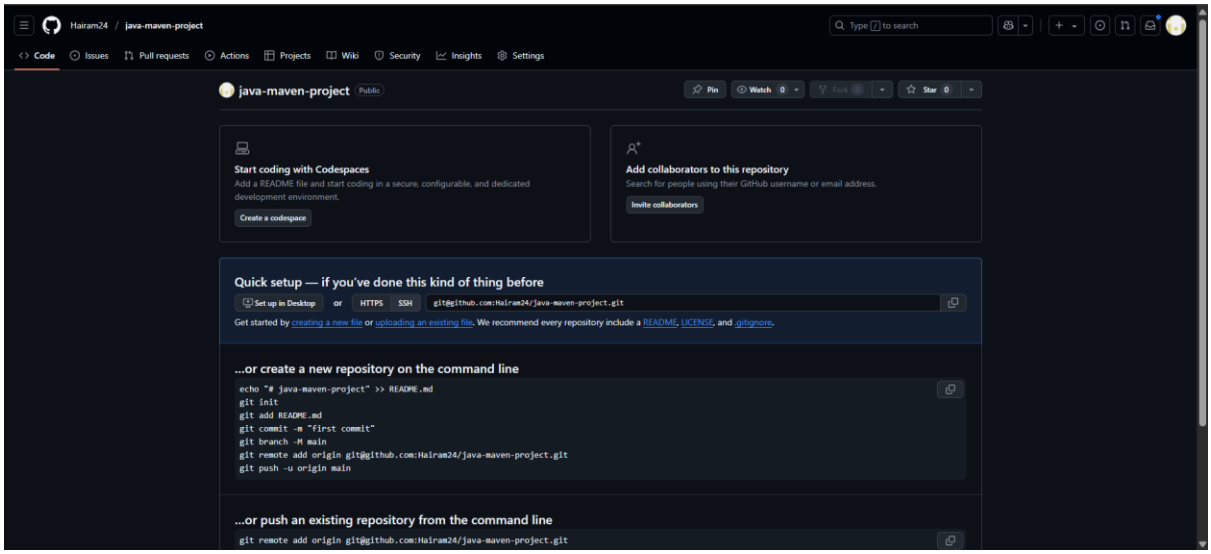
Instance is connected via SSH client.



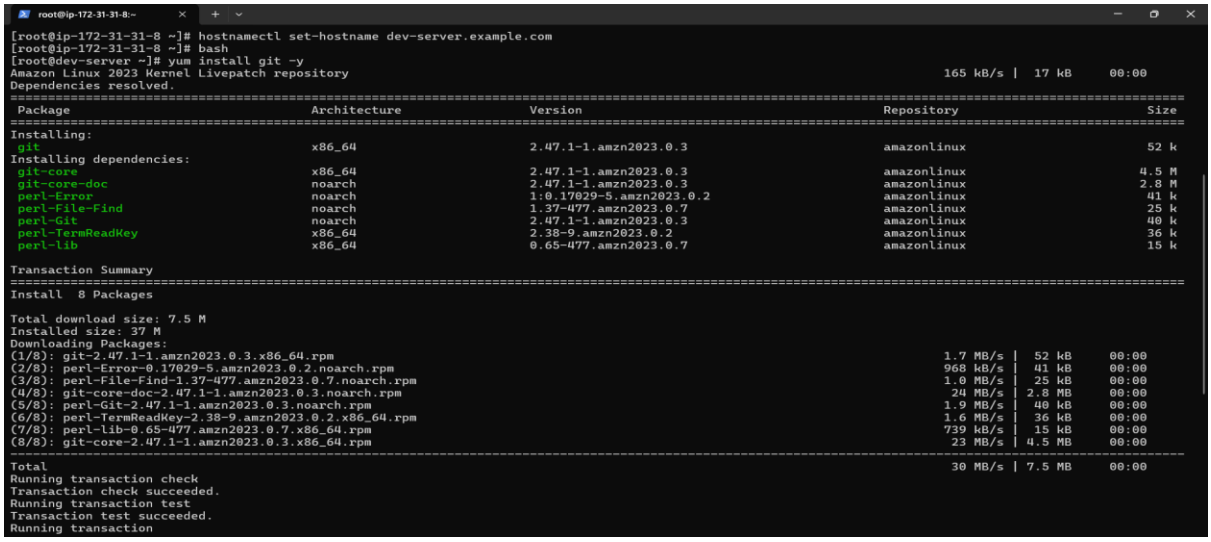
Instance is successfully connected.



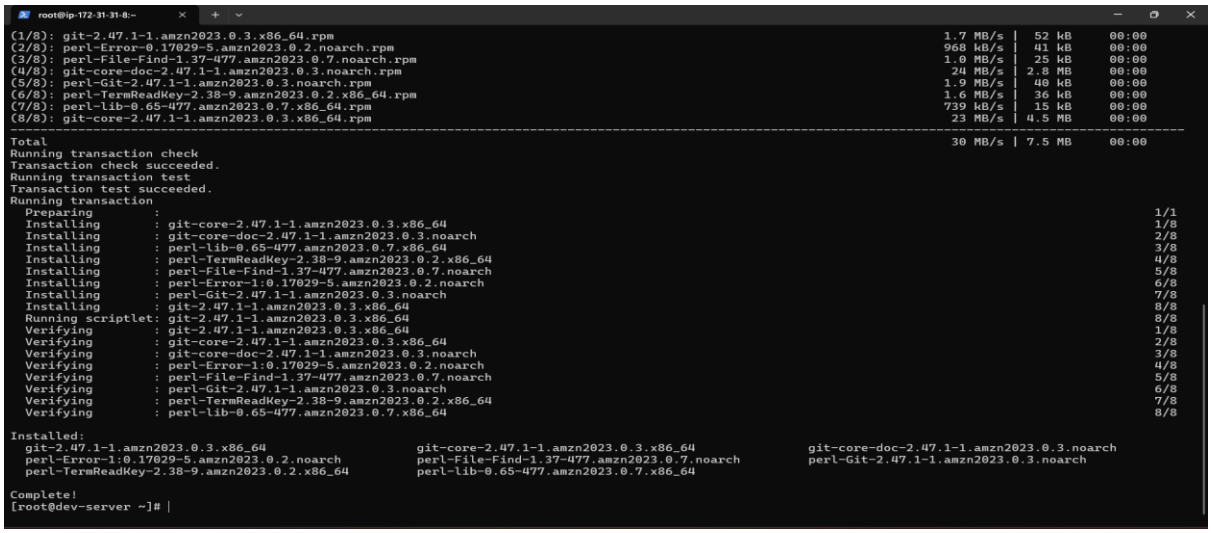
New public repository is created as java-maven project.



Public repository is created successfully.



Hostname is set and git package is installed.



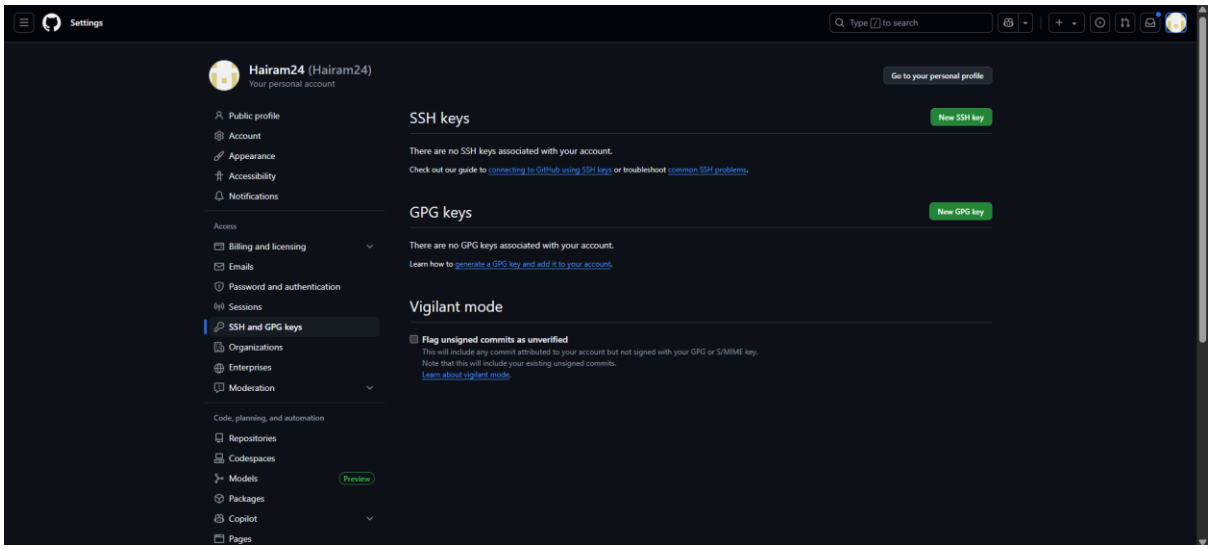
Git package is successfully installed.

```
[root@dev-server ~]#  ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/root/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/id_rsa
Your public key has been saved in /root/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:60RtT/+Vmve741l4RKlbu7xYzinz8dZS2Kfa64vysR8 root@dev-server.example.com
The key's randomart image is:
+---[RSA 3072]----+
|                 |
|                .|
|               ..|
|          .    ..|
|        S o .. +.|
|       . o o .+o*|
|        o   o.EB*|
|       o  .  B@*@|
|          .  o=B%^X|
+----[SHA256]-----+
[root@dev-server ~]# |
```

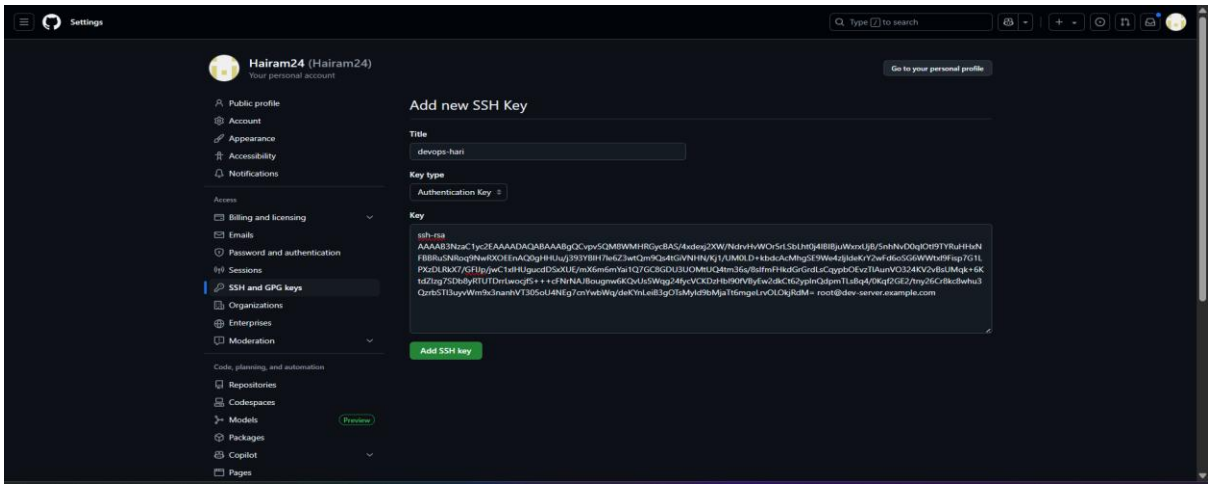
SSH key is generated.

```
[root@dev-server ~]# cd .ssh/
[root@dev-server .ssh]# cat id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABgQCvpv5QM8WMHRGycBAS/4xdexj2XW/NdrvHvWOr5rLSbLht0j4IBIBjuWxrxUjB/5nhNvD0qIOtI9TYRuHHxNFBBRuSNRoq9NwRXOEEnAQ0gHHUu/j393YB
lH7le6Z3wtQm9Qs4tGiVNHN/Kj1/UM0LD+kbdcAcMhgSE9We4zljIdeKrY2wFd6oSG6WWtxl9Fisp7G1LPXzDLRkX7/GFlJp/jwC1xlHUgucdDSxXUE/mX6m6mYai1Q7GC8GDU3UOMtUQ4tm36s/8sIfmFHk
dGrGrdLsCqypbOEvzTlAunVO324KV2vBsUMqk+6KtdZIzg7SDb8yRTUTDrrLwocjfS+++cFNrNAJBougnw6KQvUs5Wgg24fycVCKDzHbl90fVByEw2dkCt62yplnQdpmTLsBq4/0Kqf2GE2/tny26Cr8kc8w
hu3QzrbSTI3uyvWm9x3nanhVT305oU4NEg7cnYwbWq/deKYnLeiB3gOTsMyId9bMjaTt6mgeLrvOLOkjRdM= root@dev-server.example.com
[root@dev-server .ssh]# |
```

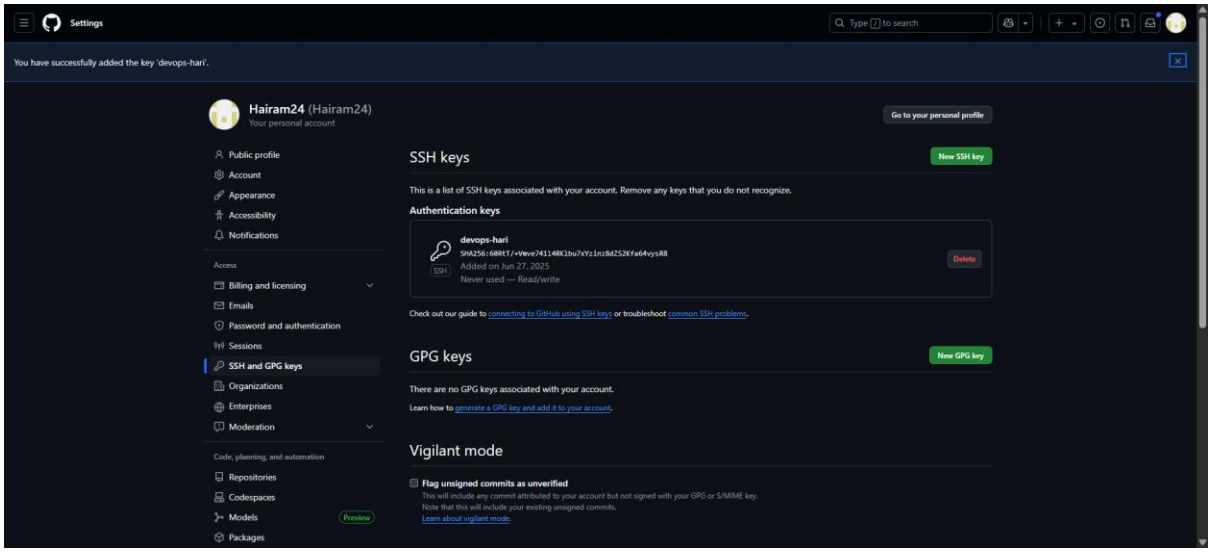
Public key is copied.



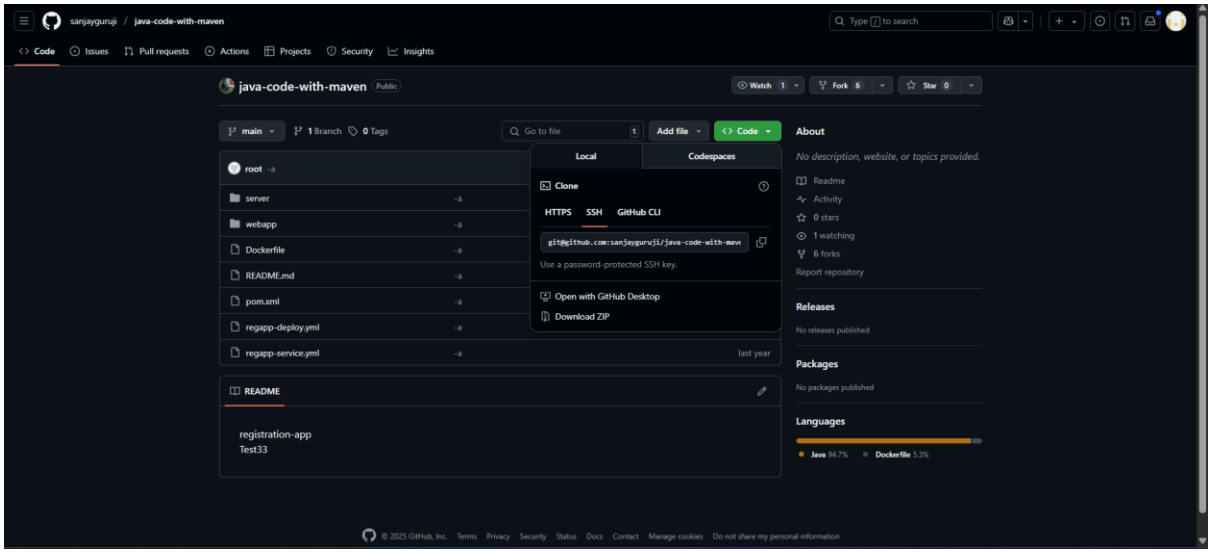
To establish connection to git through ssh key.



SSH key is added.



SSH key added successfully.



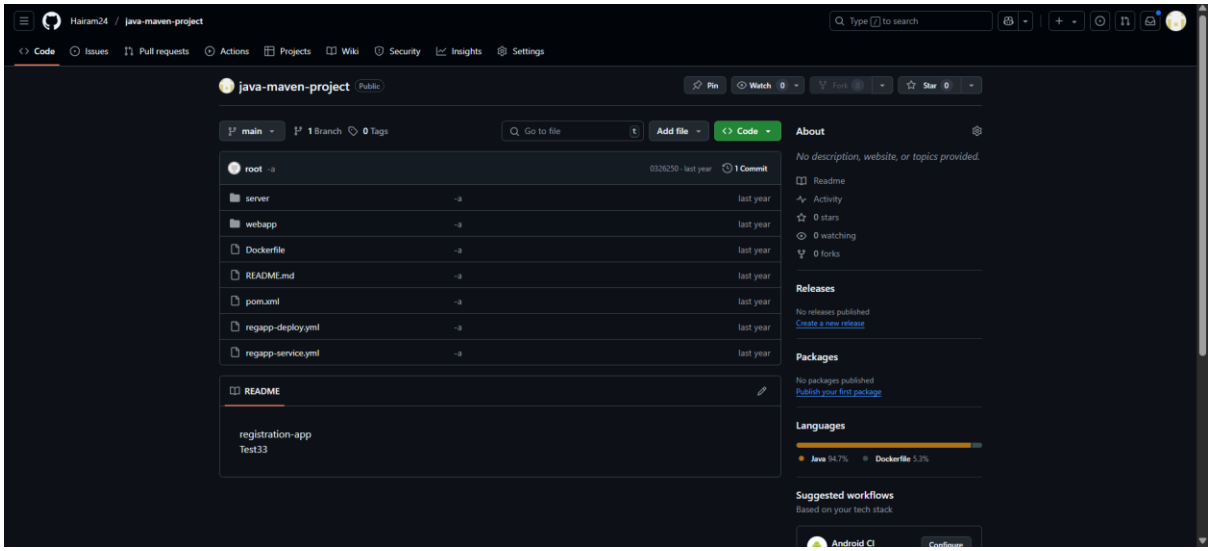
Copied the repo ssh link to clone it.

```
[root@dev-server .ssh]# cd
[root@dev-server ~]# git clone git@github.com:sanjayguruji/java-code-with-maven.git
Cloning into 'java-code-with-maven'...
The authenticity of host 'github.com (140.82.112.4)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
remote: Enumerating objects: 31, done.
remote: Counting objects: 100% (2/2), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 31 (delta 0), reused 0 (delta 0), pack-reused 29 (from 2)
Receiving objects: 100% (31/31), 4.87 KiB | 4.88 MiB/s, done.
Resolving deltas: 100% (1/1), done.
[root@dev-server ~]#
```

Git repository is cloned successfully.

```
[root@dev-server ~]# ll
total 0
drwxr-xr-x. 5 root root 147 Jun 27 04:34 java-code-with-maven
[root@dev-server ~]# cd java-code-with-maven
[root@dev-server java-code-with-maven]# git init
Reinitialized existing Git repository in /root/java-code-with-maven/.git/
[root@dev-server java-code-with-maven]# git add .
[root@dev-server java-code-with-maven]# git commit -m "first commit"
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
[root@dev-server java-code-with-maven]# git branch -M main
[root@dev-server java-code-with-maven]# git remote add origin git@github.com:Hairam24/java-maven-project.git
error: remote origin already exists.
[root@dev-server java-code-with-maven]# git push -u origin main
ERROR: Permission to sanjayguruji/java-code-with-maven.git denied to Hairam24.
fatal: Could not read from remote repository.

Please make sure you have the correct access rights
and the repository exists.
[root@dev-server java-code-with-maven]# git remote remove origin
[root@dev-server java-code-with-maven]# git remote add origin git@github.com:Hairam24/java-maven-project.git
[root@dev-server java-code-with-maven]# git push -u origin main
Enumerating objects: 31, done.
Counting objects: 100% (31/31), done.
Delta compression using up to 2 threads
Compressing objects: 100% (15/15), done.
Writing objects: 100% (31/31), 4.87 KiB | 4.88 MiB/s, done.
Total 31 (delta 1), reused 31 (delta 1), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To github.com:Hairam24/java-maven-project.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
[root@dev-server java-code-with-maven]#
```

Cloned repo is copied to the java-maven project repo.



Successfully add to the project repository.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\10835505\OneDrive - LTIMindtree\Desktop> ssh -i "aws_Ltimindtree.pem" ec2-user@ec2-54-164-19-42.compute-1.amazonaws.com
The authenticity of host 'ec2-54-164-19-42.compute-1.amazonaws.com (54.164.19.42)' can't be established.
ED25519 key fingerprint is SHA256:Cls3O3IsIyJSB23PX+QBoAY1lKI0CU393ZgMO+f+lt8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-54-164-19-42.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-172-31-25-209 ~]$ sudo su -
[root@ip-172-31-25-209 ~]# hostnamectl set-hostname jenkins.example.com
[root@ip-172-31-25-209 ~]#  bash
[root@jenkins ~]#
```

Jenkins server is connected and hostname is changed.

```
root@ip-172-31-25-209:~
[root@jenkins ~]# wget -O /etc/yum.repos.d/jenkins.repo \
     https://pkg.jenkins.io/redhat-stable/jenkins.repo
--2025-06-27 04:45:34--  https://pkg.jenkins.io/redhat-stable/jenkins.repo
Resolving pkg.jenkins.io (pkg.jenkins.io)... 146.75.34.133, 2a04:4e42:77::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)|146.75.34.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 85
Saving to: '/etc/yum.repos.d/jenkins.repo'

/etc/yum.repos.d/jenkins.repo         100%[===================================================================>]      85  --.-KB/s    in 0s

2025-06-27 04:45:34 (3.44 MB/s) - '/etc/yum.repos.d/jenkins.repo' saved [85/85]

[root@jenkins ~]# rpm --import https://pkg.jenkins.io/redhat-stable/jenkins.io-2023.key
[root@jenkins ~]#  yum upgrade
Jenkins-stable                                                                                                   676 kB/s |  31 kB     00:00
Dependencies resolved.
Nothing to do.
Complete!
[root@jenkins ~]#
```

Jenkins packages installed.

```
root@ip-172-31-25-209:~
  Verifying        : libICE-1.1.1-3.amzn2023.0.1.x86_64                                                          18/35
  Verifying        : libSM-1.2.4-3.amzn2023.0.1.x86_64                                                           19/35
  Verifying        : libX11-1.8.10-2.amzn2023.0.1.x86_64                                                         20/35
  Verifying        : libX11-common-1.8.10-2.amzn2023.0.1.noarch                                                  21/35
  Verifying        : libXau-1.0.11-6.amzn2023.0.1.x86_64                                                         22/35
  Verifying        : libXext-1.3.6-1.amzn2023.0.1.x86_64                                                         23/35
  Verifying        : libXi-1.8.2-1.amzn2023.0.1.x86_64                                                           24/35
  Verifying        : libXinerama-1.1.5-6.amzn2023.0.1.x86_64                                                     25/35
  Verifying        : libXrandr-1.5.4-3.amzn2023.0.1.x86_64                                                       26/35
  Verifying        : libXrender-0.9.11-6.amzn2023.0.1.x86_64                                                     27/35
  Verifying        : libXt-1.3.0-3.amzn2023.0.1.x86_64                                                           28/35
  Verifying        : libXtst-1.2.5-1.amzn2023.0.1.x86_64                                                         29/35
  Verifying        : libbrotli-1.0.9-4.amzn2023.0.2.x86_64                                                       30/35
  Verifying        : libjpeg-turbo-2.1.4-2.amzn2023.0.5.x86_64                                                   31/35
  Verifying        : libpng-2:1.6.37-10.amzn2023.0.6.x86_64                                                      32/35
  Verifying        : libxcb-1.17.0-1.amzn2023.0.1.x86_64                                                         33/35
  Verifying        : pixman-0.43.4-1.amzn2023.0.4.x86_64                                                         34/35
  Verifying        : xml-common-0.6.3-56.amzn2023.0.2.noarch                                                     35/35

Installed:
  alsa-lib-1.2.7.2-1.amzn2023.0.2.x86_64                                cairo-1.18.0-4.amzn2023.0.1.x86_64
  dejavu-sans-fonts-2.37-16.amzn2023.0.2.noarch                         dejavu-sans-mono-fonts-2.37-16.amzn2023.0.2.noarch
  dejavu-serif-fonts-2.37-16.amzn2023.0.2.noarch                        fontconfig-2.13.94-2.amzn2023.0.2.x86_64
  fonts-filesystem-1:2.0.5-12.amzn2023.0.2.noarch                       freetype-2.13.2-5.amzn2023.0.1.x86_64
  giflib-5.2.1-9.amzn2023.0.1.x86_64                                    google-noto-fonts-common-20201206-2.amzn2023.0.2.noarch
  google-noto-sans-vf-fonts-20201206-2.amzn2023.0.2.noarch              graphite2-1.3.14-7.amzn2023.0.2.x86_64
  harfbuzz-7.0.0-2.amzn2023.0.2.x86_64                                  java-17-amazon-corretto-1:17.0.15+6-1.amzn2023.1.x86_64
  java-17-amazon-corretto-headless-1:17.0.15+6-1.amzn2023.1.x86_64      javapackages-filesystem-6.0.0-7.amzn2023.0.6.noarch
  langpacks-core-font-en-3.0-21.amzn2023.0.4.noarch                     libICE-1.1.1-3.amzn2023.0.1.x86_64
  libSM-1.2.4-3.amzn2023.0.1.x86_64                                     libX11-1.8.10-2.amzn2023.0.1.x86_64
  libX11-common-1.8.10-2.amzn2023.0.1.noarch                            libXau-1.0.11-6.amzn2023.0.1.x86_64
  libXext-1.3.6-1.amzn2023.0.1.x86_64                                   libXi-1.8.2-1.amzn2023.0.1.x86_64
  libXinerama-1.1.5-6.amzn2023.0.1.x86_64                               libXrandr-1.5.4-3.amzn2023.0.1.x86_64
  libXrender-0.9.11-6.amzn2023.0.1.x86_64                               libXt-1.3.0-3.amzn2023.0.1.x86_64
  libXtst-1.2.5-1.amzn2023.0.1.x86_64                                   libbrotli-1.0.9-4.amzn2023.0.2.x86_64
  libjpeg-turbo-2.1.4-2.amzn2023.0.5.x86_64                             libpng-2:1.6.37-10.amzn2023.0.6.x86_64
  libxcb-1.17.0-1.amzn2023.0.1.x86_64                                   pixman-0.43.4-1.amzn2023.0.4.x86_64
  xml-common-0.6.3-56.amzn2023.0.2.noarch

Complete!
[root@jenkins ~]#
```

Installed successfully.

```
[root@jenkins ~]# yum install jenkins -y
Last metadata expiration check: 0:01:50 ago on Fri Jun 27 04:45:59 2025.
Dependencies resolved.
================================================================================================================================
 Package                    Architecture                Version                          Repository                        Size
================================================================================================================================
Installing:
 jenkins                    noarch                      2.504.3-1.1                      jenkins                           90 M

Transaction Summary
================================================================================================================================
Install  1 Package

Total download size: 90 M
Installed size: 90 M
Downloading Packages:
jenkins-2.504.3-1.1.noarch.rpm                                                                  30 MB/s |  90 MB     00:02
--------------------------------------------------------------------------------------------------------------------------------
Total                                                                                           30 MB/s |  90 MB     00:02
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                                                          1/1
  Running scriptlet: jenkins-2.504.3-1.1.noarch                                                                               1/1
  Installing       : jenkins-2.504.3-1.1.noarch                                                                               1/1
  Running scriptlet: jenkins-2.504.3-1.1.noarch                                                                               1/1
  Verifying        : jenkins-2.504.3-1.1.noarch                                                                               1/1

Installed:
  jenkins-2.504.3-1.1.noarch

Complete!
[root@jenkins ~]#
```

Jenkins is installed successfully.

```
[root@jenkins ~]# systemctl enable Jenkins
Failed to enable unit: Unit file Jenkins.service does not exist.
[root@jenkins ~]# systemctl enable jenkins
Created symlink /etc/systemd/system/multi-user.target.wants/jenkins.service → /usr/lib/systemd/system/jenkins.service.
[root@jenkins ~]# systemctl start jenkins
[root@jenkins ~]# systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: disabled)
     Active: active (running) since Fri 2025-06-27 04:49:44 UTC; 7s ago
   Main PID: 26444 (java)
      Tasks: 48 (limit: 4656)
     Memory: 714.9M
        CPU: 17.075s
     CGroup: /system.slice/jenkins.service
             └─26444 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

Jun 27 04:49:40 jenkins.example.com jenkins[26444]: 63458ca172624c49811206630784a40f
Jun 27 04:49:40 jenkins.example.com jenkins[26444]: This may also be found at: /var/lib/jenkins/secrets/initialAdminPassword
Jun 27 04:49:40 jenkins.example.com jenkins[26444]: *************************************************************
Jun 27 04:49:40 jenkins.example.com jenkins[26444]: *************************************************************
Jun 27 04:49:40 jenkins.example.com jenkins[26444]: *************************************************************
Jun 27 04:49:44 jenkins.example.com jenkins[26444]: 2025-06-27 04:49:44.409+0000 [id=31]        INFO        jenkins.InitReactorRunner$1#onAttained: Complet>
Jun 27 04:49:44 jenkins.example.com jenkins[26444]: 2025-06-27 04:49:44.420+0000 [id=23]        INFO        hudson.lifecycle.Lifecycle#onReady: Jenkins is >
Jun 27 04:49:44 jenkins.example.com systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Jun 27 04:49:44 jenkins.example.com jenkins[26444]: 2025-06-27 04:49:44.456+0000 [id=49]        INFO        h.m.DownloadService$Downloadable#load: Obtained>
Jun 27 04:49:44 jenkins.example.com jenkins[26444]: 2025-06-27 04:49:44.457+0000 [id=49]        INFO        hudson.util.Retrier#start: Performed the action>
lines 1-20/20 (END)
```

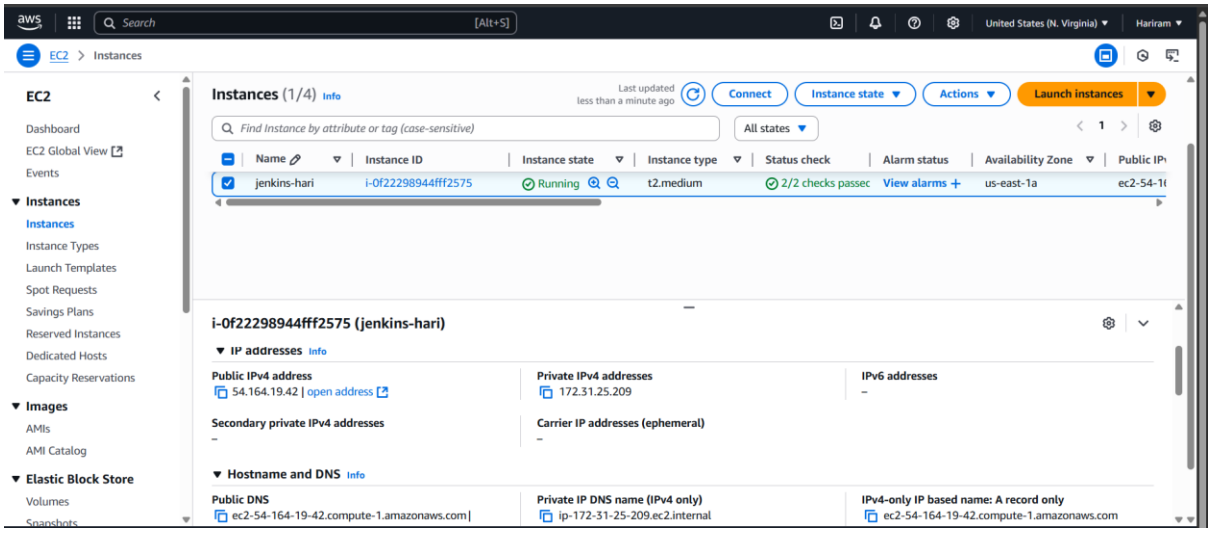
Jenkins is enabled.

```
root@ip-172-31-25-209:~
[root@jenkins ~]# yum install git -y
Last metadata expiration check: 0:04:23 ago on Fri Jun 27 04:45:59 2025.
Dependencies resolved.
==============================================================================================================
 Package                Architecture       Version                        Repository                 Size
==============================================================================================================
Installing:
 git                    x86_64             2.47.1-1.amzn2023.0.3          amazonlinux                52 k
Installing dependencies:
 git-core               x86_64             2.47.1-1.amzn2023.0.3          amazonlinux               4.5 M
 git-core-doc           noarch             2.47.1-1.amzn2023.0.3          amazonlinux               2.8 M
 perl-Error             noarch             1:0.17029-5.amzn2023.0.2       amazonlinux                41 k
 perl-File-Find         noarch             1.37-477.amzn2023.0.7          amazonlinux                25 k
 perl-Git               noarch             2.47.1-1.amzn2023.0.3          amazonlinux                40 k
 perl-TermReadKey       x86_64             2.38-9.amzn2023.0.2            amazonlinux                36 k
 perl-lib               x86_64             0.65-477.amzn2023.0.7          amazonlinux                15 k

Transaction Summary
==============================================================================================================
Install  8 Packages

Total download size: 7.5 M
Installed size: 37 M
Downloading Packages:
(1/8): git-2.47.1-1.amzn2023.0.3.x86_64.rpm                                  1.1 MB/s |  52 kB     00:00
(2/8): perl-Error-0.17029-5.amzn2023.0.2.noarch.rpm                          1.5 MB/s |  41 kB     00:00
(3/8): git-core-doc-2.47.1-1.amzn2023.0.3.noarch.rpm                          25 MB/s | 2.8 MB     00:00
(4/8): perl-File-Find-1.37-477.amzn2023.0.7.noarch.rpm                       672 kB/s |  25 kB     00:00
(5/8): perl-Git-2.47.1-1.amzn2023.0.3.noarch.rpm                             1.8 MB/s |  40 kB     00:00
(6/8): perl-TermReadKey-2.38-9.amzn2023.0.2.x86_64.rpm                       1.6 MB/s |  36 kB     00:00
(7/8): perl-lib-0.65-477.amzn2023.0.7.x86_64.rpm                             628 kB/s |  15 kB     00:00
(8/8): git-core-2.47.1-1.amzn2023.0.3.x86_64.rpm                              25 MB/s | 4.5 MB     00:00
--------------------------------------------------------------------------------------------------------------
Total                                                                         35 MB/s | 7.5 MB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                                      1/1
  Installing       : git-core-2.47.1-1.amzn2023.0.3.x86_64                                                1/8
```

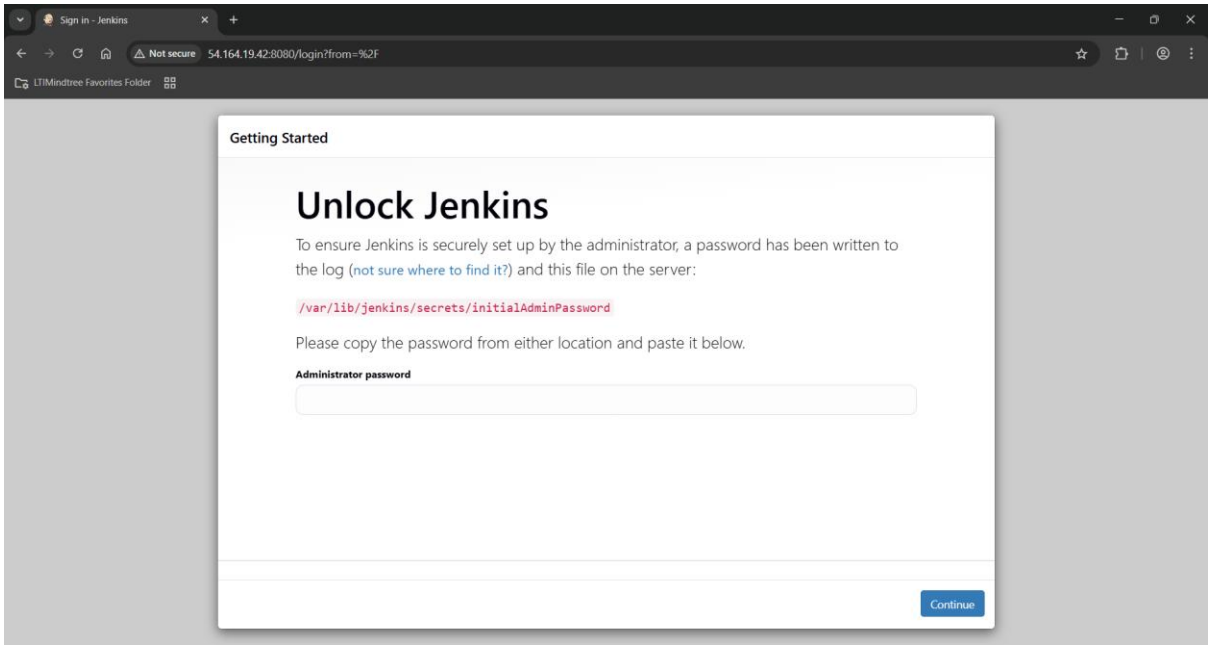
Git package is installed.

```
root@ip-172-31-25-209:~
[root@jenkins ~]# yum install maven
Last metadata expiration check: 0:05:07 ago on Fri Jun 27 04:45:59 2025.
Dependencies resolved.
==============================================================================================================
 Package                                   Architecture   Version                       Repository       Size
==============================================================================================================
Installing:
 maven                                     noarch         1:3.8.4-3.amzn2023.0.5        amazonlinux      18 k
Installing dependencies:
 apache-commons-cli                        noarch         1.5.0-3.amzn2023.0.3          amazonlinux      76 k
 apache-commons-codec                      noarch         1.15-6.amzn2023.0.3           amazonlinux     303 k
 apache-commons-io                         noarch         1:2.8.0-7.amzn2023.0.5        amazonlinux     283 k
 apache-commons-lang3                      noarch         3.12.0-7.amzn2023.0.3         amazonlinux     559 k
 atinject                                  noarch         1.0.5-3.amzn2023.0.3          amazonlinux      23 k
 cdi-api                                   noarch         2.0.2-6.amzn2023.0.3          amazonlinux      54 k
 google-guice                              noarch         4.2.3-8.amzn2023.0.6          amazonlinux     473 k
 guava                                     noarch         31.0.1-3.amzn2023.0.6         amazonlinux     2.4 M
 httpcomponents-client                     noarch         4.5.13-4.amzn2023.0.4         amazonlinux     657 k
 httpcomponents-core                       noarch         4.4.13-6.amzn2023.0.3         amazonlinux     632 k
 jakarta-annotations                       noarch         1.3.5-13.amzn2023.0.3         amazonlinux      46 k
 jansi                                     x86_64         2.4.0-3.amzn2023.0.3          amazonlinux     113 k
 java-17-amazon-corretto-devel             x86_64         1:17.0.15+6-1.amzn2023.1      amazonlinux     142 k
 jcl-over-slf4j                            noarch         1.7.32-3.amzn2023.0.4         amazonlinux      25 k
 jsoup                                     noarch         1.16.1-4.amzn2023.0.2         amazonlinux     433 k
 jsr-305                                   noarch         3.0.2-5.amzn2023.0.4          amazonlinux      32 k
 maven-amazon-corretto17                   noarch         1:3.8.4-3.amzn2023.0.5        amazonlinux     9.4 k
 maven-lib                                 noarch         1:3.8.4-3.amzn2023.0.5        amazonlinux     1.5 M
 maven-resolver                            noarch         1:1.7.3-3.amzn2023.0.4        amazonlinux     557 k
 maven-shared-utils                        noarch         3.3.4-4.amzn2023.0.3          amazonlinux     152 k
 maven-wagon                               noarch         3.4.2-6.amzn2023.0.4          amazonlinux     113 k
 plexus-cipher                             noarch         1.8-3.amzn2023.0.3            amazonlinux      27 k
 plexus-classworlds                        noarch         2.6.0-10.amzn2023.0.4         amazonlinux      61 k
 plexus-containers-component-annotations   noarch         2.1.0-9.amzn2023.0.4          amazonlinux      19 k
 plexus-interpolation                      noarch         1.26-10.amzn2023.0.4          amazonlinux      80 k
 plexus-sec-dispatcher                     noarch         2.0-3.amzn2023.0.3            amazonlinux      34 k
 plexus-utils                              noarch         3.3.0-9.amzn2023.0.4          amazonlinux     254 k
 publicsuffix-list                         noarch         20240212-61.amzn2023          amazonlinux      89 k
 sisu                                      noarch         1:0.3.4-9.amzn2023.0.4        amazonlinux     510 k
 slf4j                                     noarch         1.7.32-3.amzn2023.0.4         amazonlinux      70 k

Transaction Summary
```

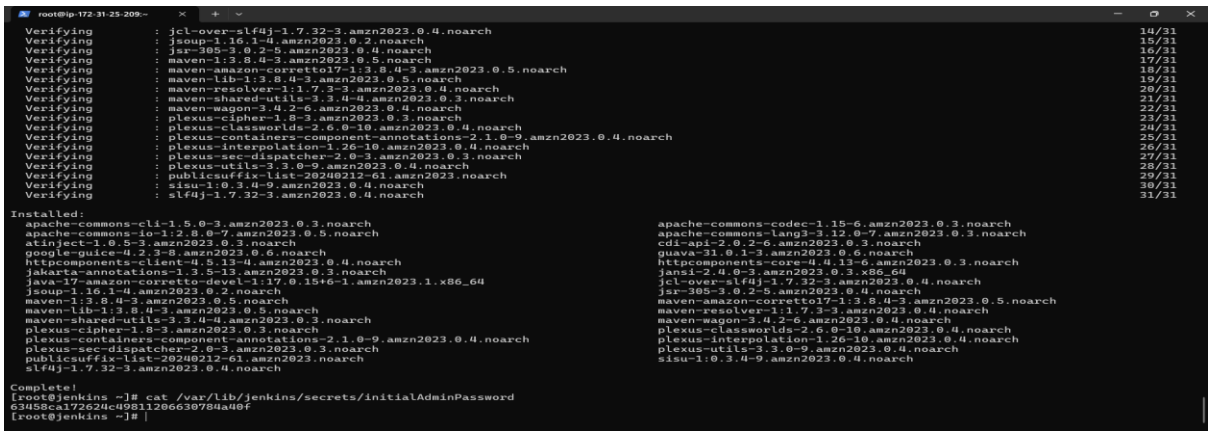
Maven package is installed.



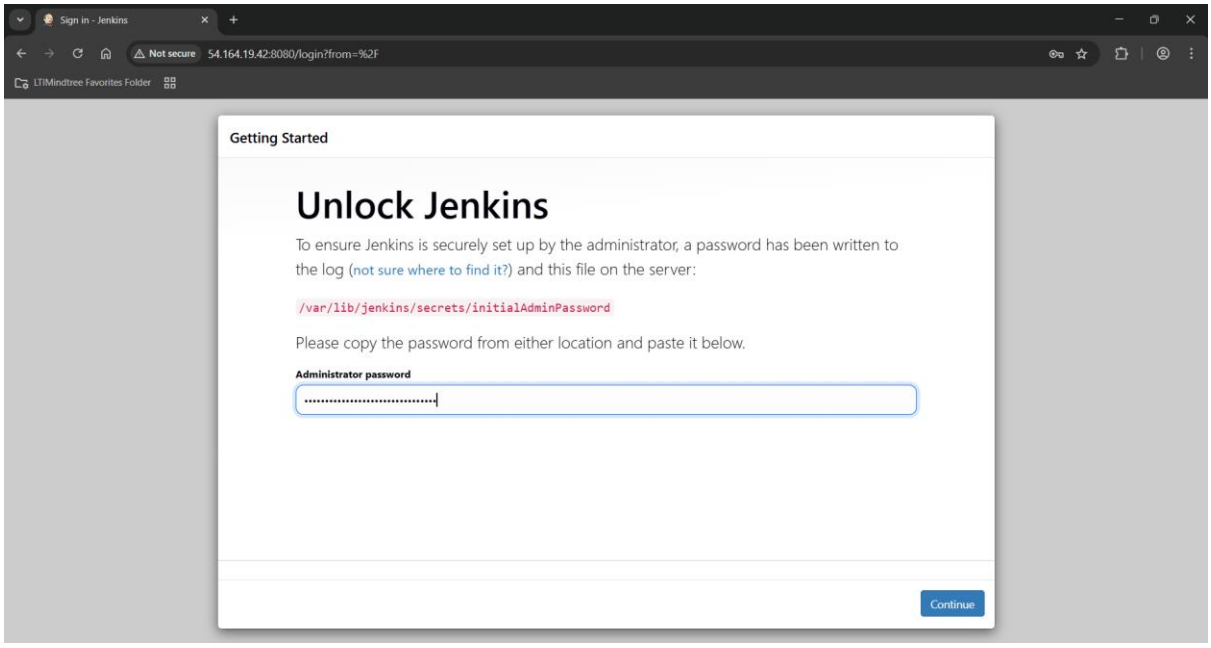
Jenkins public IP is copied.



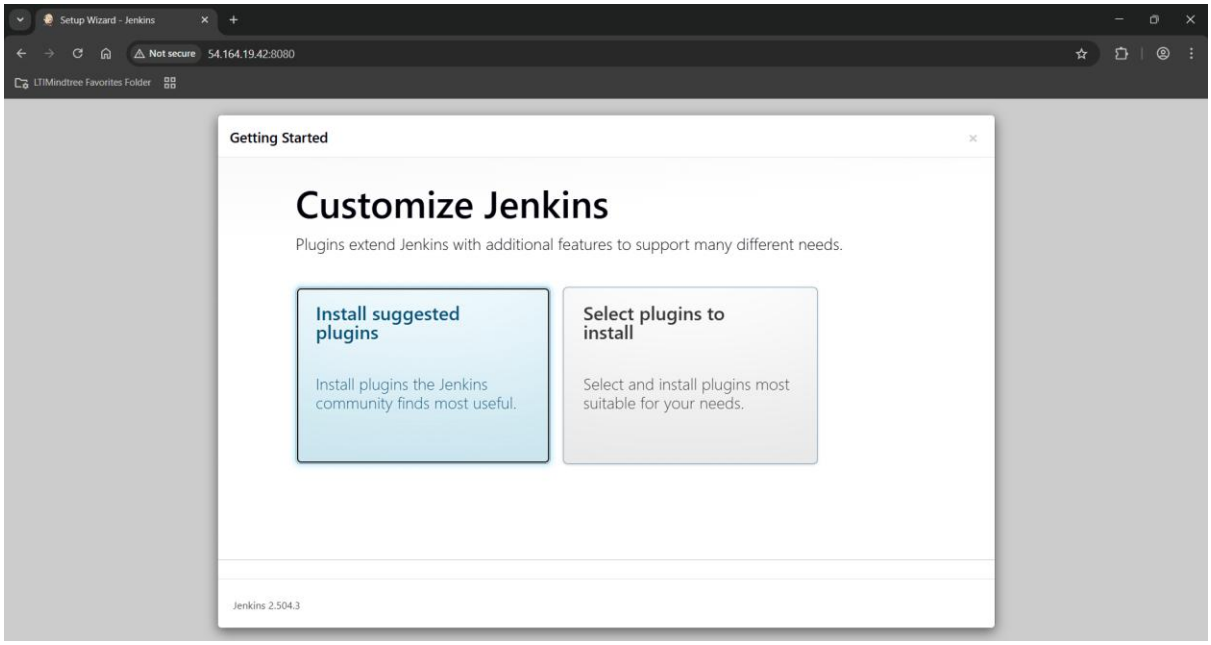
Through Jenkins public IP and port 8080, Jenkins dashboard is opened.



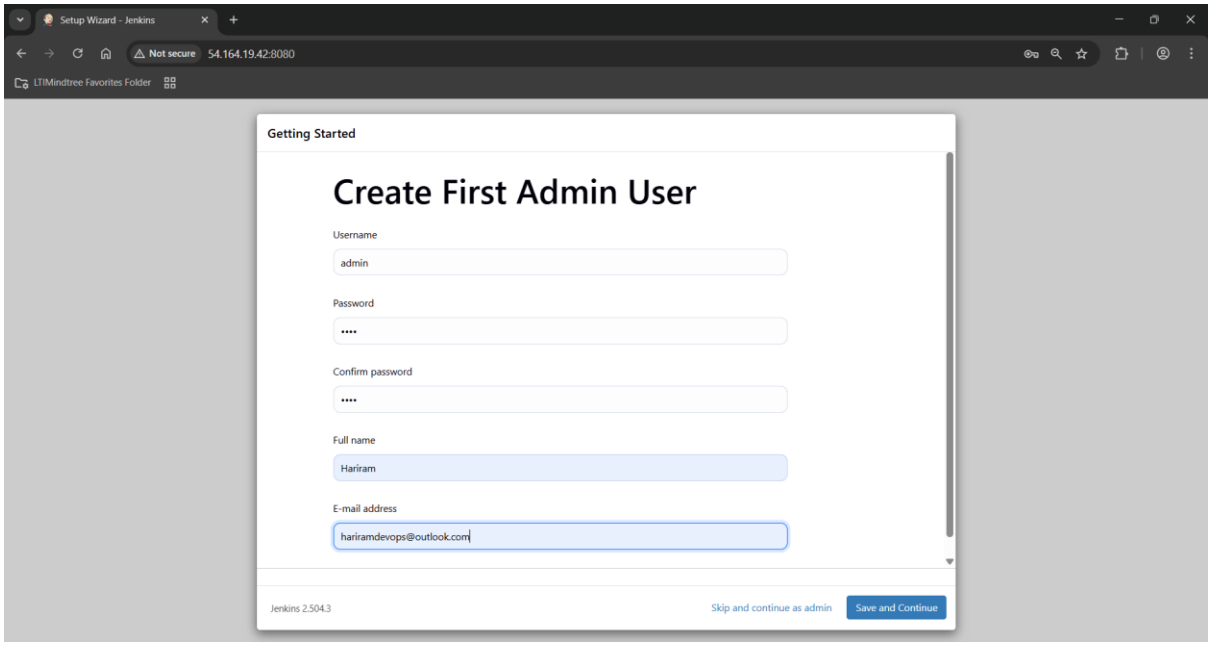
Administrator password is taken from Jenkins machine.



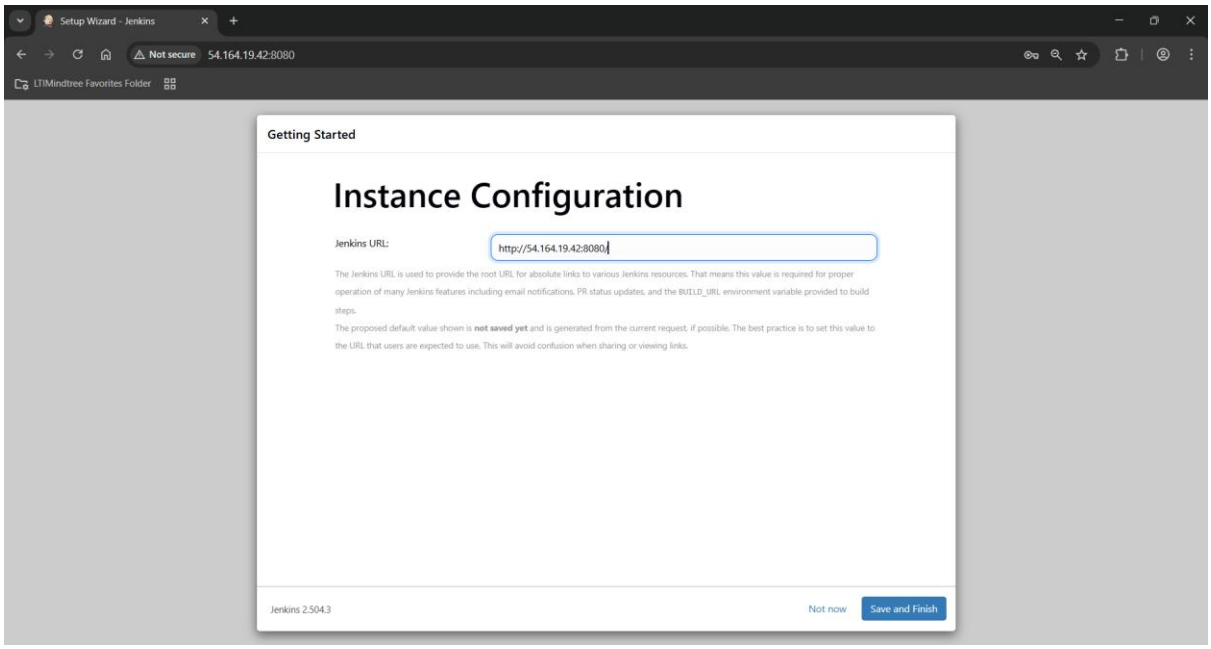
Pasted in Jenkins dashboard to access it.



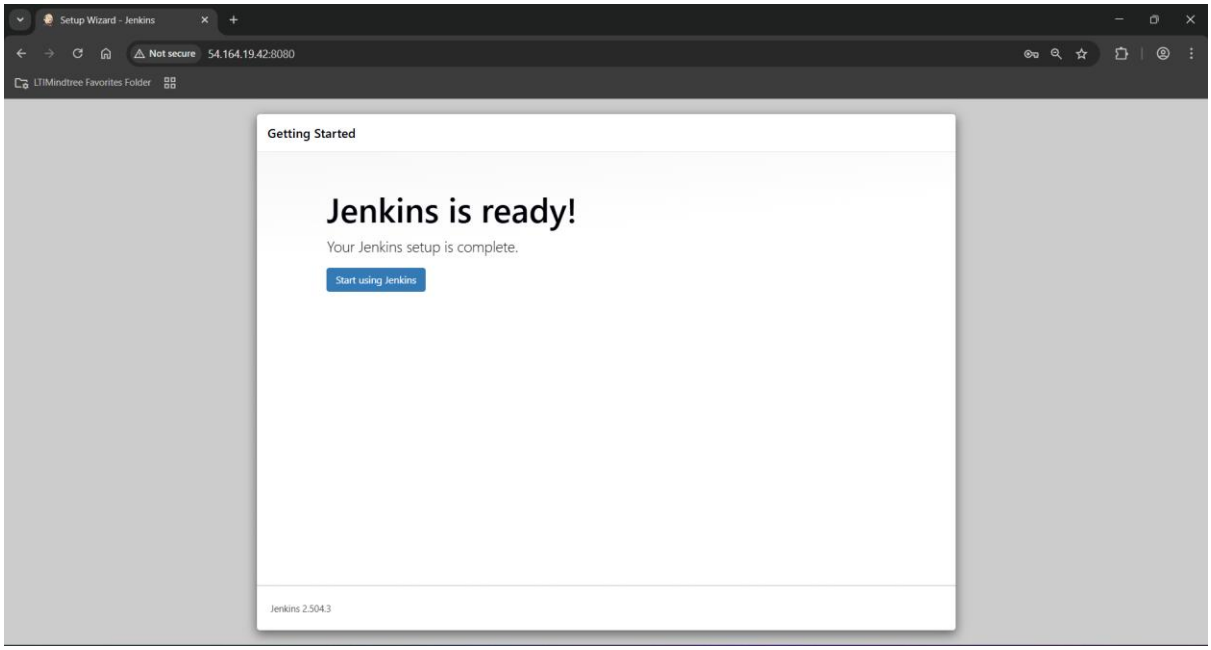
Install suggested plugins.



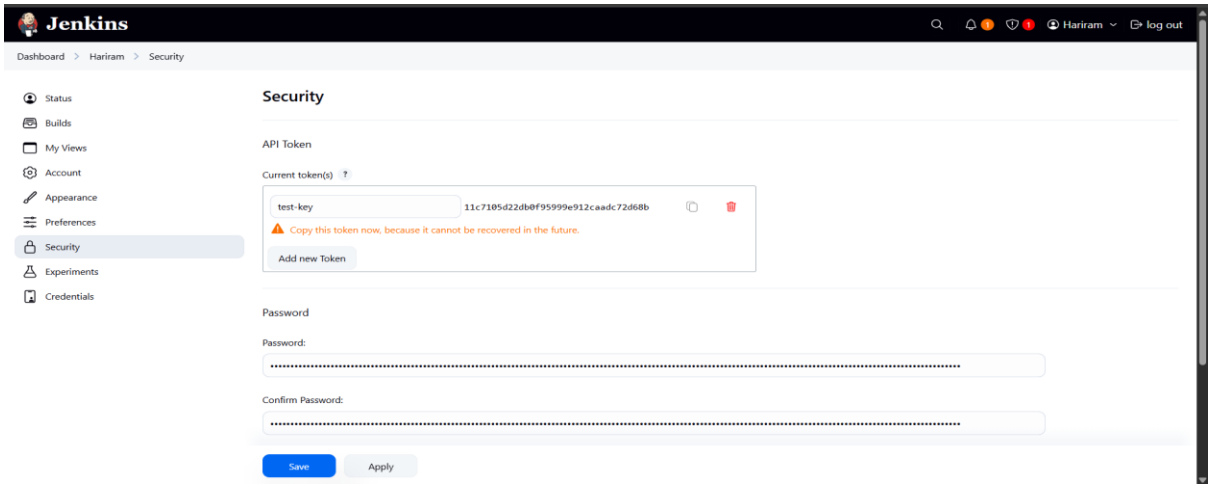
Created admin user with username: admin and password: admin.



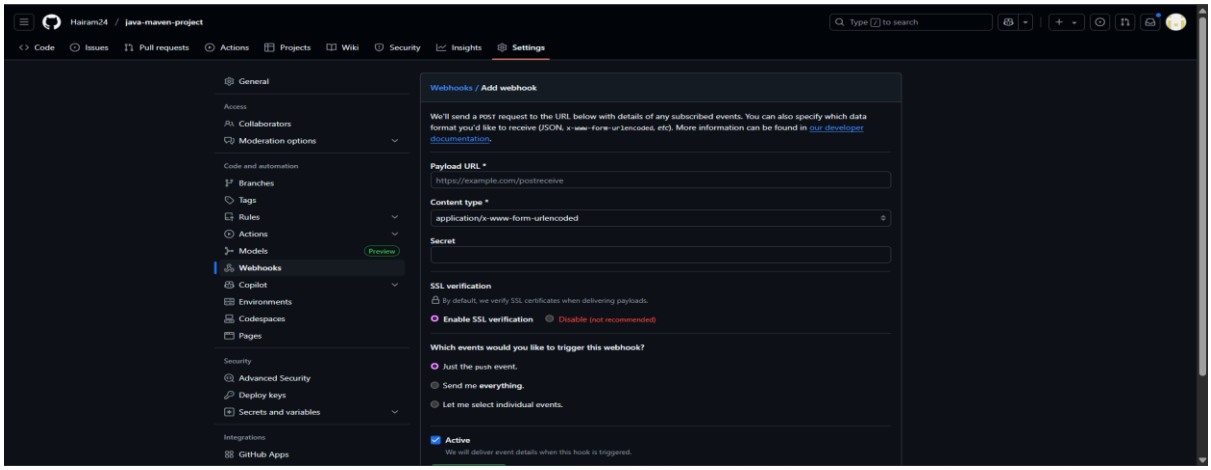
This is the Jenkins url.



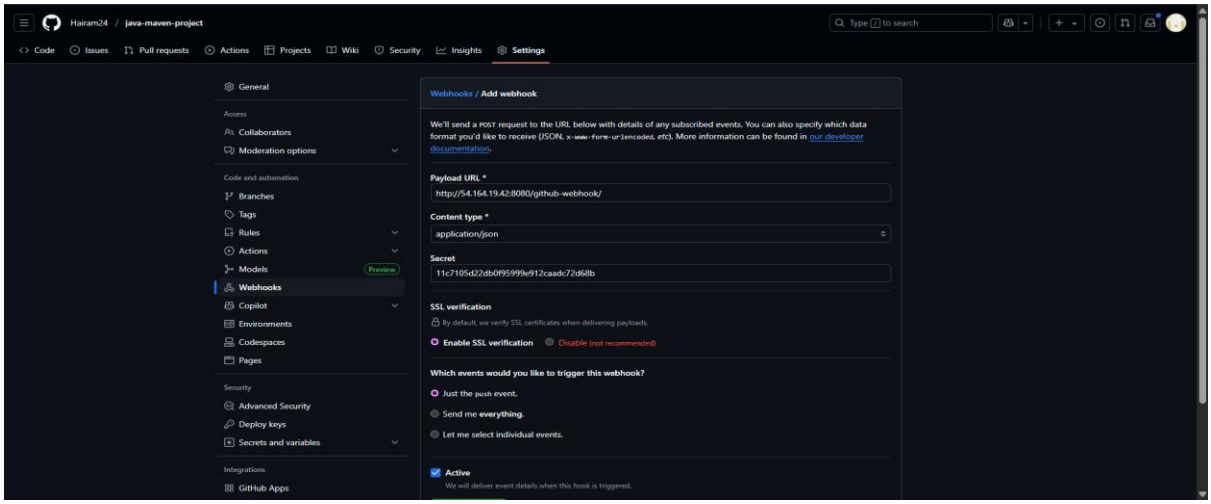
Jenkins is ready to use.



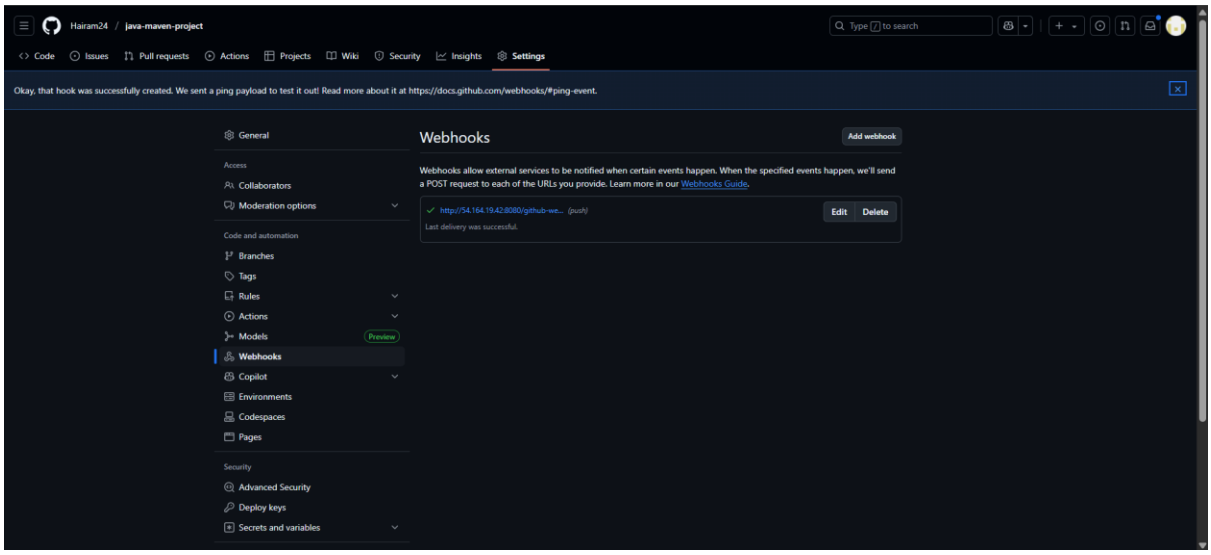
In dashboard > profile > security , create a API token.



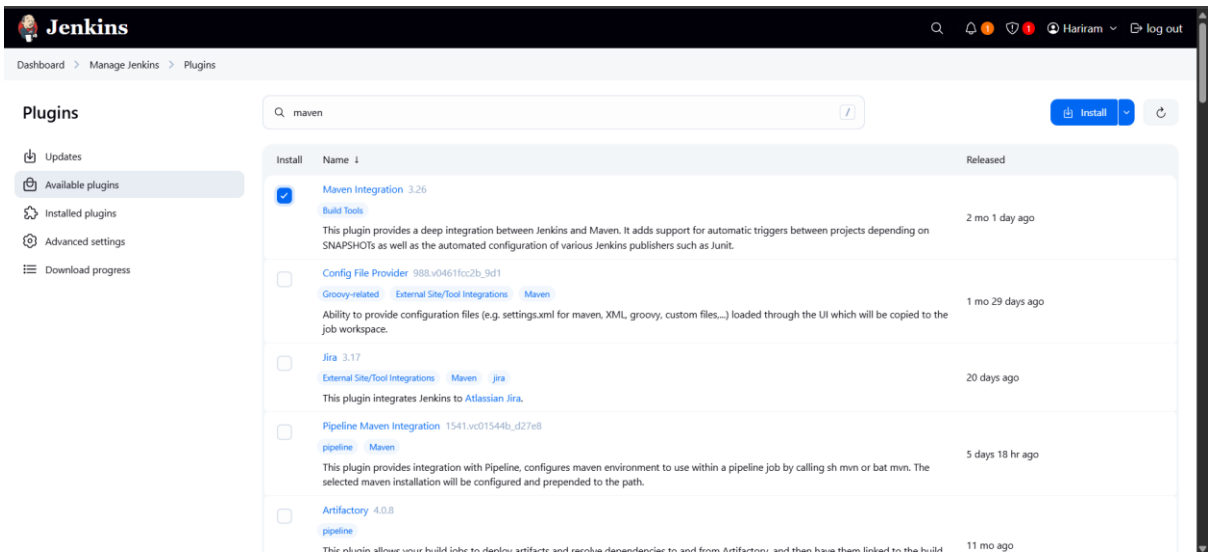
API key pasted in repo setting > webhook.



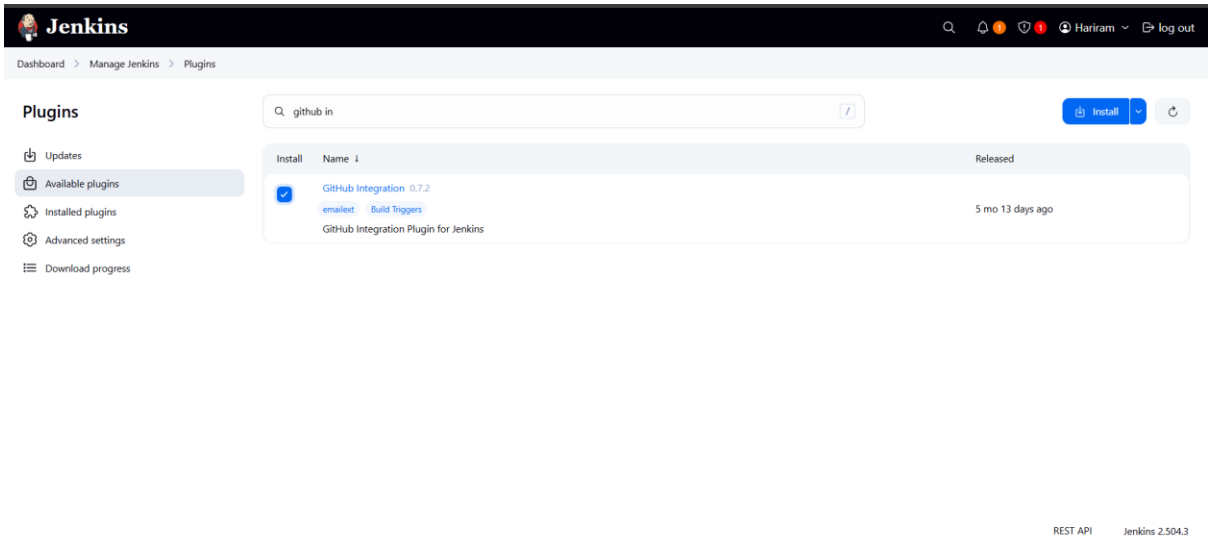
Connection type : application/json is selected.



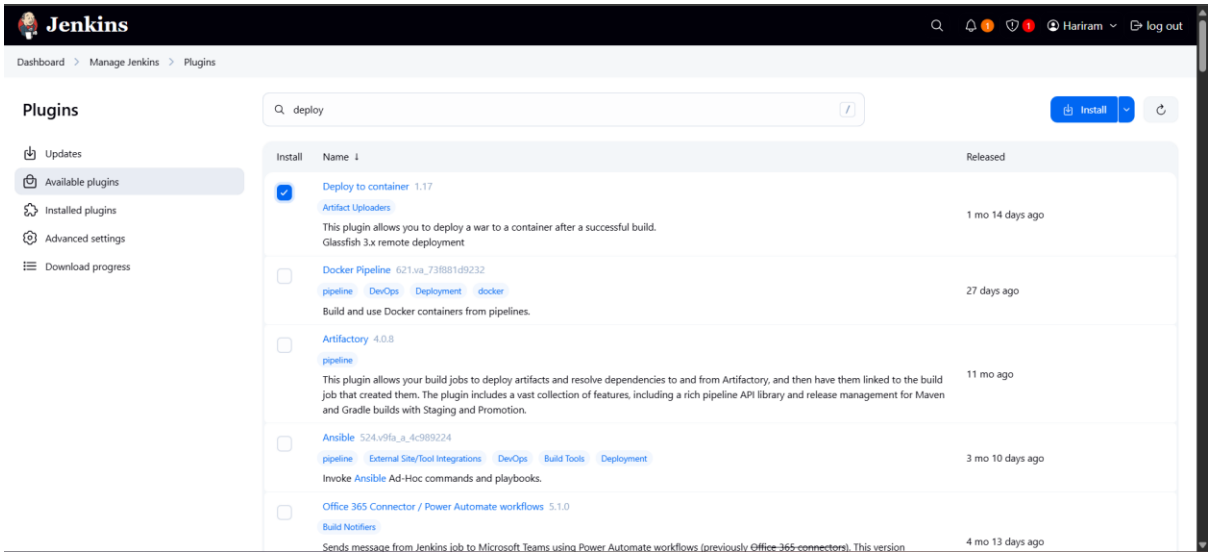
Webhook is created successfully.



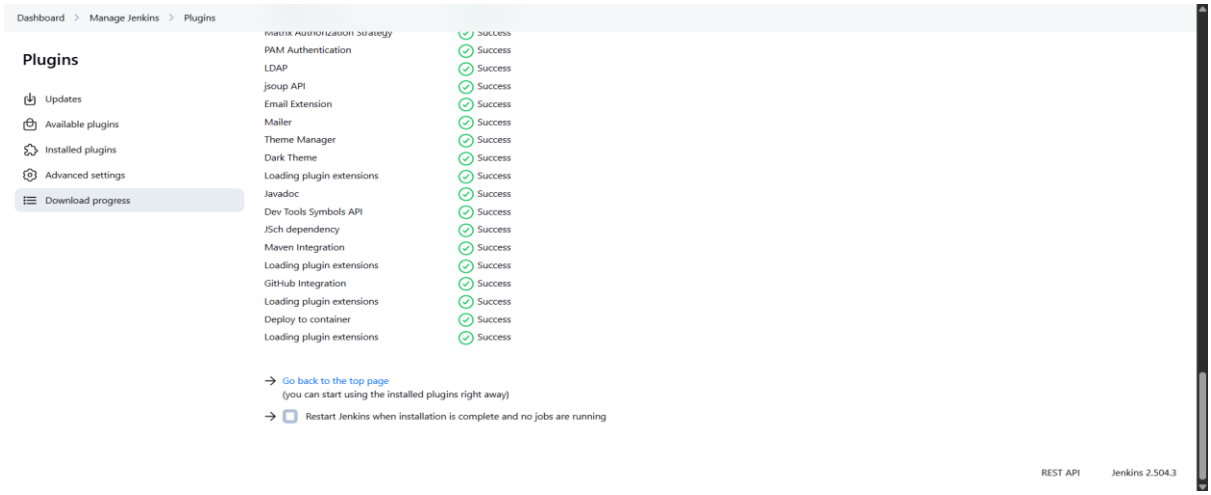
Maven plugin is installed.



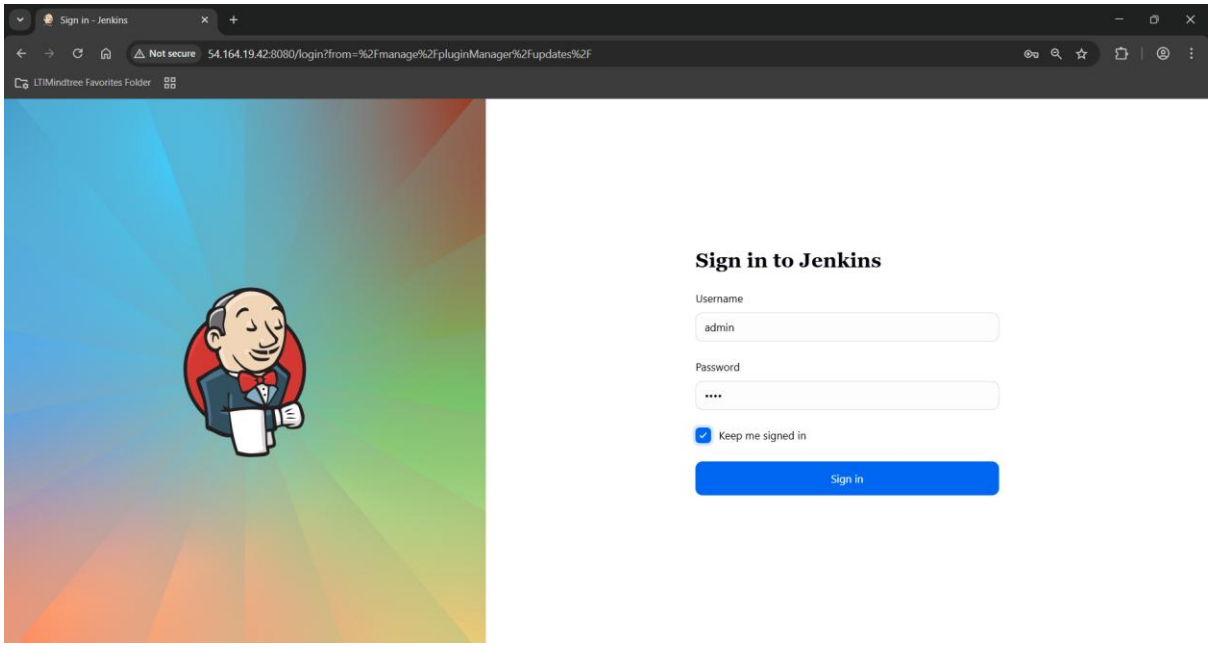
GitHub integration plugin is installed.



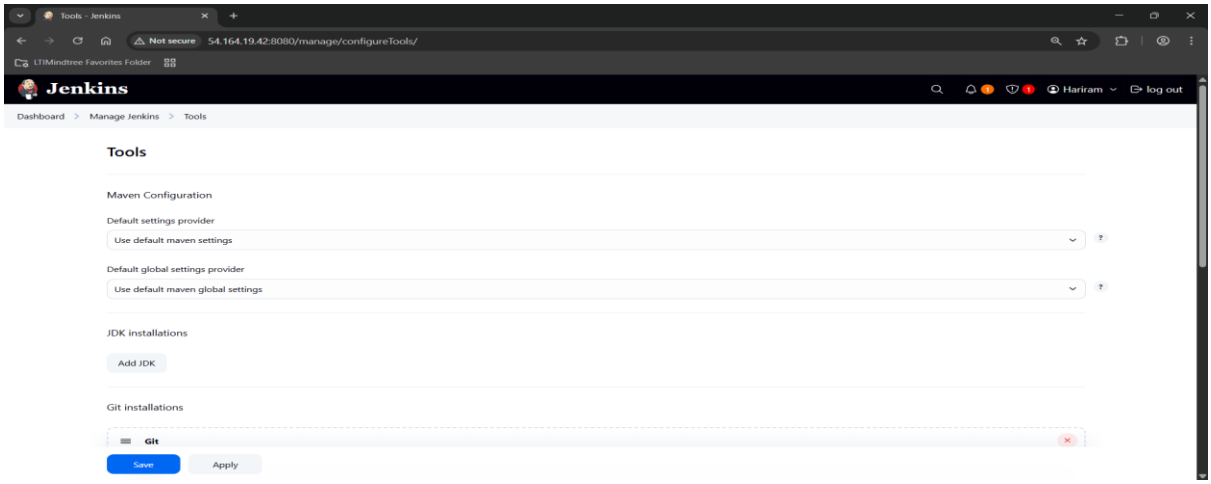
Deploy to container plugin is installed.



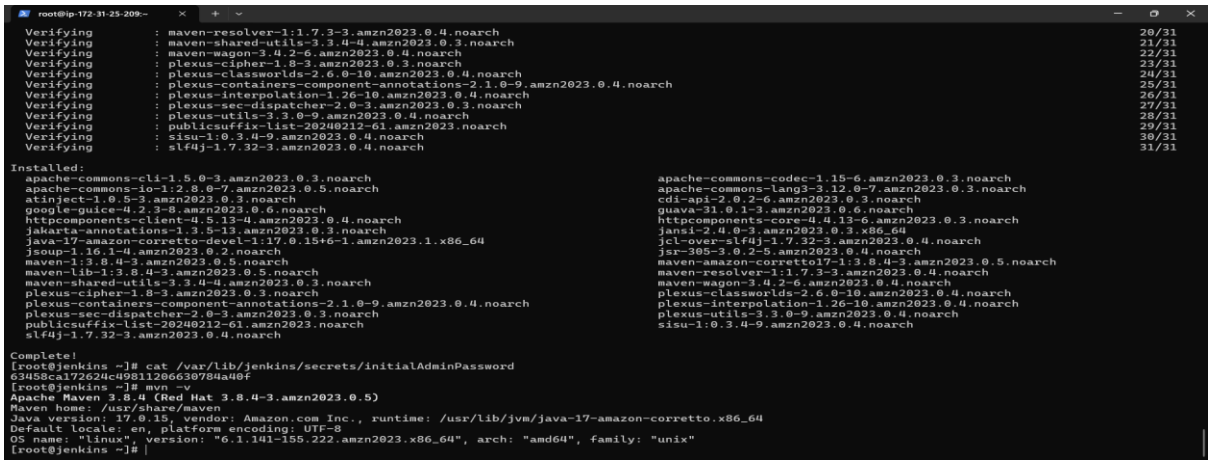
Restarted the Jenkins.



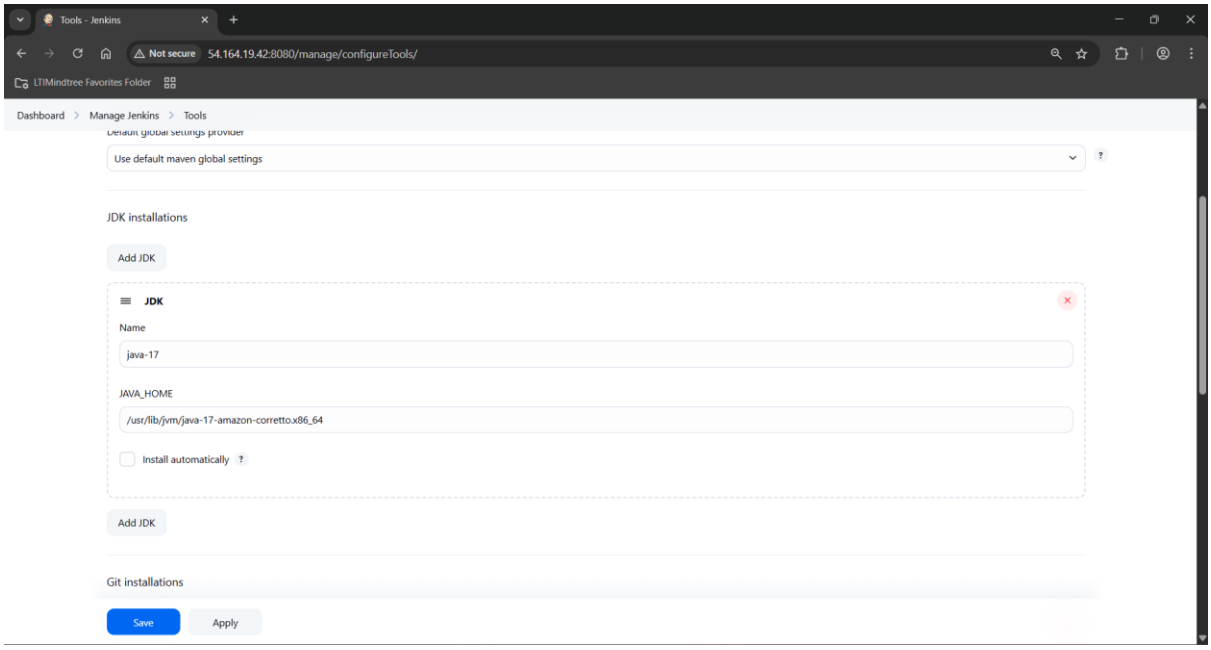
Logged in to the Jenkins.



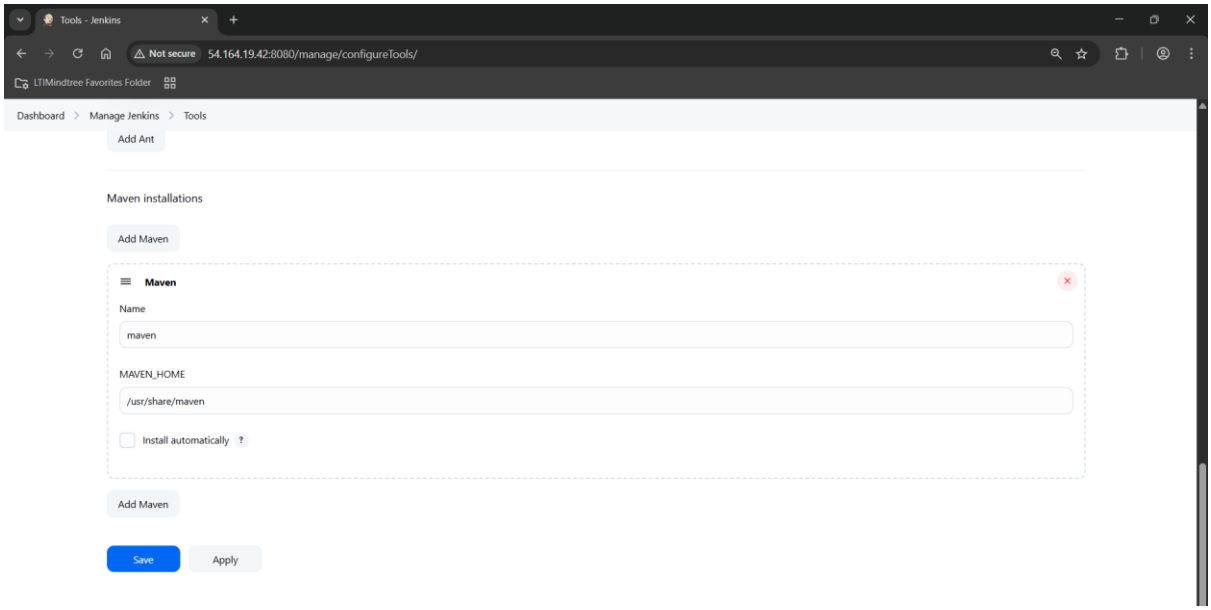
In manage Jenkins > tools add the JDK and Maven path.



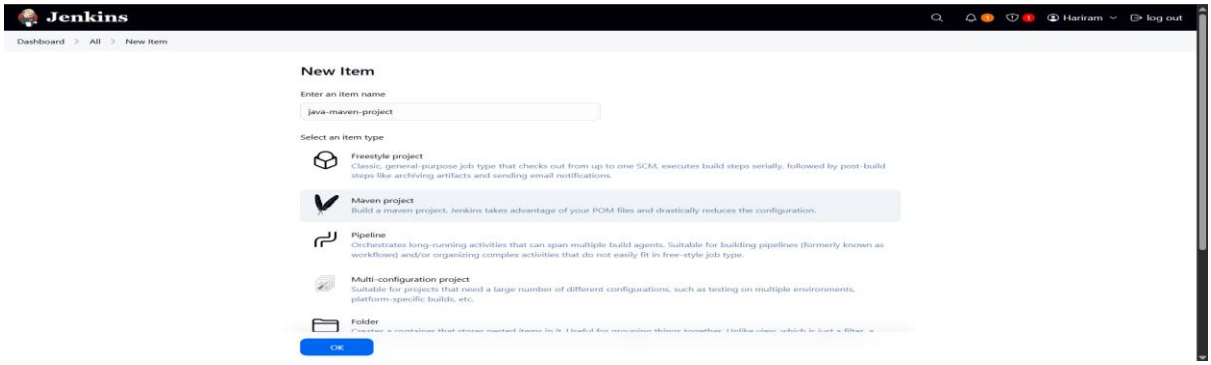
Using mvn -v java and maven path is copied.



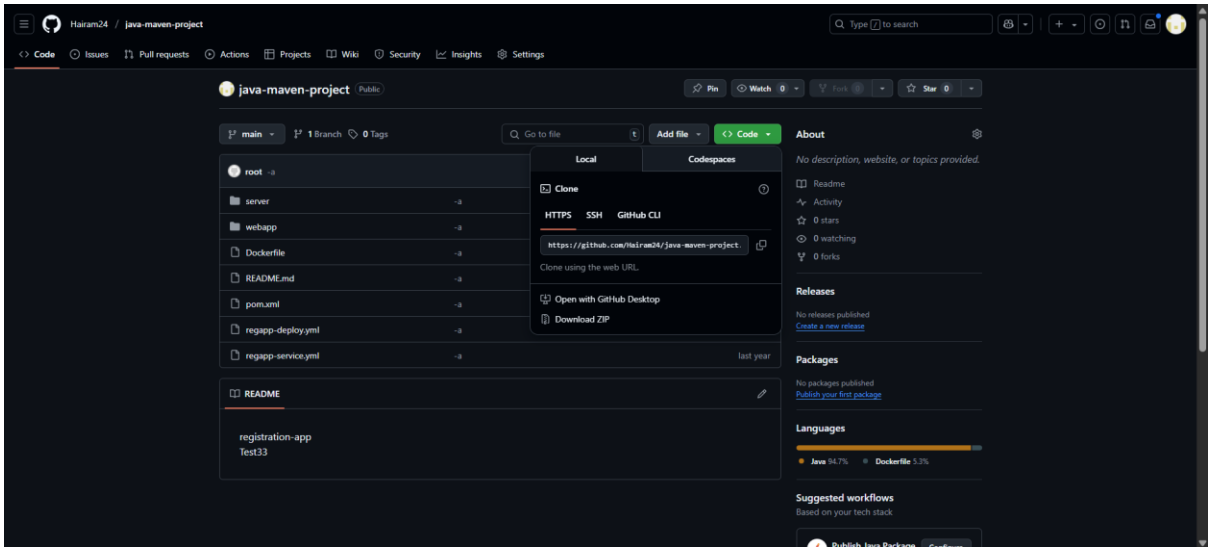
JDK path is added.



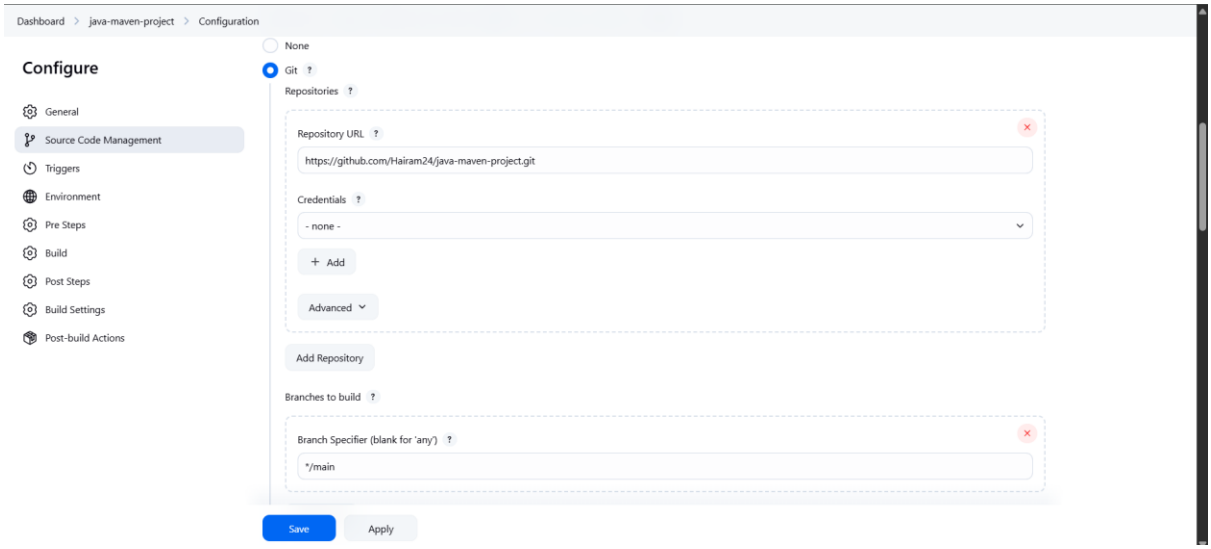
Maven path is added.



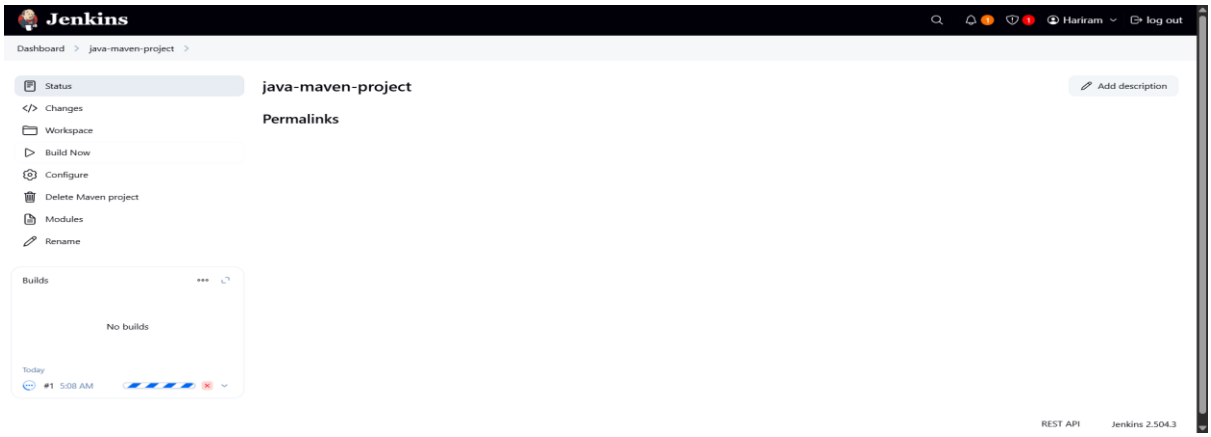
Create a new maven project.



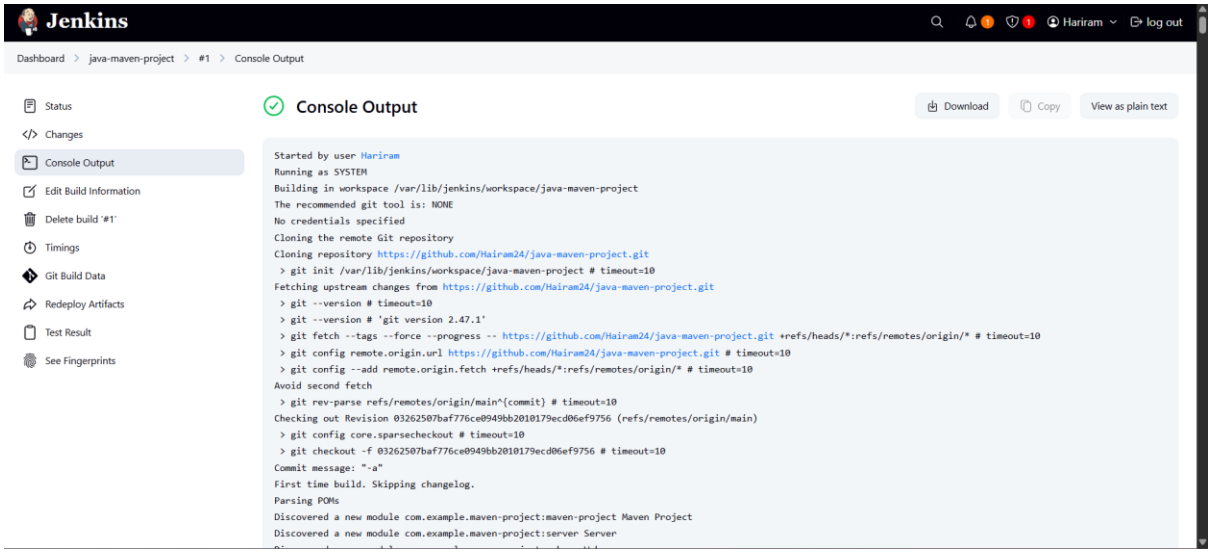
Copy the https repo link.



Add the git repository.

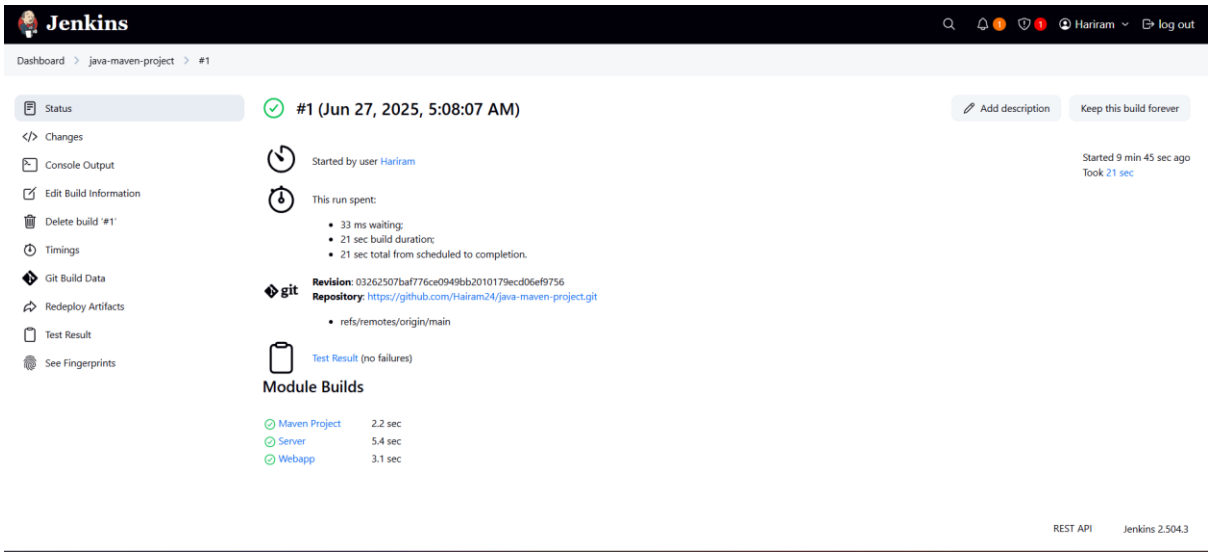


Build now to check.



Build successfully.

Build finished success.



```
root@jenkins:/var/lib/jenkins
-rw-r--r--.   1 jenkins jenkins   246 Jun 28 12:21 org.jenkinsci.plugins.workflow.libs.GlobalUntrustedLibraries.xml
drwxr-xr-x. 102 jenkins jenkins 16384 Jun 27 10:59 plugins
-rw-r--r--.   1 jenkins jenkins   259 Jun 28 12:46 queue.xml
-rw-r--r--.   1 jenkins jenkins   259 Jun 28 04:37 queue.xml.bak
-rw-r--r--.   1 jenkins jenkins    64 Jun 27 04:49 secret.key
-rw-r--r--.   1 jenkins jenkins     0 Jun 27 04:49 secret.key.not-so-secret
drwx------.   2 jenkins jenkins 16384 Jun 27 05:16 secrets
drwxr-xr-x.   2 jenkins jenkins   182 Jun 30 04:37 updates
drwxr-xr-x.   2 jenkins jenkins    24 Jun 27 04:49 userContent
drwxr-xr-x.   4 jenkins jenkins    92 Jun 27 06:11 users
drwxr-xr-x.   3 jenkins jenkins    32 Jun 27 05:08 workspace
[root@jenkins jenkins]# cd workspace
[root@jenkins workspace]# ll
total 0
drwxr-xr-x. 5 jenkins jenkins 147 Jun 28 12:34 java-maven-project
[root@jenkins workspace]# cd java-maven-project
[root@jenkins java-maven-project]# llk
-bash: llk: command not found
[root@jenkins java-maven-project]# ll
total 24
-rw-r--r--. 1 jenkins jenkins  143 Jun 28 12:34 Dockerfile
-rw-r--r--. 1 jenkins jenkins   29 Jun 27 05:08 README.md
-rw-r--r--. 1 jenkins jenkins 6333 Jun 27 05:08 pom.xml
-rw-r--r--. 1 jenkins jenkins  488 Jun 27 05:08 regapp-deploy.yml
-rw-r--r--. 1 jenkins jenkins  196 Jun 27 05:08 regapp-service.yml
drwxr-xr-x. 4 jenkins jenkins   46 Jun 27 05:08 server
drwxr-xr-x. 4 jenkins jenkins   46 Jun 27 05:08 webapp
[root@jenkins java-maven-project]# cd webapp
[root@jenkins webapp]# ll
total 4
-rw-r--r--. 1 jenkins jenkins 1141 Jun 27 05:08 pom.xml
drwxr-xr-x. 3 jenkins jenkins   18 Jun 27 05:08 src
drwxr-xr-x. 5 jenkins jenkins   76 Jun 27 05:08 target
[root@jenkins webapp]# cd  target
[root@jenkins target]# ll
total 4
drwxr-xr-x. 2 jenkins jenkins   28 Jun 27 05:08 maven-archiver
drwxr-xr-x. 2 jenkins jenkins    6 Jun 28 12:46 surefire
drwxr-xr-x. 4 jenkins jenkins   54 Jun 28 12:46 webapp
-rw-r--r--. 1 jenkins jenkins 2421 Jun 28 12:46 webapp.war
[root@jenkins target]#
```

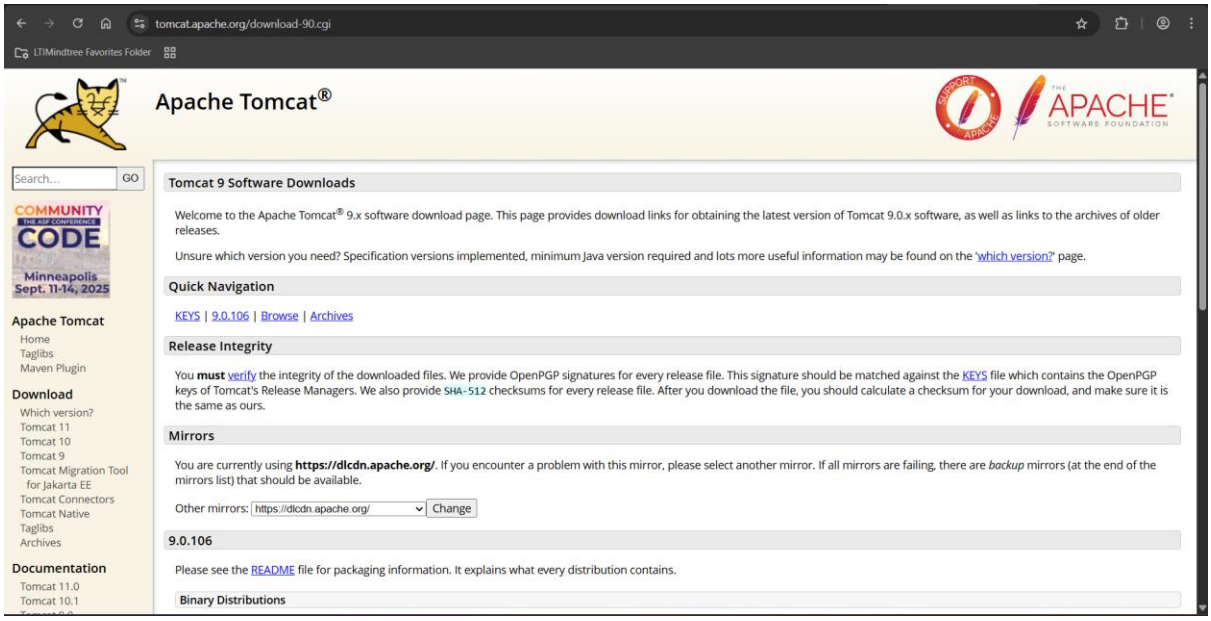
WAR file created successfully.

```
root@ip-172-31-17-39:~
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\10835505\OneDrive - LTIMindtree\Desktop> ssh -i "aws_Ltimindtree.pem" ec2-user@ec2-18-234-223-7.compute-1.amazonaws.com
The authenticity of host 'ec2-18-234-223-7.compute-1.amazonaws.com (18.234.223.7)' can't be established.
ED25519 key fingerprint is SHA256:jl/WvEHkJs4W6x/h7lJavUnmbrL8MT+xS5fVD8DGzsI.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-18-234-223-7.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
       #_
       ~\_  ####_        Amazon Linux 2023
      ~~  \_#####\
      ~~     \###|
      ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
       ~~       V~' '->
        ~~~         /
          ~~._.   _/
             _/ _/
           _/m/'
[ec2-user@ip-172-31-17-39 ~]$ sudo su -
[root@ip-172-31-17-39 ~]# hostnamectl set-hostname tomcat.example.com
[root@ip-172-31-17-39 ~]# bash
[root@tomcat ~]#
```

Tomcat machine is connected, and hostname is changed.

```
root@ip-172-31-17-39:~
[root@tomcat ~]# yum install java-1.8*
Amazon Linux 2023 Kernel Livepatch repository                                                    176 kB/s |  17 kB     00:00
Dependencies resolved.
=================================================================================================================================
 Package                                  Architecture        Version                                  Repository          Size
=================================================================================================================================
Installing:
 java-1.8.0-amazon-corretto               x86_64              1:1.8.0_452.b09-2.amzn2023               amazonlinux          38 M
 java-1.8.0-amazon-corretto-devel         x86_64              1:1.8.0_452.b09-2.amzn2023               amazonlinux          63 M
Installing dependencies:
 adwaita-cursor-theme                     noarch              47.0-1.amzn2023.0.1                      amazonlinux         325 k
 adwaita-icon-theme                       noarch              47.0-1.amzn2023.0.1                      amazonlinux         285 k
 adwaita-icon-theme-legacy                noarch              46.2-2.amzn2023                          amazonlinux         2.2 M
 alsa-lib                                 x86_64              1.2.7.2-1.amzn2023.0.2                   amazonlinux         504 k
 at-spi2-atk                              x86_64              2.54.0-1.amzn2023.0.1                    amazonlinux          90 k
 at-spi2-core                             x86_64              2.54.0-1.amzn2023.0.1                    amazonlinux         363 k
 atk                                      x86_64              2.54.0-1.amzn2023.0.1                    amazonlinux          85 k
 avahi-glib                               x86_64              0.8-14.amzn2023.0.14                     amazonlinux          15 k
 avahi-libs                               x86_64              0.8-14.amzn2023.0.14                     amazonlinux          68 k
 bluez-libs                               x86_64              5.62-2.amzn2023.0.5                      amazonlinux          83 k
 cairo                                    x86_64              1.18.0-4.amzn2023.0.1                    amazonlinux         718 k
 cairo-gobject                            x86_64              1.18.0-4.amzn2023.0.1                    amazonlinux          20 k
 colord-libs                              x86_64              1.4.5-2.amzn2023.0.2                     amazonlinux         235 k
 cups-filesystem                          noarch              1:2.4.11-8.amzn2023.0.1                  amazonlinux          15 k
 cups-libs                                x86_64              1:2.4.11-8.amzn2023.0.1                  amazonlinux         274 k
 dejavu-sans-fonts                        noarch              2.37-16.amzn2023.0.2                     amazonlinux         1.3 M
 dejavu-sans-mono-fonts                   noarch              2.37-16.amzn2023.0.2                     amazonlinux         467 k
 dejavu-serif-fonts                       noarch              2.37-16.amzn2023.0.2                     amazonlinux         1.0 M
 duktape                                  x86_64              2.7.0-21.amzn2023                        amazonlinux         171 k
 exiv2-libs                               x86_64              0.28.5-128.amzn2023                      amazonlinux         904 k
 fdk-aac-free                             x86_64              2.0.0-14.amzn2023.0.1                    amazonlinux         338 k
 flac-libs                                x86_64              1.3.4-1.amzn2023.0.2                     amazonlinux         234 k
 fontconfig                               x86_64              2.13.94-2.amzn2023.0.2                   amazonlinux         273 k
 fonts-filesystem                         noarch              1:2.0.5-12.amzn2023.0.2                  amazonlinux         9.5 k
 freetype                                 x86_64              2.13.2-5.amzn2023.0.1                    amazonlinux         423 k
 fribidi                                  x86_64              1.0.11-3.amzn2023.0.2                    amazonlinux          54 k
 fuse-common                              x86_64              3.10.4-1.amzn2023.0.2                    amazonlinux         8.5 k
 fuse3                                    x86_64              3.10.4-1.amzn2023.0.2                    amazonlinux          54 k
 fuse3-libs                               x86_64              3.10.4-1.amzn2023.0.2                    amazonlinux          92 k
 gdk-pixbuf2                              x86_64              2.42.12-180.amzn2023                     amazonlinux         471 k
 gdk-pixbuf2-modules                      x86_64              2.42.12-180.amzn2023                     amazonlinux          29 k
```

Java package is installed.



Copy the tomcat 9 tar file link.

```
[root@tomcat ~]# wget https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.106/bin/apache-tomcat-9.0.106.tar.gz
--2025-06-27 05:26:35--  https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.106/bin/apache-tomcat-9.0.106.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 13007554 (12M) [application/x-gzip]
Saving to: 'apache-tomcat-9.0.106.tar.gz'

apache-tomcat-9.0.106.tar.gz          100%[===================================================================>]  12.40M  --.-KB/s    in 0.09s

2025-06-27 05:26:35 (131 MB/s) - 'apache-tomcat-9.0.106.tar.gz' saved [13007554/13007554]

[root@tomcat ~]#
```

Download the tar file.

```
[root@tomcat ~]# tar -zvxf apache-tomcat-9.0.106.tar.gz
apache-tomcat-9.0.106/conf/
apache-tomcat-9.0.106/conf/catalina.policy
apache-tomcat-9.0.106/conf/catalina.properties
apache-tomcat-9.0.106/conf/context.xml
apache-tomcat-9.0.106/conf/jaspic-providers.xml
apache-tomcat-9.0.106/conf/jaspic-providers.xsd
apache-tomcat-9.0.106/conf/logging.properties
apache-tomcat-9.0.106/conf/server.xml
apache-tomcat-9.0.106/conf/tomcat-users.xml
apache-tomcat-9.0.106/conf/tomcat-users.xsd
apache-tomcat-9.0.106/conf/web.xml
apache-tomcat-9.0.106/bin/
apache-tomcat-9.0.106/lib/
apache-tomcat-9.0.106/logs/
apache-tomcat-9.0.106/temp/
apache-tomcat-9.0.106/webapps/
apache-tomcat-9.0.106/webapps/ROOT/
apache-tomcat-9.0.106/webapps/ROOT/WEB-INF/
apache-tomcat-9.0.106/webapps/docs/
apache-tomcat-9.0.106/webapps/docs/META-INF/
apache-tomcat-9.0.106/webapps/docs/WEB-INF/
apache-tomcat-9.0.106/webapps/docs/WEB-INF/jsp/
apache-tomcat-9.0.106/webapps/docs/annotationapi/
apache-tomcat-9.0.106/webapps/docs/api/
apache-tomcat-9.0.106/webapps/docs/appdev/
apache-tomcat-9.0.106/webapps/docs/appdev/sample/
apache-tomcat-9.0.106/webapps/docs/appdev/sample/docs/
apache-tomcat-9.0.106/webapps/docs/appdev/sample/src/
apache-tomcat-9.0.106/webapps/docs/appdev/sample/src/mypackage/
apache-tomcat-9.0.106/webapps/docs/appdev/sample/web/
apache-tomcat-9.0.106/webapps/docs/appdev/sample/web/WEB-INF/
apache-tomcat-9.0.106/webapps/docs/appdev/sample/web/images/
apache-tomcat-9.0.106/webapps/docs/architecture/
```

Unzip the tar file.

```
[root@tomcat ~]# cd apache-tomcat-9.0.106
[root@tomcat apache-tomcat-9.0.106]# cd bin
[root@tomcat bin]# chmod +x startup.sh
[root@tomcat bin]# chmod +x shutdown.sh
[root@tomcat bin]# ln -s /opt/apache-tomcat-9.0.106/bin/startup.sh /usr/local/bin/tomcatup
[root@tomcat bin]# ln -s /opt/apache-tomcat-9.0.106/bin/shutdown.sh /usr/local/bin/tomcatdown
```

Tomcat installation is started.

```
[root@tomcat ~]# cd apache-tomcat-9.0.106
[root@tomcat apache-tomcat-9.0.106]# cd bin
[root@tomcat bin]# ./startup.sh
Using CATALINA_BASE:   /root/apache-tomcat-9.0.106
Using CATALINA_HOME:   /root/apache-tomcat-9.0.106
Using CATALINA_TMPDIR: /root/apache-tomcat-9.0.106/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /root/apache-tomcat-9.0.106/bin/bootstrap.jar:/root/apache-tomcat-9.0.106/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
[root@tomcat bin]#
```

Tomcat is started.

```
[root@tomcat apache-tomcat-9.0.106]# vi ./webapps/examples/META-INF/context.xml
```

```
<?xml version="1.0" encoding="UTF-8"?>
<!--
  Licensed to the Apache Software Foundation (ASF) under one or more
  contributor license agreements.  See the NOTICE file distributed with
  this work for additional information regarding copyright ownership.
  The ASF licenses this file to You under the Apache License, Version 2.0
  (the "License"); you may not use this file except in compliance with
  the License.  You may obtain a copy of the License at

      http://www.apache.org/licenses/LICENSE-2.0

  Unless required by applicable law or agreed to in writing, software
  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License.
-->
<Context>
  <CookieProcessor className="org.apache.tomcat.util.http.Rfc6265CookieProcessor"
                   sameSiteCookies="strict" />
  <!--  <Valve className="org.apache.catalina.valves.RemoteAddrValve"
  allow="127\.\d+\.\d+\.\d+|::1|0:0:0:0:0:0:0:1" />  -->
</Context>
~
~
~
~
~
~
~
~
~
~
-- INSERT --                                                    22,57         All
```

Commented the IP part.

```
<?xml version="1.0" encoding="UTF-8"?>
<!--
  Licensed to the Apache Software Foundation (ASF) under one or more
  contributor license agreements.  See the NOTICE file distributed with
  this work for additional information regarding copyright ownership.
  The ASF licenses this file to You under the Apache License, Version 2.0
  (the "License"); you may not use this file except in compliance with
  the License.  You may obtain a copy of the License at

      http://www.apache.org/licenses/LICENSE-2.0

  Unless required by applicable law or agreed to in writing, software
  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License.
-->
<Context antiResourceLocking="false" privileged="true" >
  <CookieProcessor className="org.apache.tomcat.util.http.Rfc6265CookieProcessor"
                   sameSiteCookies="strict" />
  <!-- <Valve className="org.apache.catalina.valves.RemoteAddrValve"
  allow="127\.\d+\.\d+\.\d+|::1|0:0:0:0:0:0:0:1" />  -->
  <Manager sessionAttributeValueClassNameFilter="java\.lang\.(?:Boolean|Integer|Long|Number|String)|org\.apache\.catalina\.filters\.Cs
rfPreventionFilter\$LruCache(?:\$1)?|java\.util\.(?:Linked)?HashMap"/>
</Context>
~
~
~
~
~
~
~
~
-- INSERT --                                                    22,57         All
```

Commented the IP part.

```
<?xml version="1.0" encoding="UTF-8"?>
<!--
  Licensed to the Apache Software Foundation (ASF) under one or more
  contributor license agreements.  See the NOTICE file distributed with
  this work for additional information regarding copyright ownership.
  The ASF licenses this file to You under the Apache License, Version 2.0
  (the "License"); you may not use this file except in compliance with
  the License.  You may obtain a copy of the License at

      http://www.apache.org/licenses/LICENSE-2.0

  Unless required by applicable law or agreed to in writing, software
  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License.
-->
<Context antiResourceLocking="false" privileged="true" >
  <CookieProcessor className="org.apache.tomcat.util.http.Rfc6265CookieProcessor"
                   sameSiteCookies="strict" />
  <!--  <Valve className="org.apache.catalina.valves.RemoteAddrValve"
  allow="127\.\d+\.\d+\.\d+|::1|0:0:0:0:0:0:0:1" />  -->
  <Manager sessionAttributeValueClassNameFilter="java\.lang\.(?:Boolean|Integer|Long|Number|String)|org\.apache\.catalina\.filters\.Cs
rfPreventionFilter\$LruCache(?:\$1)?|java\.util\.(?:Linked)?HashMap"/>
</Context>
~
~
~
~
~
~
~
~
~
-- INSERT --                                                                    22,57         All
```

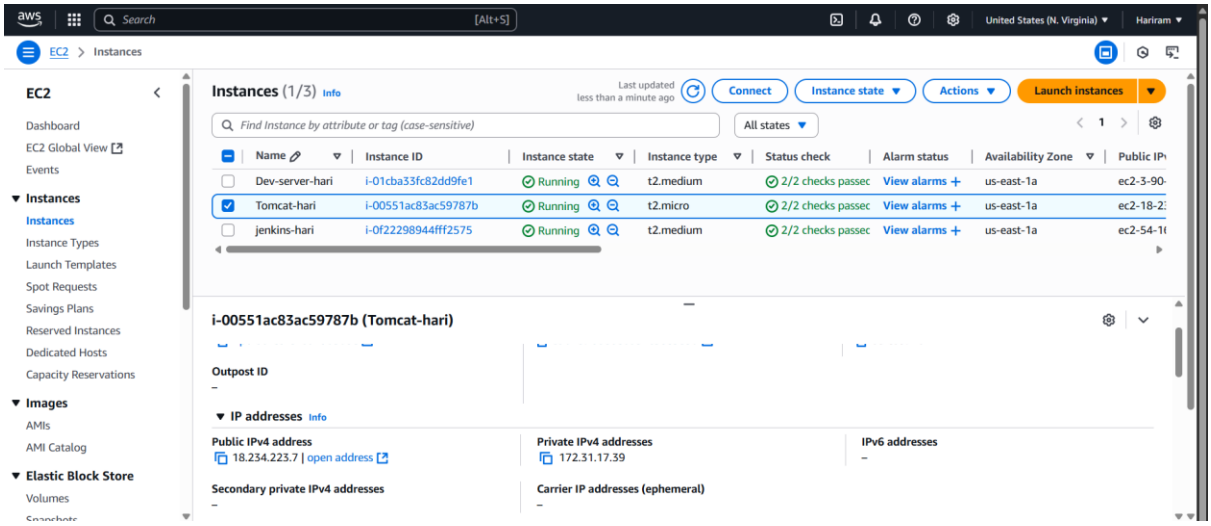
Commented the IP part.

```
[root@tomcat apache-tomcat-9.0.106]# vi ./webapps/examples/META-INF/context.xml
[root@tomcat apache-tomcat-9.0.106]# vi ./webapps/host-manager/META-INF/context.xml
[root@tomcat apache-tomcat-9.0.106]# vi ./webapps/manager/META-INF/context.xml
[root@tomcat apache-tomcat-9.0.106]#
```

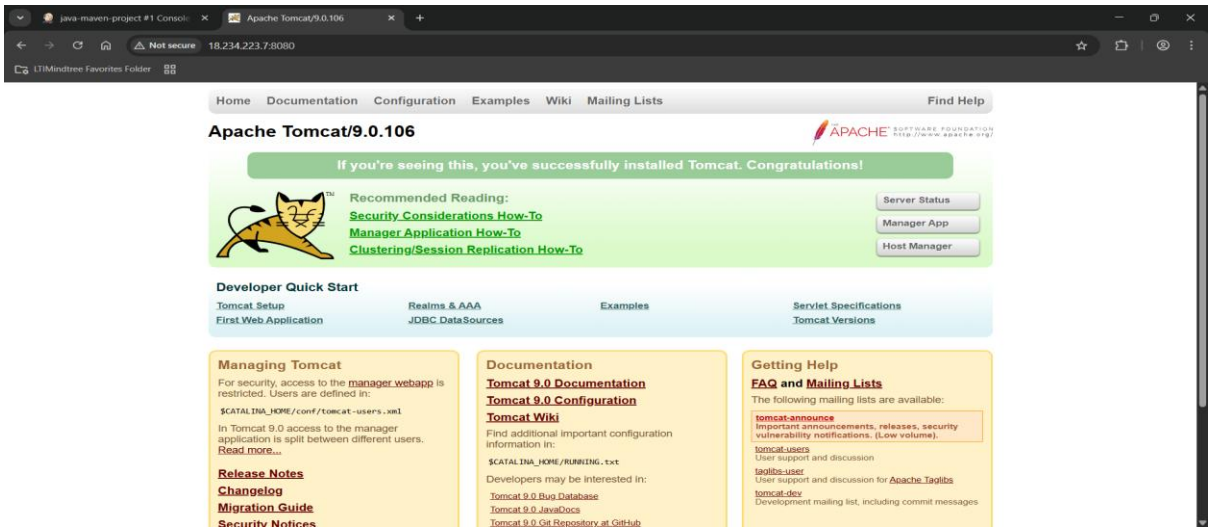
```
[root@tomcat apache-tomcat-9.0.106]# vi ./webapps/examples/META-INF/context.xml
[root@tomcat apache-tomcat-9.0.106]# vi ./webapps/host-manager/META-INF/context.xml
[root@tomcat apache-tomcat-9.0.106]# vi ./webapps/manager/META-INF/context.xml
[root@tomcat apache-tomcat-9.0.106]# cd conf
[root@tomcat conf]# ll
total 228
drwxr-x---. 3 root root     23 Jun 27 05:40 Catalina
-rw-------. 1 root root  12953 Jun  5 19:02 catalina.policy
-rw-------. 1 root root   7654 Jun  5 19:02 catalina.properties
-rw-------. 1 root root   1400 Jun  5 19:02 context.xml
-rw-------. 1 root root   1149 Jun  5 19:02 jaspic-providers.xml
-rw-------. 1 root root   2313 Jun  5 19:02 jaspic-providers.xsd
-rw-------. 1 root root   4003 Jun  5 19:02 logging.properties
-rw-------. 1 root root   8022 Jun  5 19:02 server.xml
-rw-------. 1 root root   2756 Jun  5 19:02 tomcat-users.xml
-rw-------. 1 root root   2558 Jun  5 19:02 tomcat-users.xsd
-rw-------. 1 root root 173782 Jun  5 19:02 web.xml
[root@tomcat conf]# vi tomcat-users.xml
```

```
    - manager-status - allows access to the status pages only

  The users below are wrapped in a comment and are therefore ignored. If you
  wish to configure one or more of these users for use with the manager web
  application, do not forget to remove the <!..  ..> that surrounds them. You
  will also need to set the passwords to something appropriate.
-->
<!--
  <user username="admin" password="<must-be-changed>" roles="manager-gui"/>
  <user username="robot" password="<must-be-changed>" roles="manager-script"/>
-->
<!--
  The sample user and role entries below are intended for use with the
  examples web application. They are wrapped in a comment and thus are ignored
  when reading this file. If you wish to configure these users for use with the
  examples web application, do not forget to remove the <!..  ..> that surrounds
  them. You will also need to set the passwords to something appropriate.
-->
<!--
  <role rolename="tomcat"/>
  <role rolename="role1"/>
  <user username="tomcat" password="<must-be-changed>" roles="tomcat"/>
  <user username="both" password="<must-be-changed>" roles="tomcat,role1"/>
  <user username="role1" password="<must-be-changed>" roles="role1"/>
-->

<role rolename="manager-gui"/>
<role rolename="manager-script"/>
<role rolename="manager-jmx"/>
<role rolename="manager-status"/>
<user username="admin" password="admin" roles="manager-gui,manager-script,manager-jmx,manager-status"/>
<user username="deployer" password="deployer" roles="manager-script"/>
<user username="tomcat" password="s3cret" roles="manager-gui"/>
</tomcat-users>
-- INSERT --                                                                    64,16         Bot
```

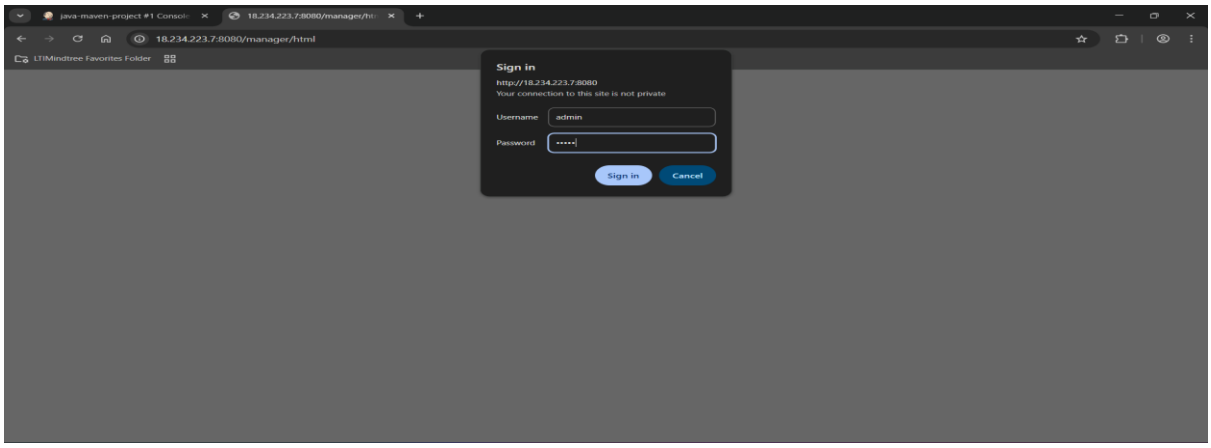
Roles are added.



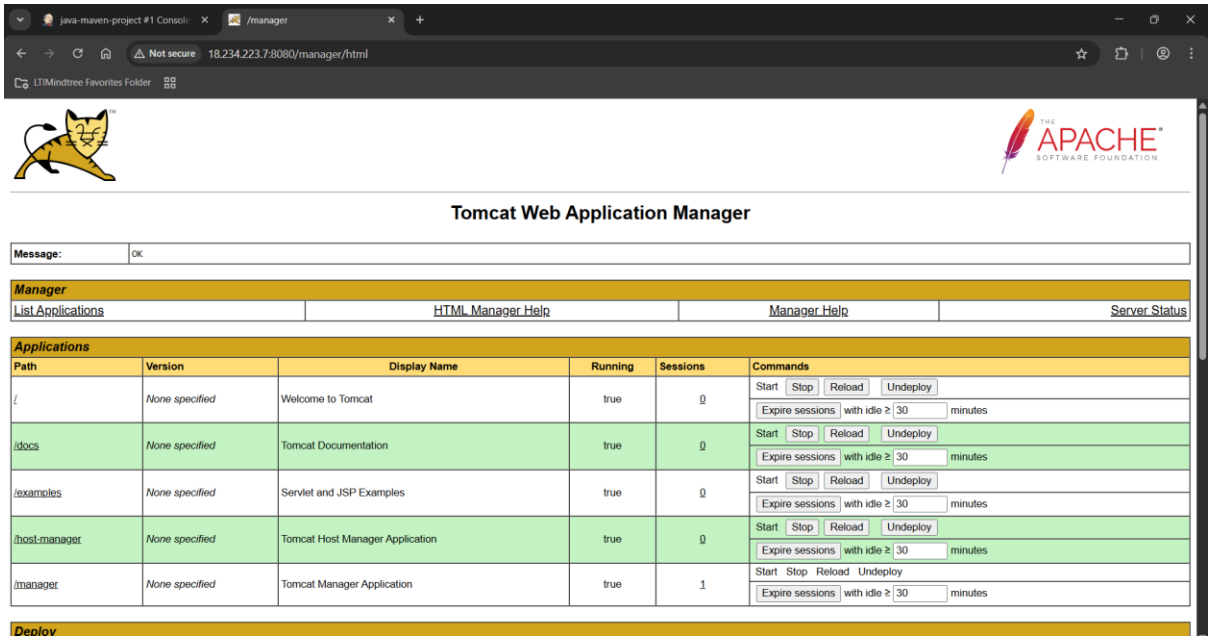
Tomcat machine public IP is copied.



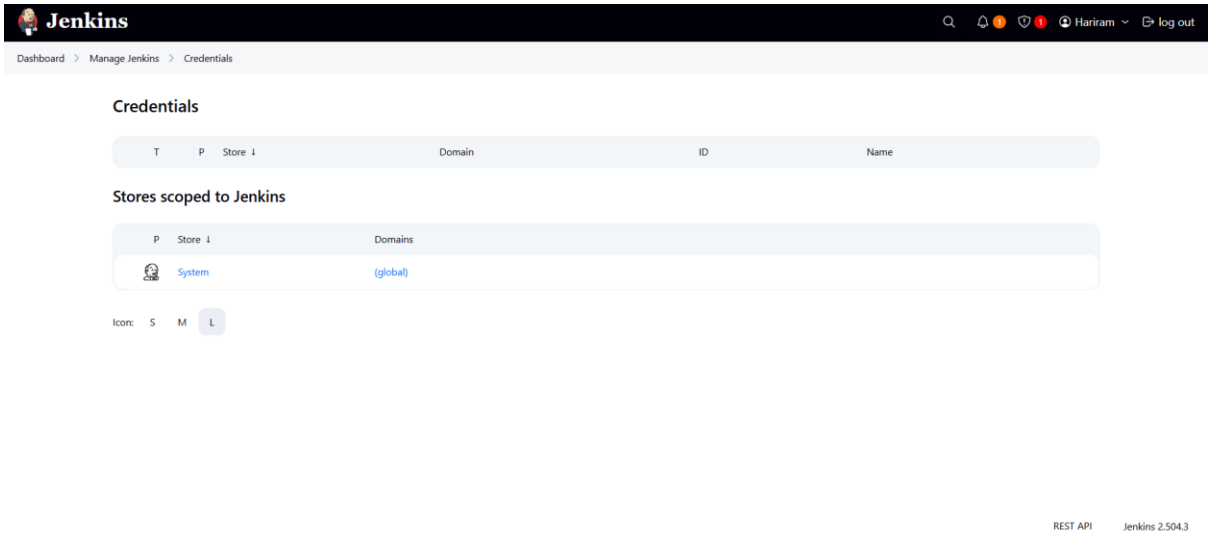
Tomcat dashboard is accessible.



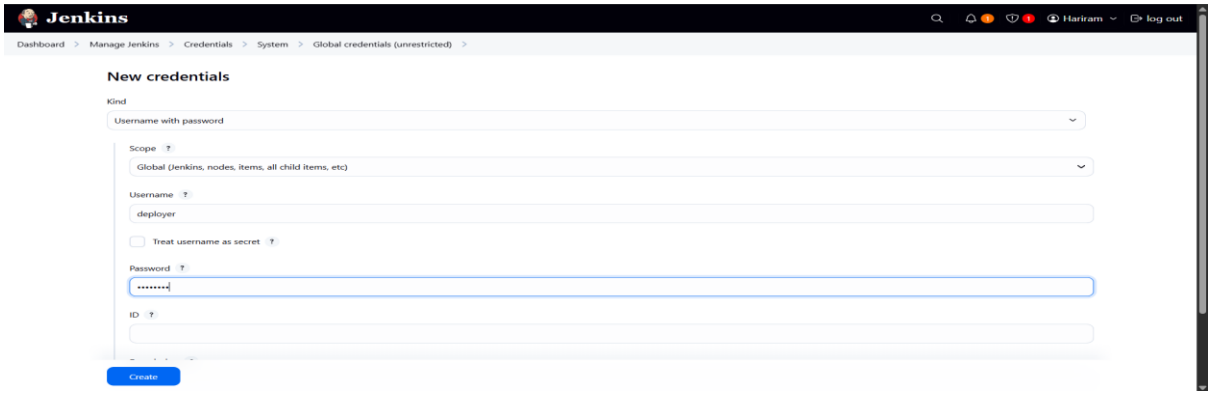
Login through role credentials.



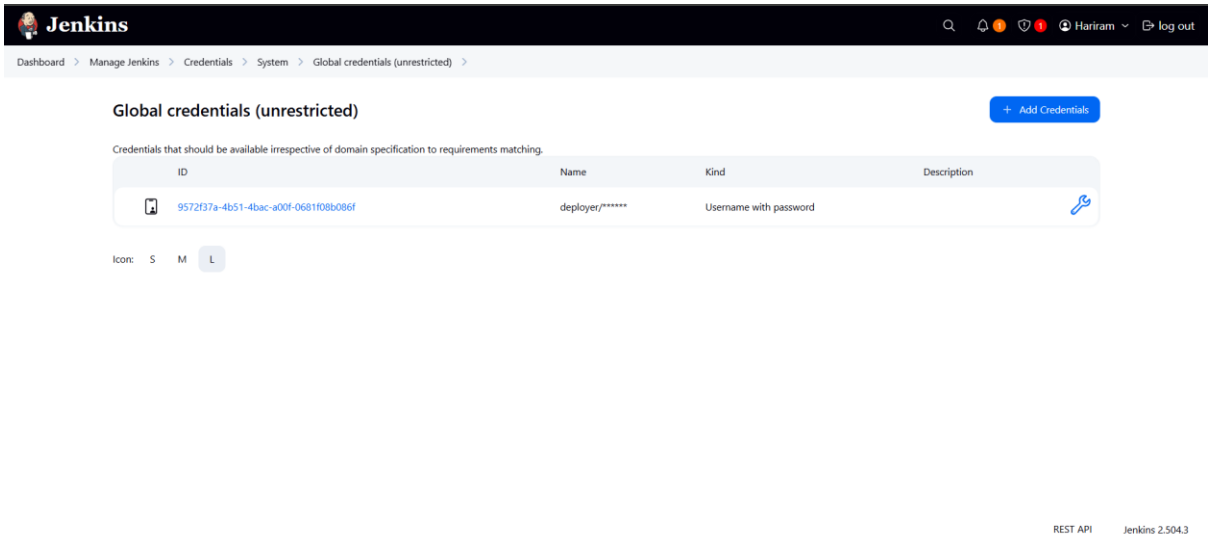
Tomcat setup is done.



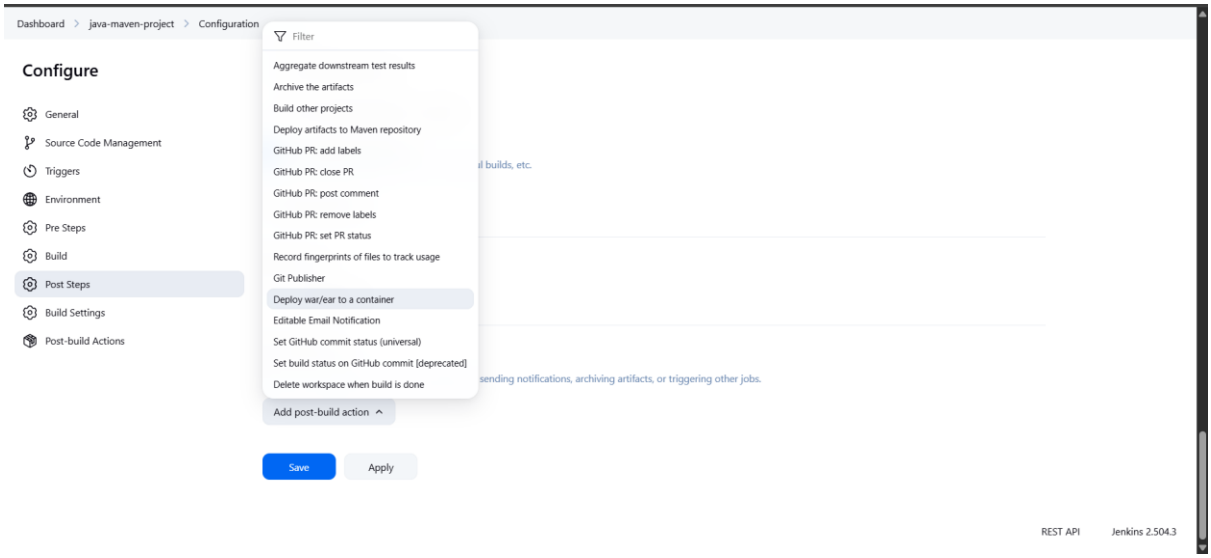
In Jenkins add global credentials.



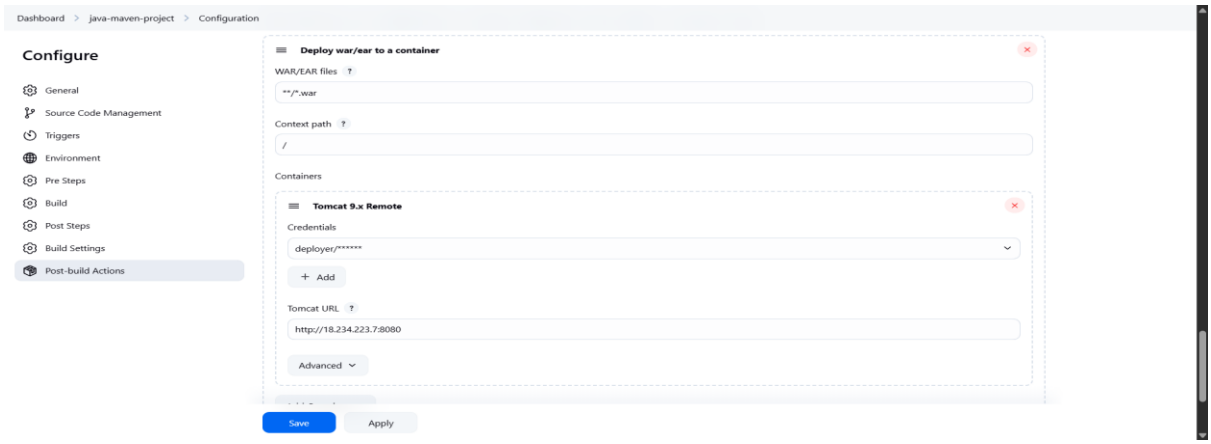
Username: deployer , Password: deployer.



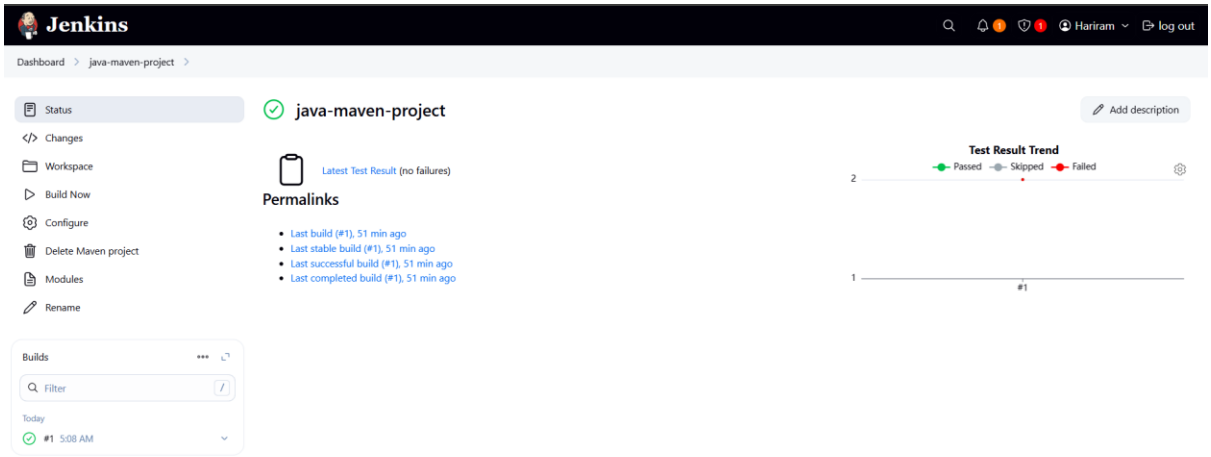
Credential is added successfully.



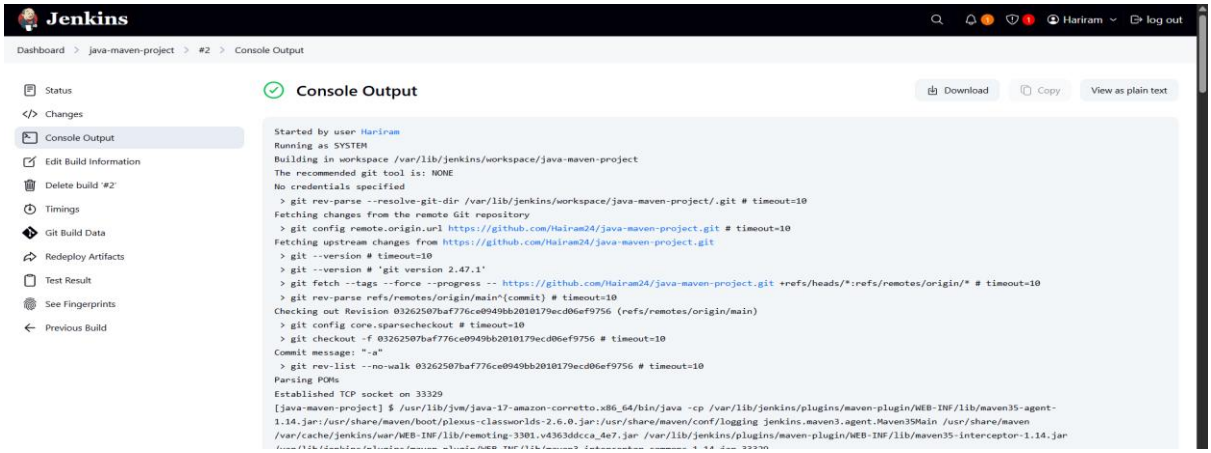
Add post build action. Use deploy war/ear to a container.



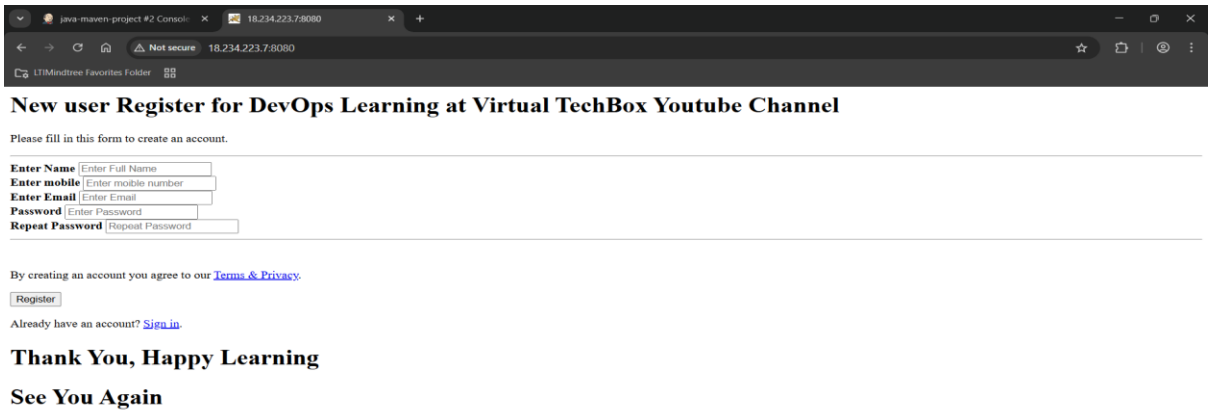
Entered the required details.



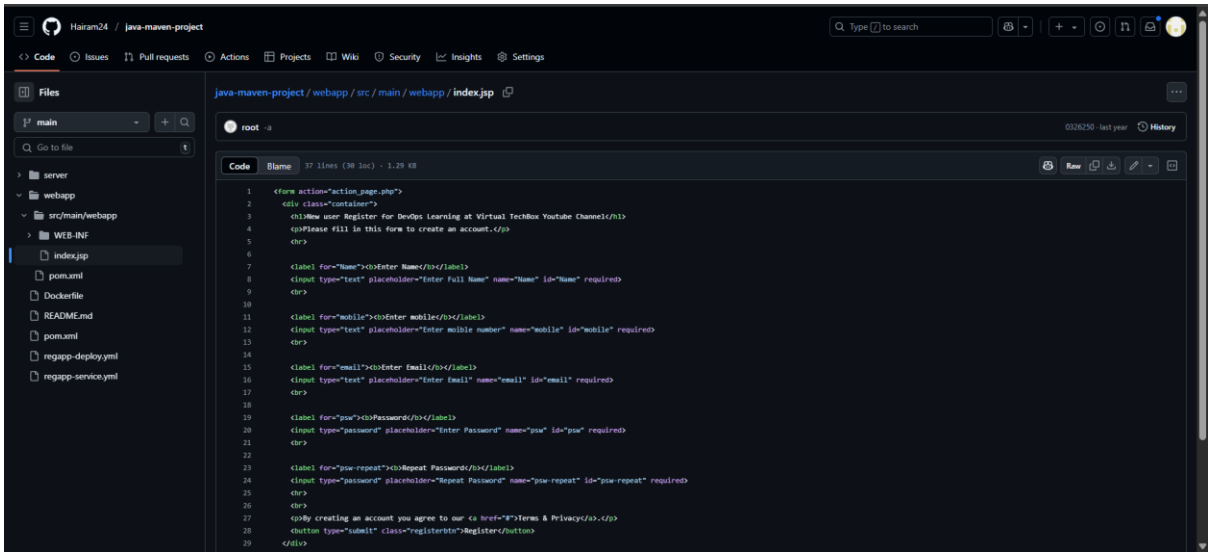
Build now.

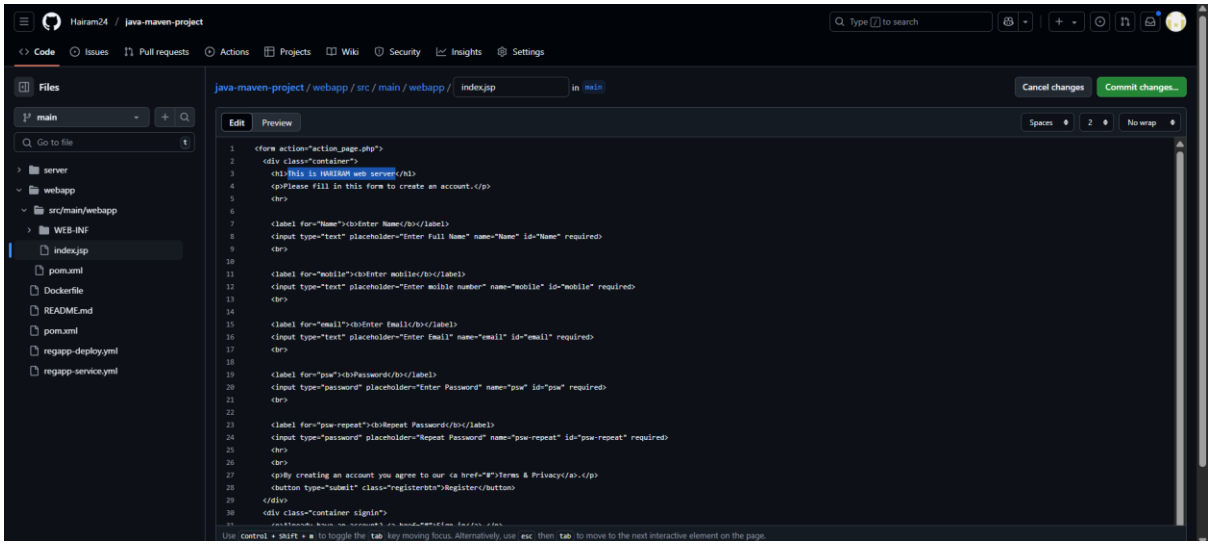


Build successful.

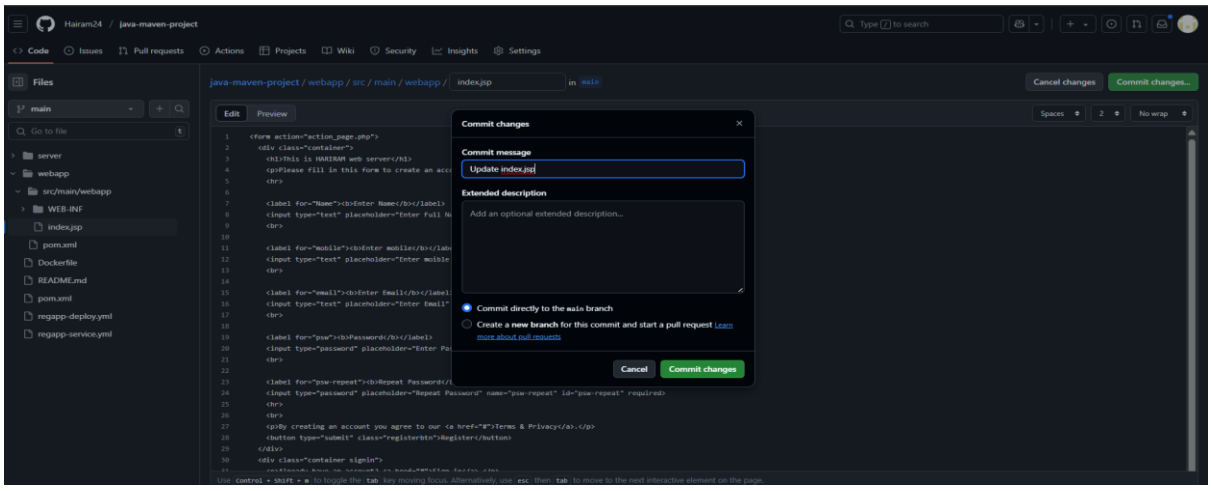


Web page is visible in tomcat.

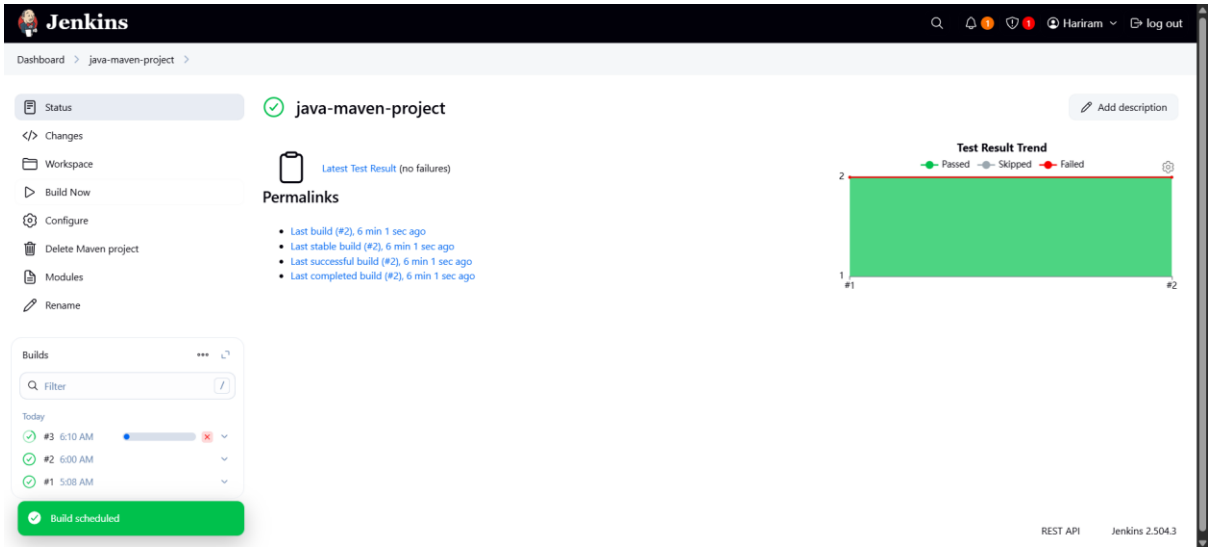




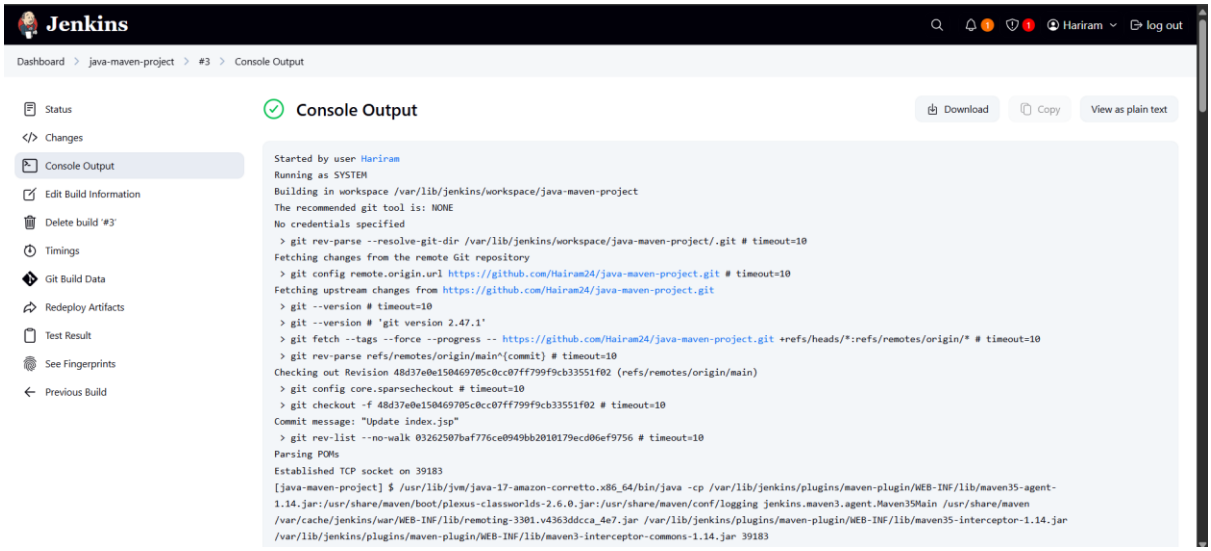
Made change in project index code.



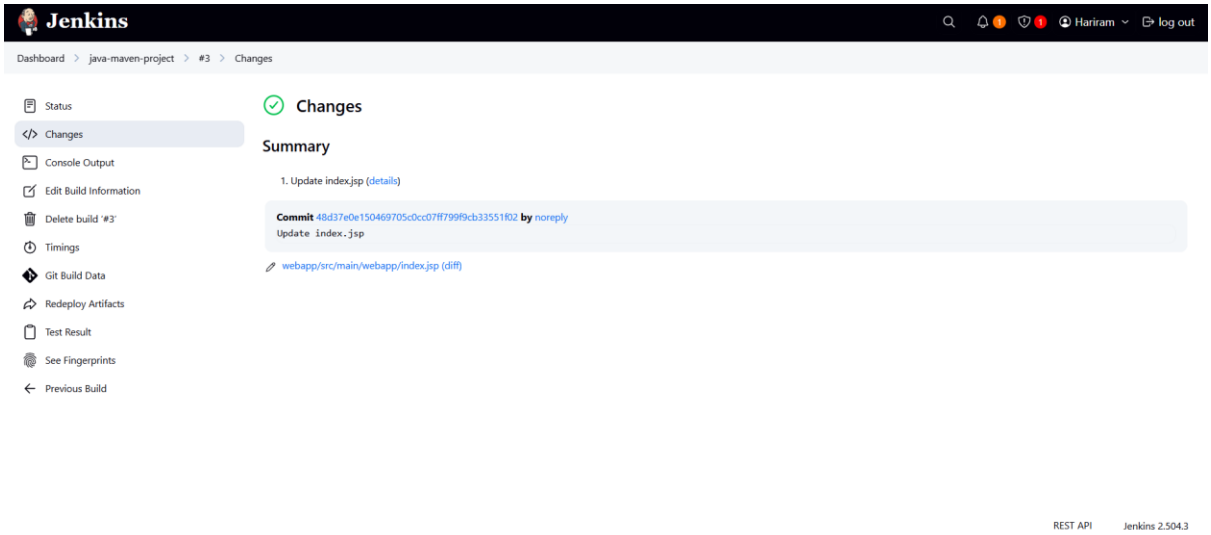
Commit the changes.



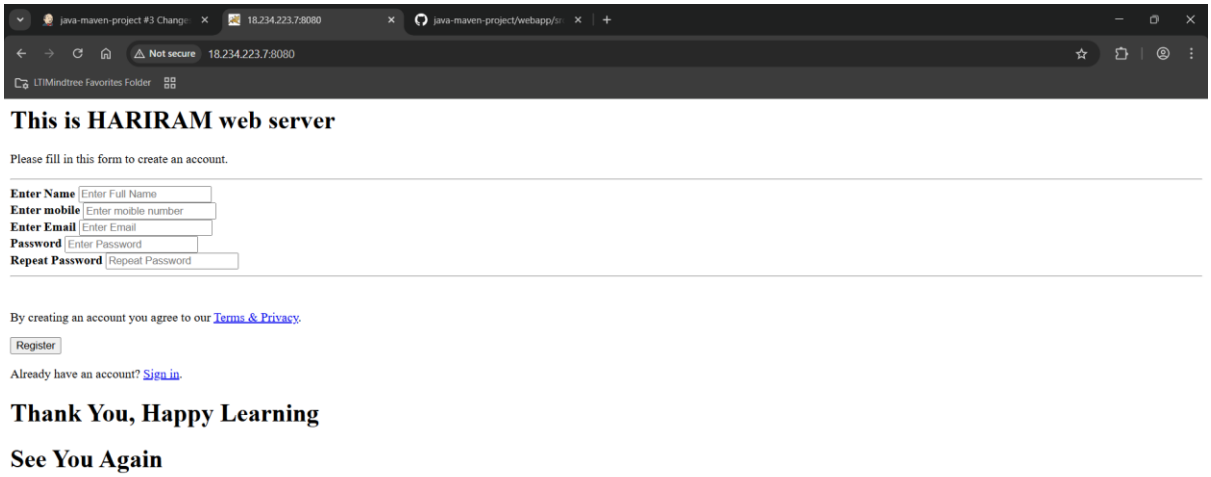
Manual Build.



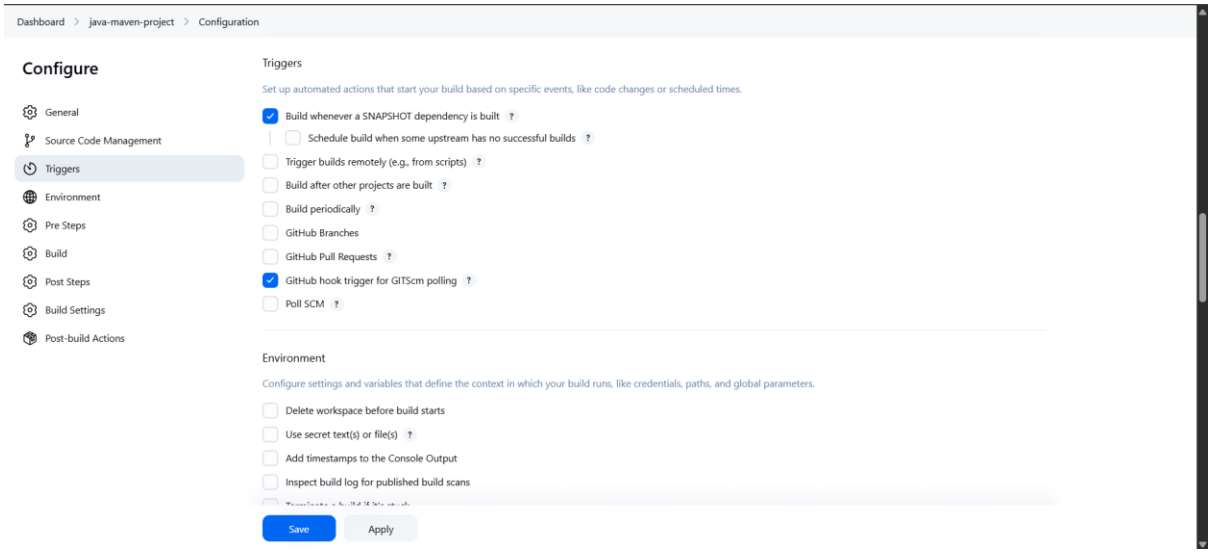
Build successfully.



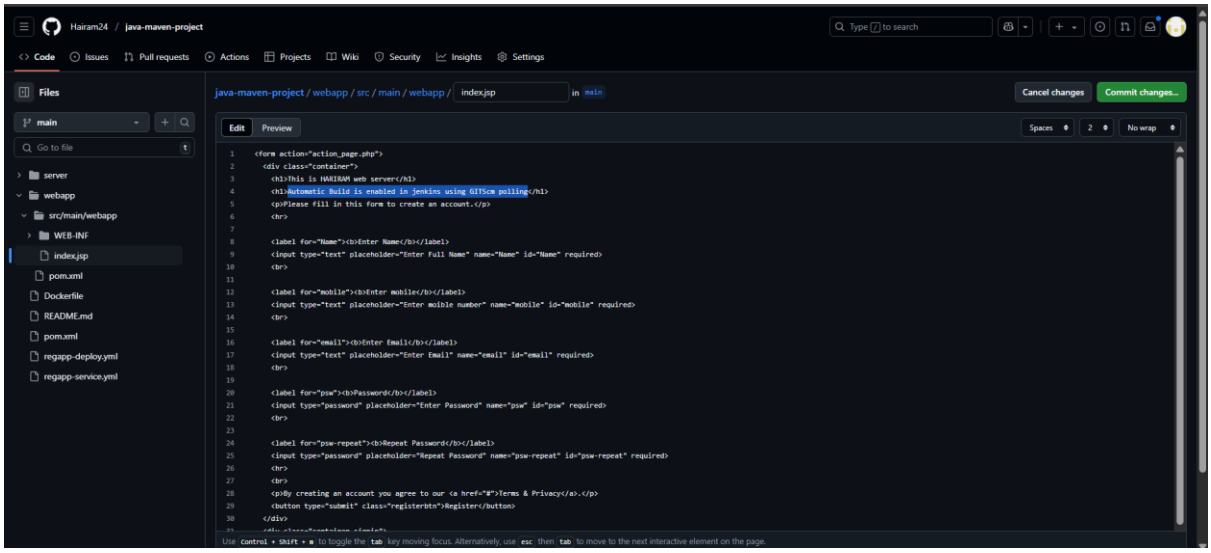
Changes made was updated.



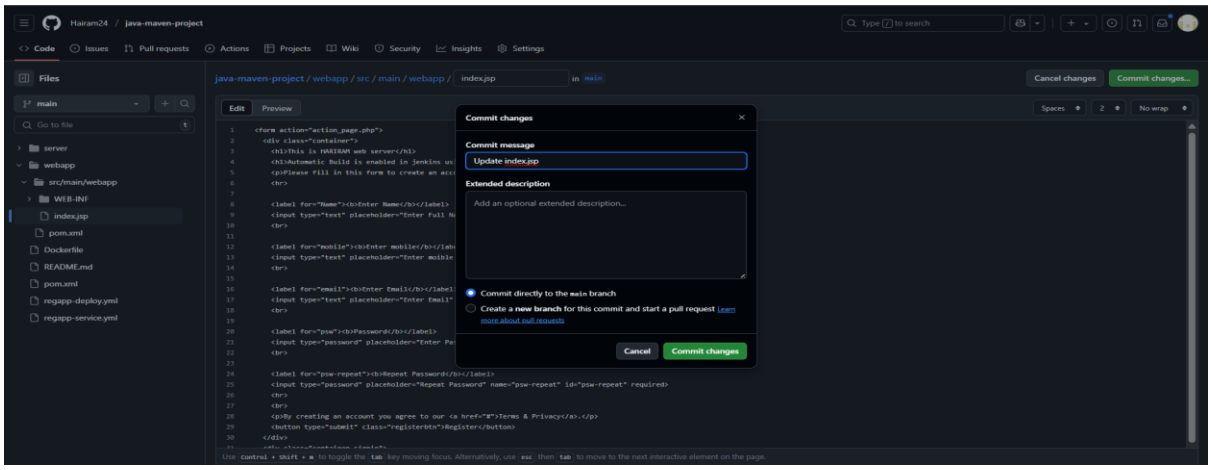
Reflected in the web page.



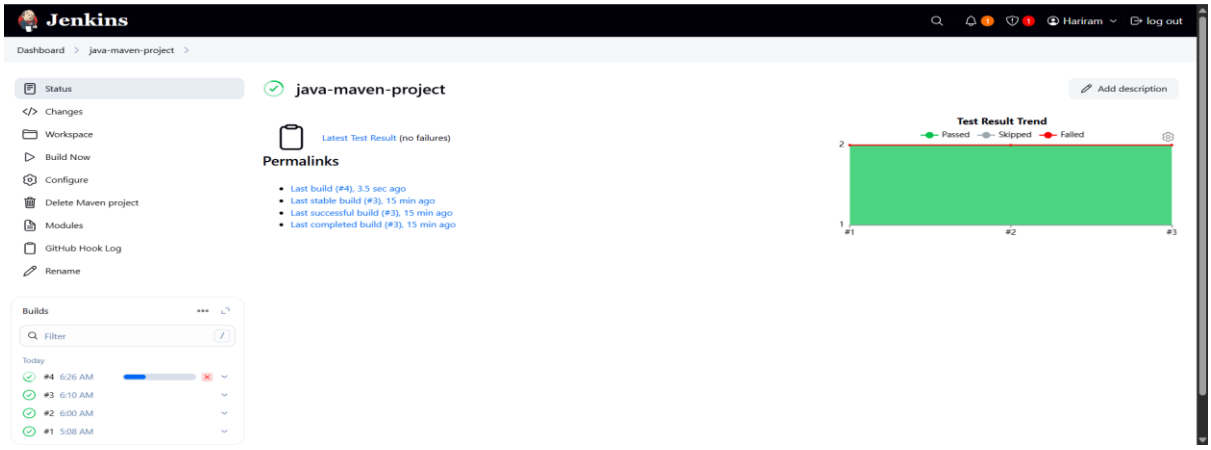
To automate build when code is changed, using GITscm polling.



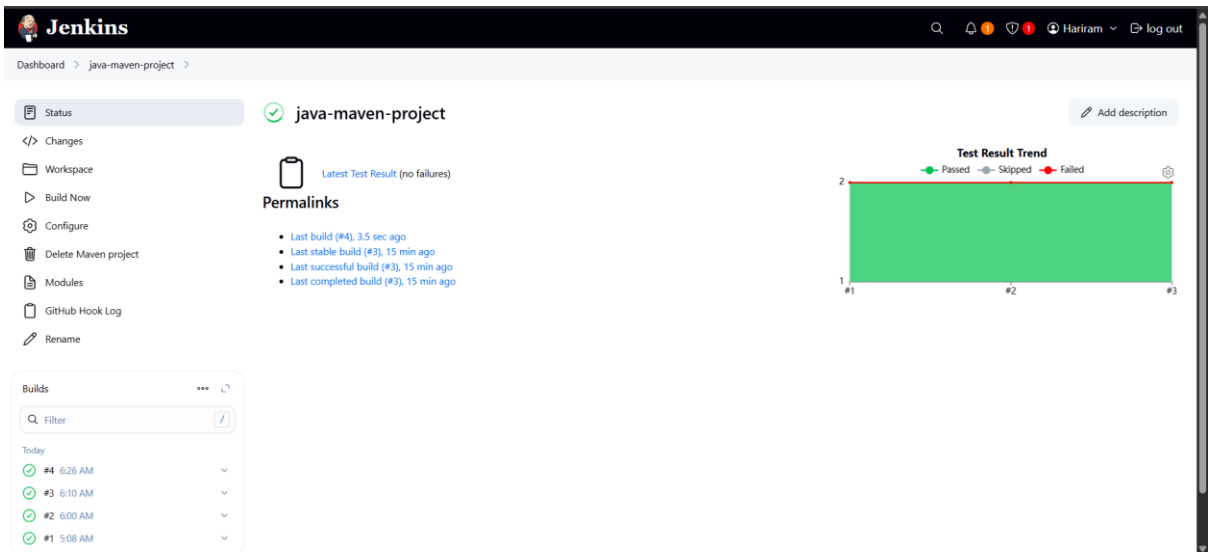
Made change in index code.



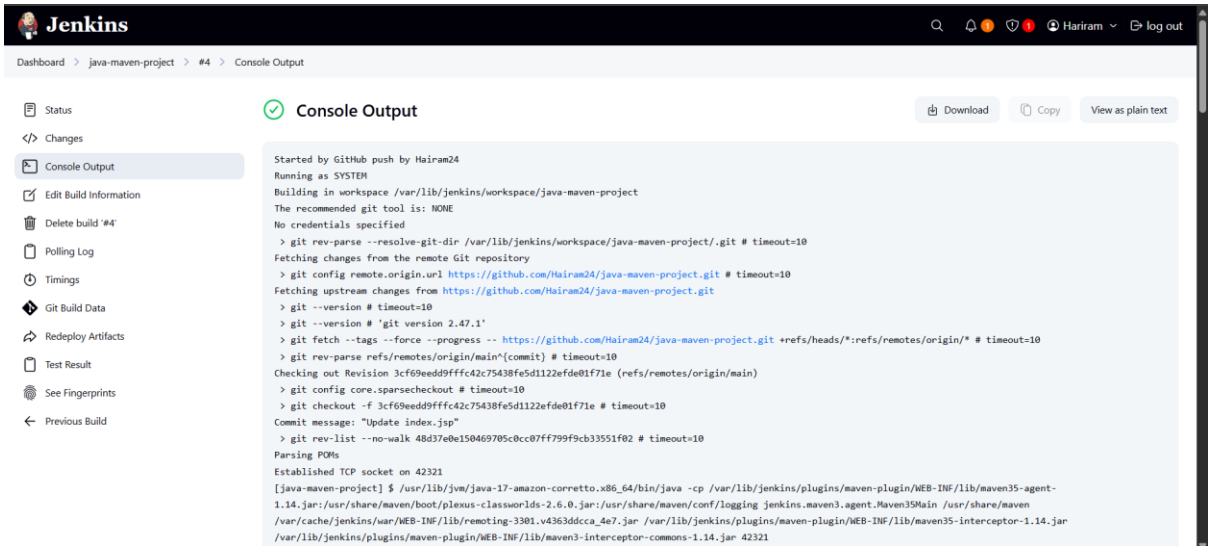
Commit the changes.



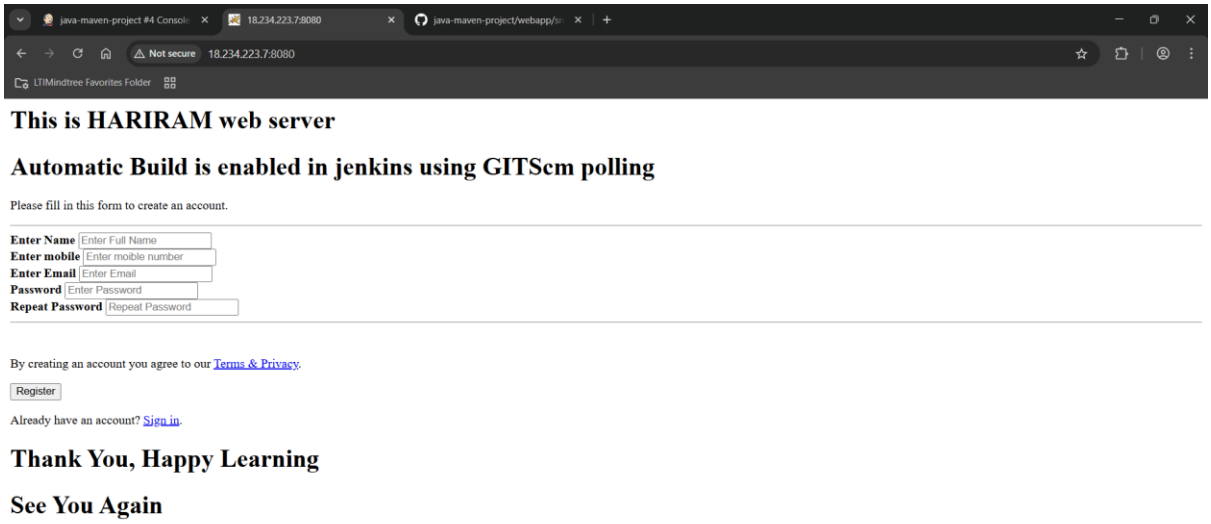
Automatically build is done.



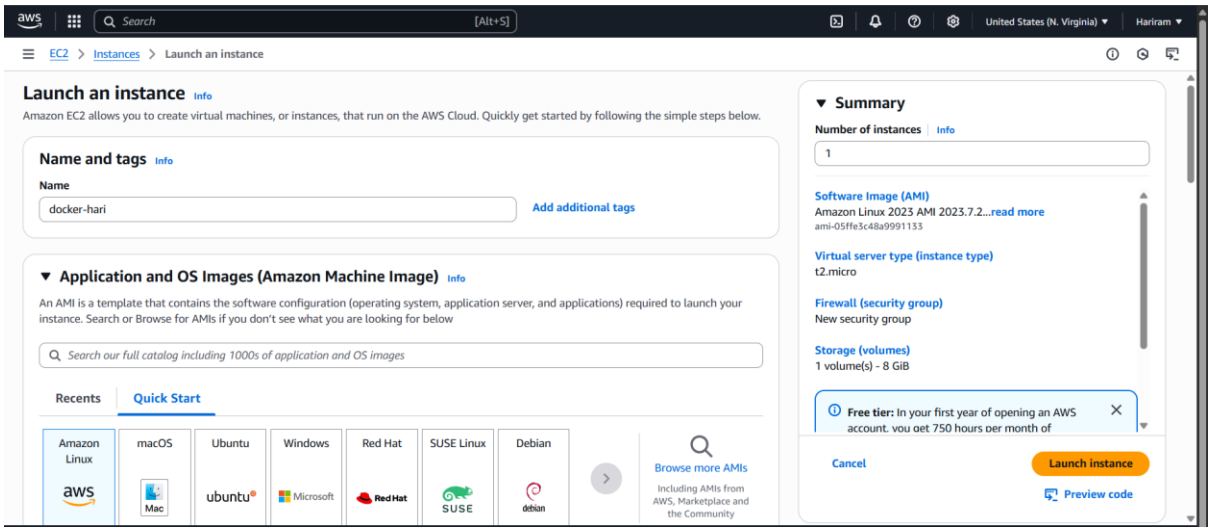
Build successfully.



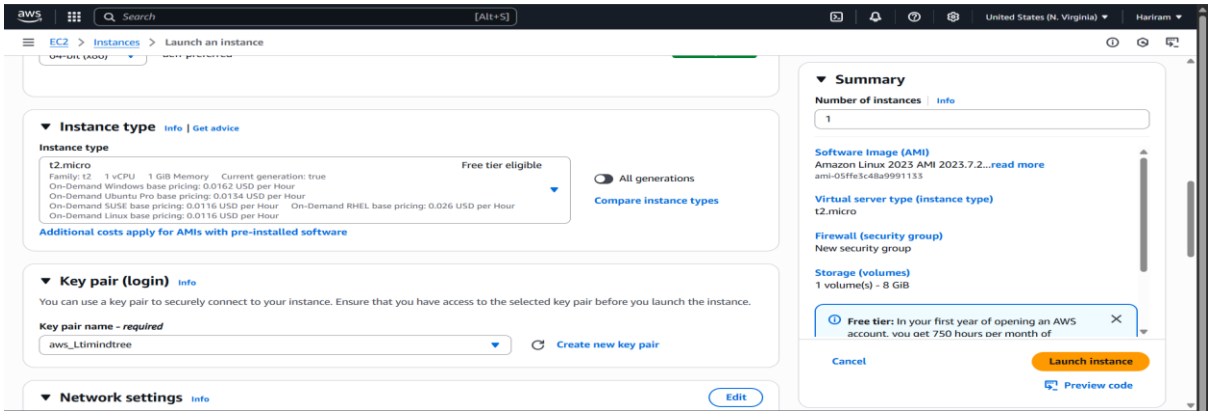
Console output.



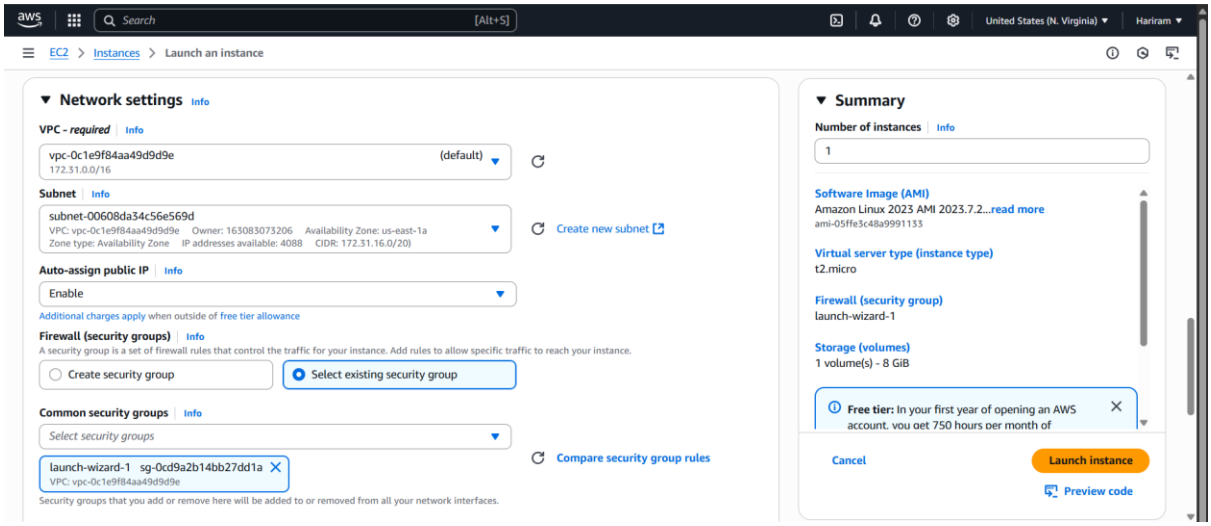
Reflected in web page.



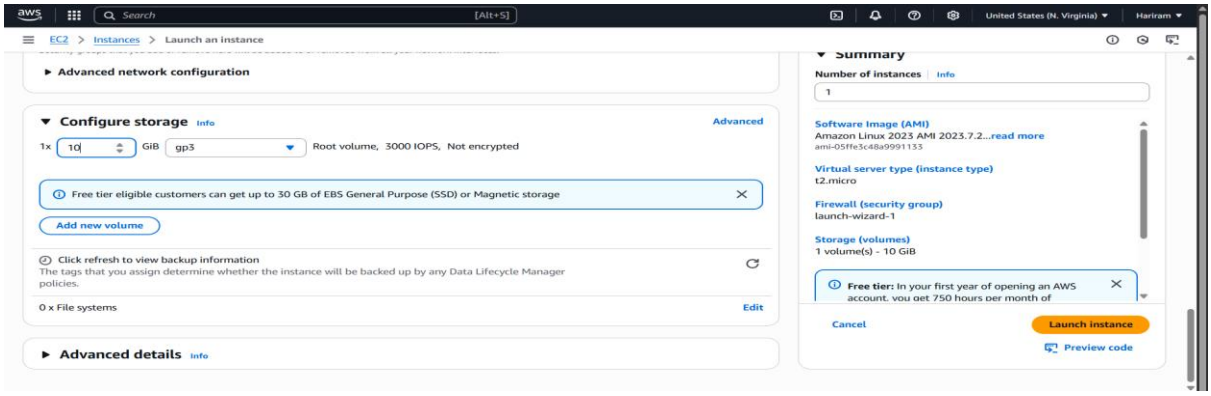
Docker machine is created with amazon Linux.



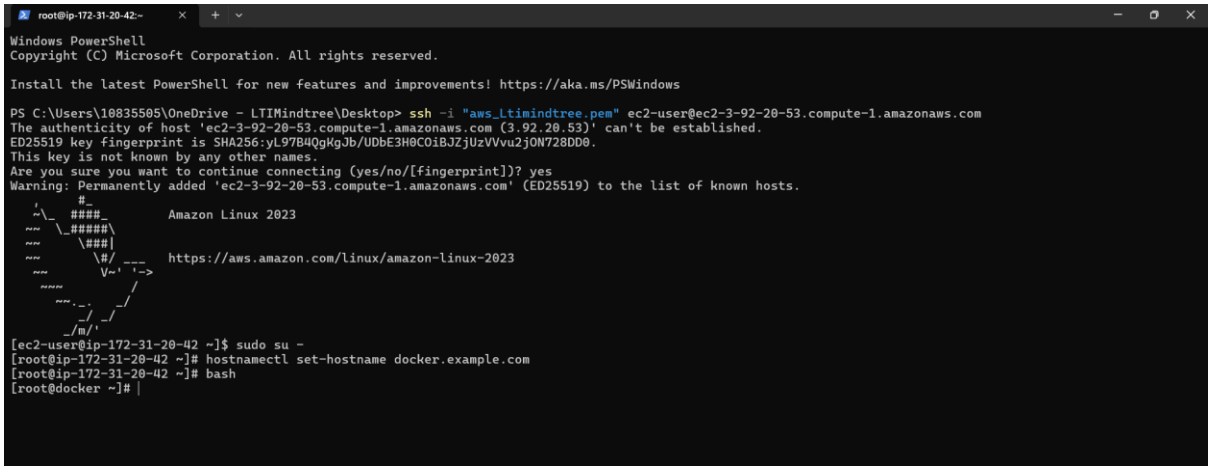
T2.micro is selected. Existing key pair is selected.



Existing security group is selected.



10 GB storage is selected.



Docker server connected and hostname is changed.

```
root@ip-172-31-20-42:~
[root@docker ~]# yum install docker*
Amazon Linux 2023 Kernel Livepatch repository                                                    133 kB/s |  17 kB     00:00
Dependencies resolved.
================================================================================================================================
 Package                                    Architecture                Version                     Repository                Size
================================================================================================================================
Installing:
 docker                                     x86_64                      25.0.8-1.amzn2023.0.4       amazonlinux               44 M
Installing dependencies:
 container-selinux                          noarch                      3:2.233.0-1.amzn2023        amazonlinux               55 k
 containerd                                 x86_64                      2.0.5-1.amzn2023.0.1        amazonlinux               25 M
 iptables-libs                              x86_64                      1.8.8-3.amzn2023.0.2        amazonlinux              401 k
 iptables-nft                               x86_64                      1.8.8-3.amzn2023.0.2        amazonlinux              183 k
 libcgroup                                  x86_64                      3.0-1.amzn2023.0.1          amazonlinux               75 k
 libnetfilter_conntrack                     x86_64                      1.0.8-2.amzn2023.0.2        amazonlinux               58 k
 libnfnetlink                               x86_64                      1.0.1-19.amzn2023.0.2       amazonlinux               30 k
 libnftnl                                   x86_64                      1.2.2-2.amzn2023.0.2        amazonlinux               84 k
 pigz                                       x86_64                      2.5-1.amzn2023.0.3          amazonlinux               83 k
 runc                                       x86_64                      1.2.4-2.amzn2023.0.1        amazonlinux              3.4 M

Transaction Summary
================================================================================================================================
Install  11 Packages

Total download size: 74 M
Installed size: 288 M
Is this ok [y/N]: y
Downloading Packages:
(1/11): container-selinux-2.233.0-1.amzn2023.noarch.rpm                                  495 kB/s |  55 kB     00:00
(2/11): iptables-libs-1.8.8-3.amzn2023.0.2.x86_64.rpm                                    7.9 MB/s | 401 kB     00:00
(3/11): iptables-nft-1.8.8-3.amzn2023.0.2.x86_64.rpm                                     3.5 MB/s | 183 kB     00:00
(4/11): libcgroup-3.0-1.amzn2023.0.1.x86_64.rpm                                          2.0 MB/s |  75 kB     00:00
(5/11): libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64.rpm                           1.4 MB/s |  58 kB     00:00
(6/11): libnfnetlink-1.0.1-19.amzn2023.0.2.x86_64.rpm                                    752 kB/s |  30 kB     00:00
(7/11): libnftnl-1.2.2-2.amzn2023.0.2.x86_64.rpm                                         2.0 MB/s |  84 kB     00:00
(8/11): pigz-2.5-1.amzn2023.0.3.x86_64.rpm                                               2.1 MB/s |  83 kB     00:00
(9/11): runc-1.2.4-2.amzn2023.0.1.x86_64.rpm                                             9.5 MB/s | 3.4 MB     00:00
(10/11): containerd-2.0.5-1.amzn2023.0.1.x86_64.rpm                                       22 MB/s |  25 MB     00:01
(11/11): docker-25.0.8-1.amzn2023.0.4.x86_64.rpm                                          25 MB/s |  44 MB     00:01
--------------------------------------------------------------------------------------------------------------------------------
Total                                                                                     42 MB/s |  74 MB     00:01
```

Docker package is installed.

```
root@jenkins:~
[root@jenkins ~]# ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/root/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/id_rsa
Your public key has been saved in /root/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:eqIEw/GnAcGb9VR+LbiXCqFIHpQpSkqOQqAb1B2mezI root@jenkins.example.com
The key's randomart image is:
+---[RSA 3072]----+
|oo+o.o...        |
|+=+o+... . .     |
|@.== o. o o .    |
|=Bo*.... o o     |
|o *E+.o S o      |
|   o++ o o       |
|    o o o        |
|   . . o         |
|    .            |
+----[SHA256]-----+
[root@jenkins ~]#
```

SSH key is generated.

```
root@ip-172-31-20-42:~
[root@docker ~]# ip a s
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enX0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc fq_codel state UP group default qlen 1000
    link/ether 0a:ff:da:92:e3:a9 brd ff:ff:ff:ff:ff:ff
    altname eni-02dd40433258b8ad7
    altname device-number-0.0
    inet 172.31.20.42/20 metric 512 brd 172.31.31.255 scope global dynamic enX0
       valid_lft 3392sec preferred_lft 3392sec
    inet6 fe80::8ff:daff:fe92:e3a9/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
[root@docker ~]# passwd root
Changing password for user root.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@docker ~]#
```

Root password is set.

```
root@docker:~
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin yes
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2
# but this is overridden so installations will only check .ssh/authorized_keys
AuthorizedKeysFile      .ssh/authorized_keys

#AuthorizedPrincipalsFile none


# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# Explicitly disable PasswordAuthentication. By presetting it, we
# avoid the cloud-init set_passwords module modifying sshd_config and
# restarting sshd in the default instance launch configuration.
PasswordAuthentication yes
PermitEmptyPasswords no

# Change to no to disable s/key passwords
#KbdInteractiveAuthentication yes

# Kerberos options
#KerberosAuthentication no
#KerberosOrLocalPasswd yes
#KerberosTicketCleanup yes
-- INSERT --                                                                    64,27         36%
```

```
root@docker:~
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin yes
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2
# but this is overridden so installations will only check .ssh/authorized_keys
AuthorizedKeysFile      .ssh/authorized_keys

#AuthorizedPrincipalsFile none


# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# Explicitly disable PasswordAuthentication. By presetting it, we
# avoid the cloud-init set_passwords module modifying sshd_config and
# restarting sshd in the default instance launch configuration.
PasswordAuthentication yes
PermitEmptyPasswords no

# Change to no to disable s/key passwords
#KbdInteractiveAuthentication yes

# Kerberos options
#KerberosAuthentication no
#KerberosOrLocalPasswd yes
#KerberosTicketCleanup yes
-- INSERT --                                                    64,27          36%
```

```
root@docker:~
[root@docker ~]# vim /etc/ssh/sshd_config
[root@docker ~]# systemctl restart sshd
[root@docker ~]# systemctl enable sshd
[root@docker ~]#
```

SSHD configuration is edited and sshd is restarted.

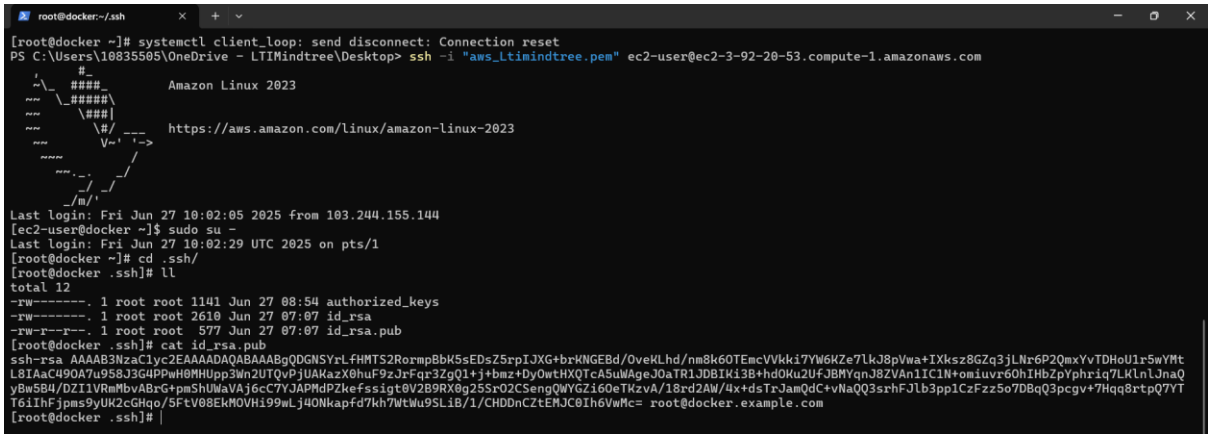
```
[root@jenkins ~]# ssh-copy-id root@172.31.20.42
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/root/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
root@172.31.20.42's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'root@172.31.20.42'"
and check to make sure that only the key(s) you wanted were added.

[root@jenkins ~]#
```

Ssh public key is added to Jenkins

```
root@jenkins:~

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin yes
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2
# but this is overridden so installations will only check .ssh/authorized_keys
AuthorizedKeysFile      .ssh/authorized_keys

#AuthorizedPrincipalsFile none


# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# Explicitly disable PasswordAuthentication. By presetting it, we
# avoid the cloud-init set_passwords module modifying sshd_config and
# restarting sshd in the default instance launch configuration.
PasswordAuthentication yes
PermitEmptyPasswords no

# Change to no to disable s/key passwords
#KbdInteractiveAuthentication yes

# Kerberos options
-- INSERT --                                                    37,18          33%
```

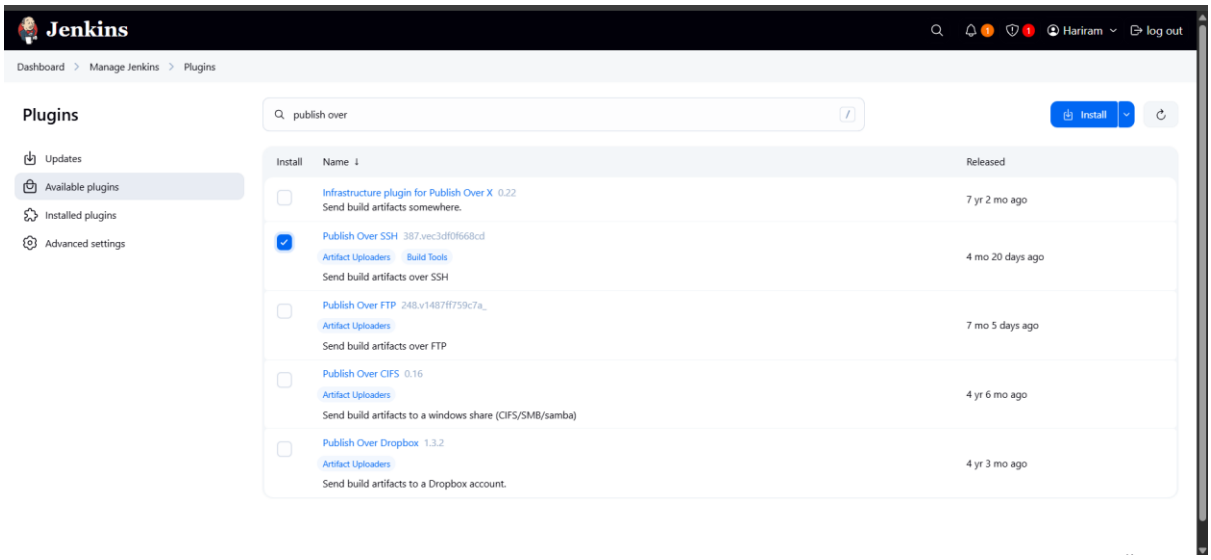
```
root@jenkins:~
[root@jenkins ~]# vim /etc/ssh/sshd_config
[root@jenkins ~]# systemctl start sshd
[root@jenkins ~]# systemctl enable sshd
[root@jenkins ~]#
```

Jenkins sshd config is edited.

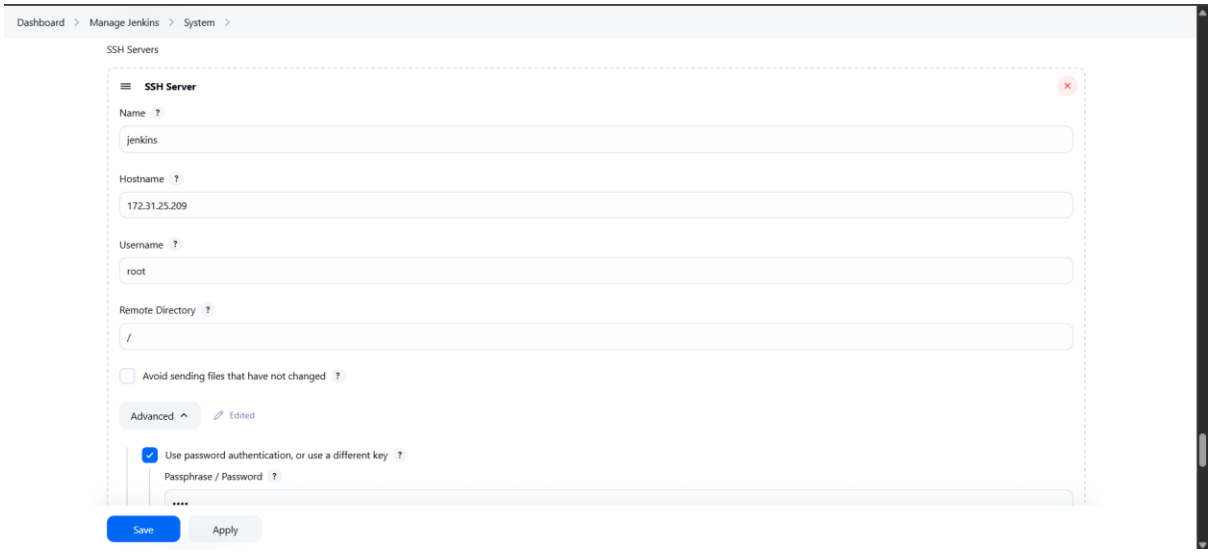


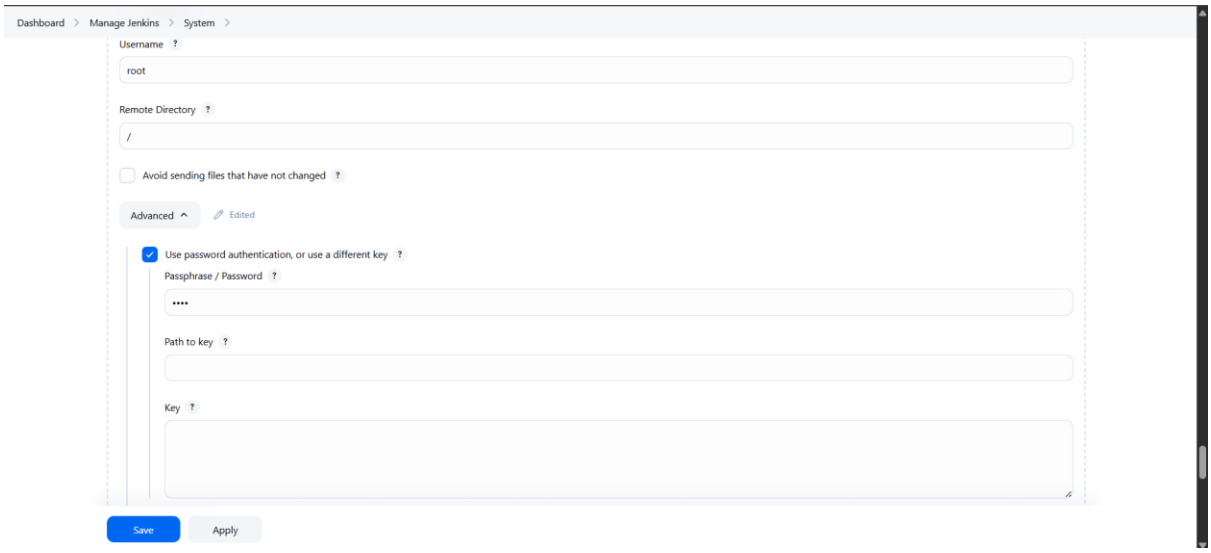
Public is copied.

Jenkins pub key is added to the docker authorized_keys.

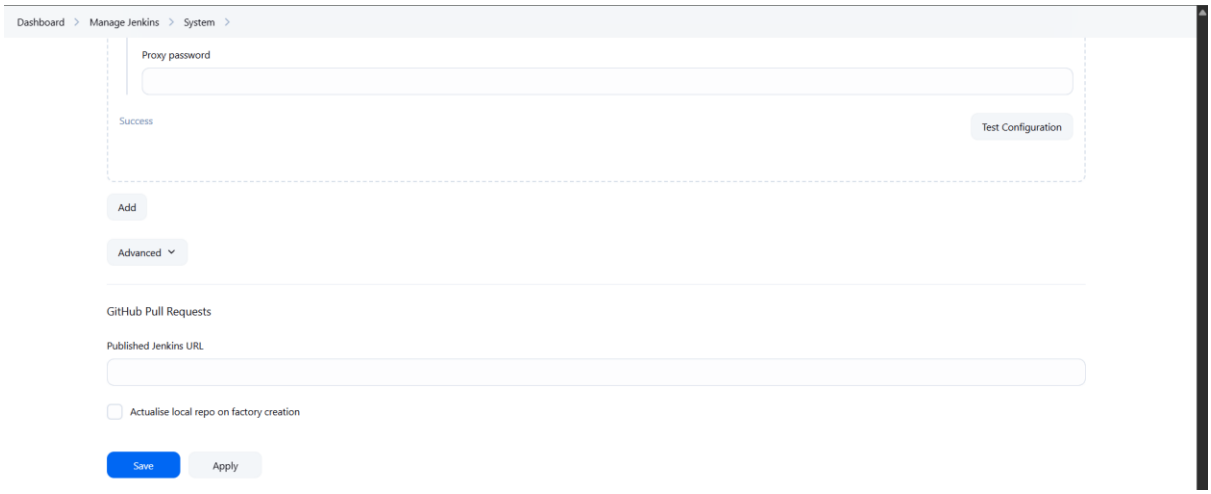


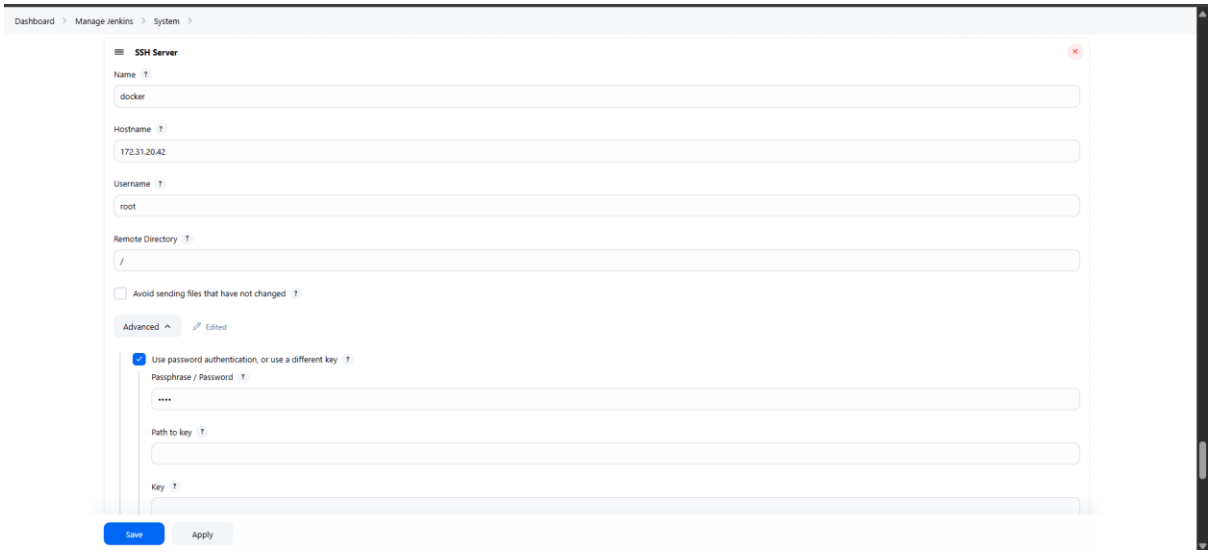
Publish over ssh plugin is installed.

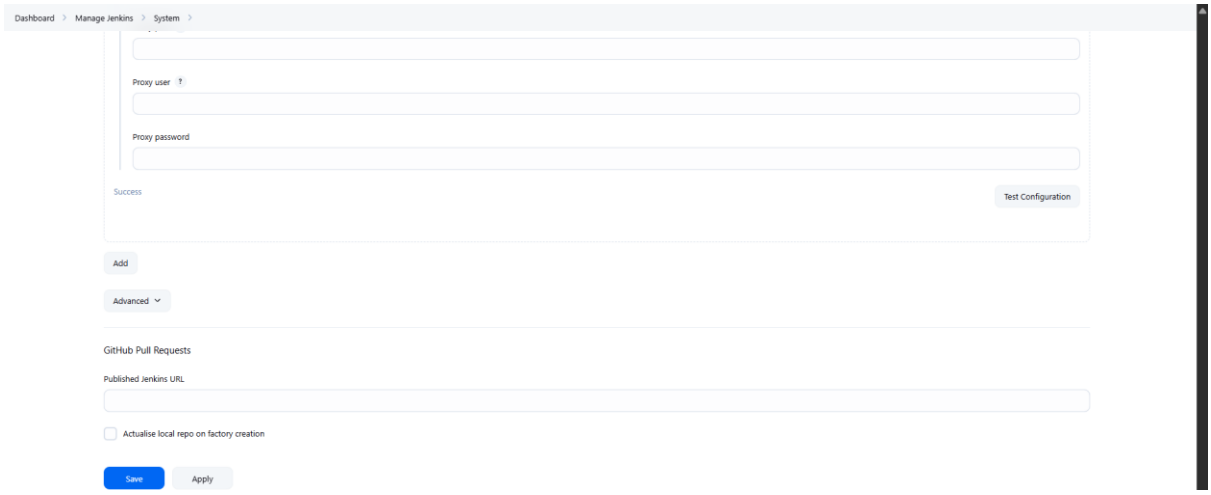




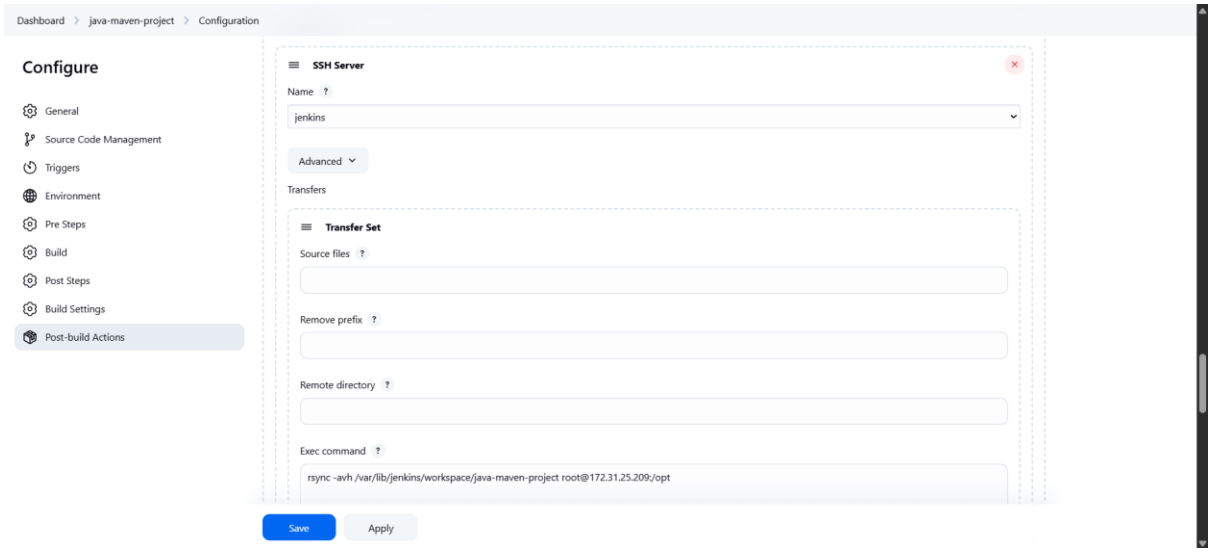
Jenkins ssh server is added.



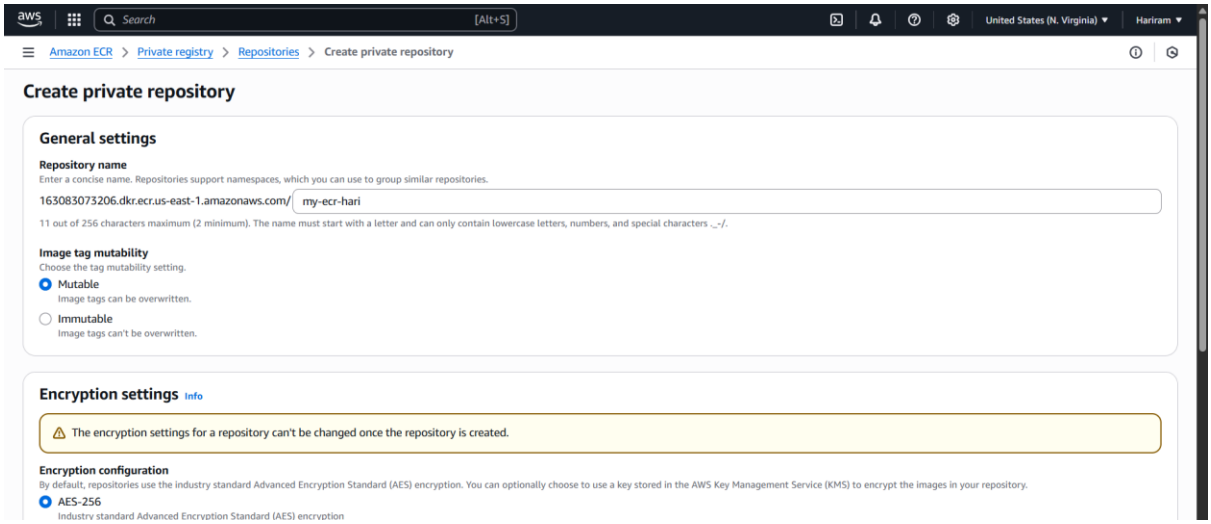




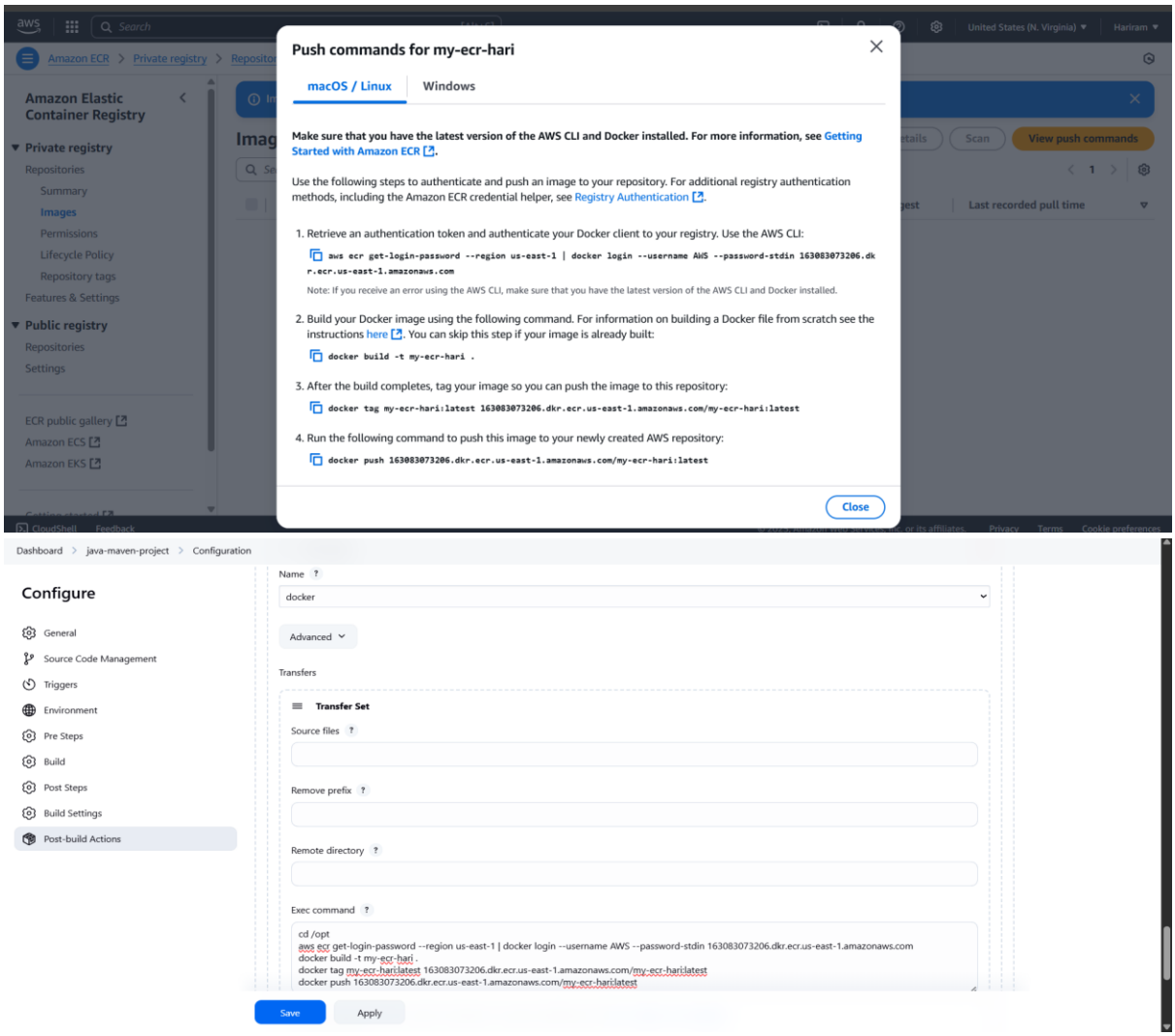
Docker ssh server is added.



Jenkins post build is added with exec commands.



Create a ECR.



Copy and paste the push commands of ECR to docker ssh server.

```
root@tomcat:~

#LoginGraceTime 2m
PermitRootLogin yes
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2
# but this is overridden so installations will only check .ssh/authorized_keys
AuthorizedKeysFile      .ssh/authorized_keys

#AuthorizedPrincipalsFile none


# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# Explicitly disable PasswordAuthentication. By presetting it, we
# avoid the cloud-init set_passwords module modifying sshd_config and
# restarting sshd in the default instance launch configuration.
PasswordAuthentication yes
PermitEmptyPasswords no

# Change to no to disable s/key passwords
#KbdInteractiveAuthentication yes

# Kerberos options
-- INSERT --                                                    60,18         37%
```

```
root@tomcat:~
[root@tomcat ~]# vim /etc/ssh/sshd_config
[root@tomcat ~]# systemctl restart sshd
[root@tomcat ~]# systemctl enable sshd
```

Tomcat sshd is edited and restarted.

```
root@tomcat:~/.ssh
no-port-forwarding,no-agent-forwarding,no-X11-forwarding,command="echo 'Please login as the user \"ec2-user\" rather than the user \"r
oot\".';echo;sleep 10;exit 142" ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQDowCDZw93pIkZ5Zyr4FMsOjcgHTQnJdrf8gRXklMU5mZUBlAqJABcR5lsyzodN+N
1iAX9Yee4zd3f+0meqX45ZxTXzDRNKGT50m7fbdB5Z4dw/IUJsRcSOICVPvJQwKqB6Fcrm1Maa3yguwvt2pBWJcqgfvdOr1yisqKOGwztIUoHUDWCrlag8Lh+gXBLZx5FRF4mp
glg67WRNP22dzE+hD5TiBe2DKQvSjU38FiiDcFRVm5X+EKa0VnqJBxVjqUspolzK/z3W25cVLqh0/NrBQrdn+0J8YNPtYzo2Np5ghhKV3ie4z0tQcNY8Aa5ZWjFlptqmcIvg9N
zvdm25QDE/ aws_Ltimindtree

ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABgQCY5oIJGoTByF5rADOTAHB44WmJmloJctqnyUx2ua6oVcQRxJkVwXkhBbgNt5oVZnfgwR8sXSGsll5yBOOhdLAhdUcTLaWnV5
pabz35wakqaT96hEYOLVFSgjbPVrMcfoW77YZVN79U5zTs4+IxorA/EUB+0T81CeP8rbFnCdLNUGbxQBk1x7i7qAPUzTbU/O2XXhKFhfQ3eRU+ZApipOJzDLTuf6pqaamfWGSp
FPL1BY4334qthfZnbA846stRL5zd5BDw5B7U5P27MA5qh/nJkmSfv8pPHQIOWAXPDKFG2fPUTCgRiJqMz6tuzCBBFJlbfCgfBAn4izGd4ygVn0qXS9axUxpnJbYpoI+y08cNHp
vfiuXXLIjbTkf+g7aHODGTeVhQ4aOjWuQJ3Flsbom/yisOChncRFM7hlb1UPnUZtRL+HJRHczy9cwB4V5vf5n+S7PNEUi8Bj4YneqNgzf5WYfgiz9JGiZfe9Tn/hkkmB6n3LRS
JPJ5VDqkWywNWeU= root@jenkins.example.com
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-- INSERT --                                                    3,578         All
```
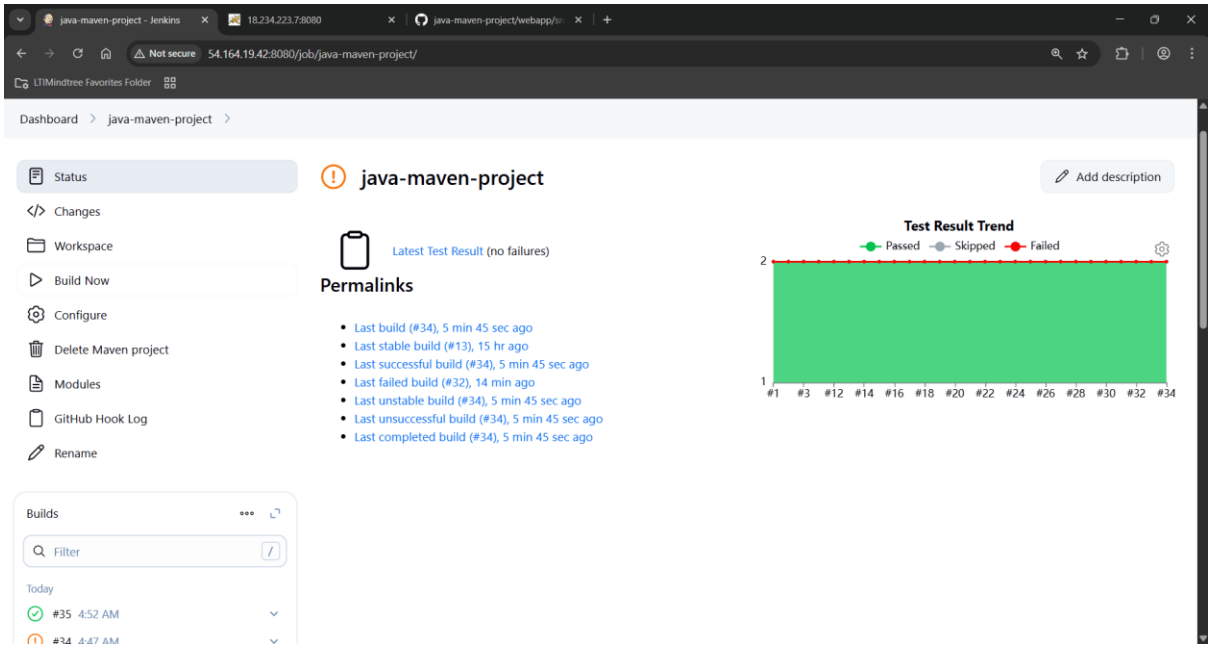
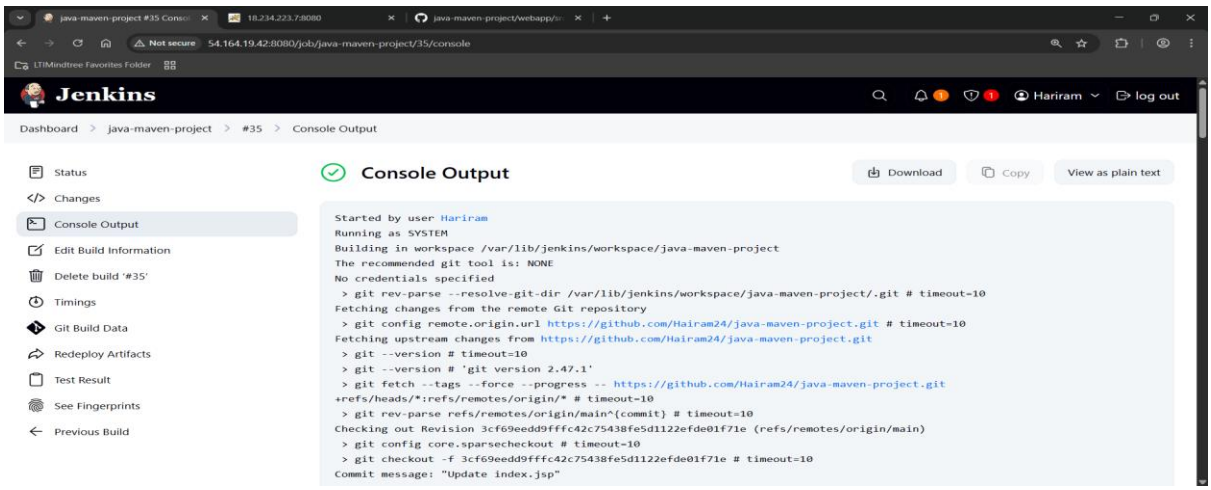
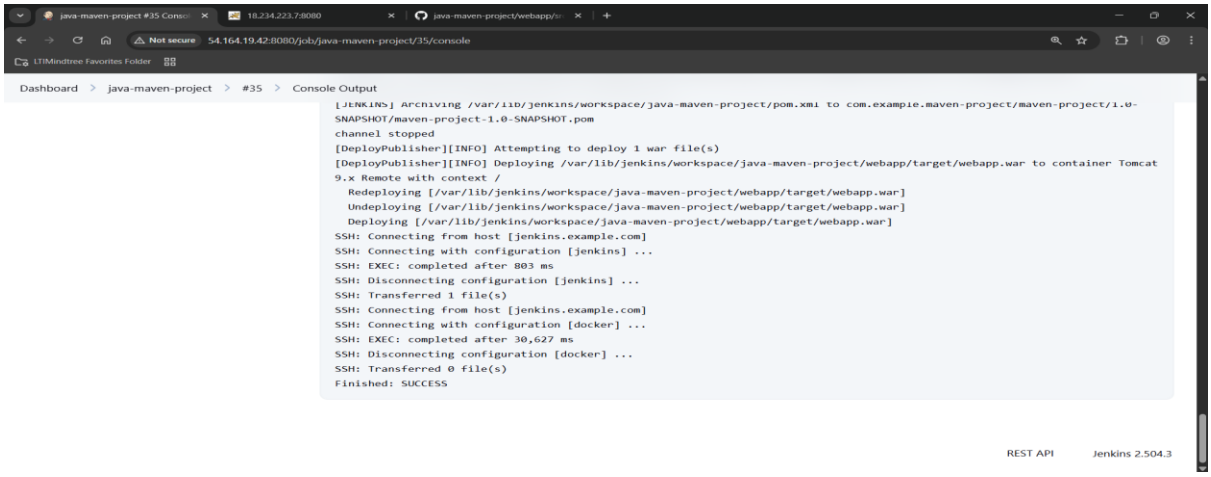

Jenkins pub key is added to the tomcat authorized_keys.



Build successful.





Console output build successfully.

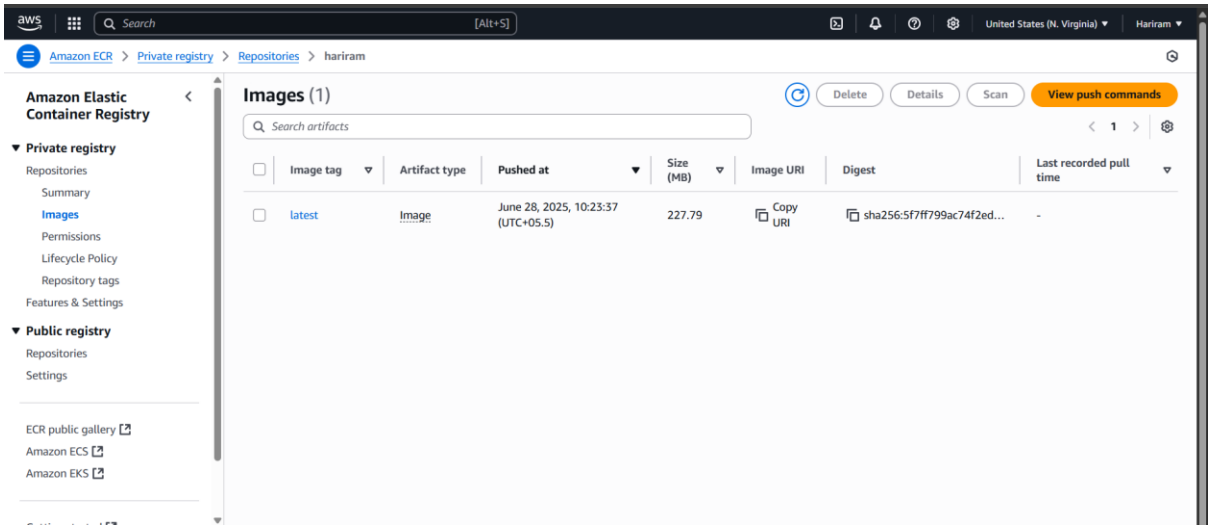
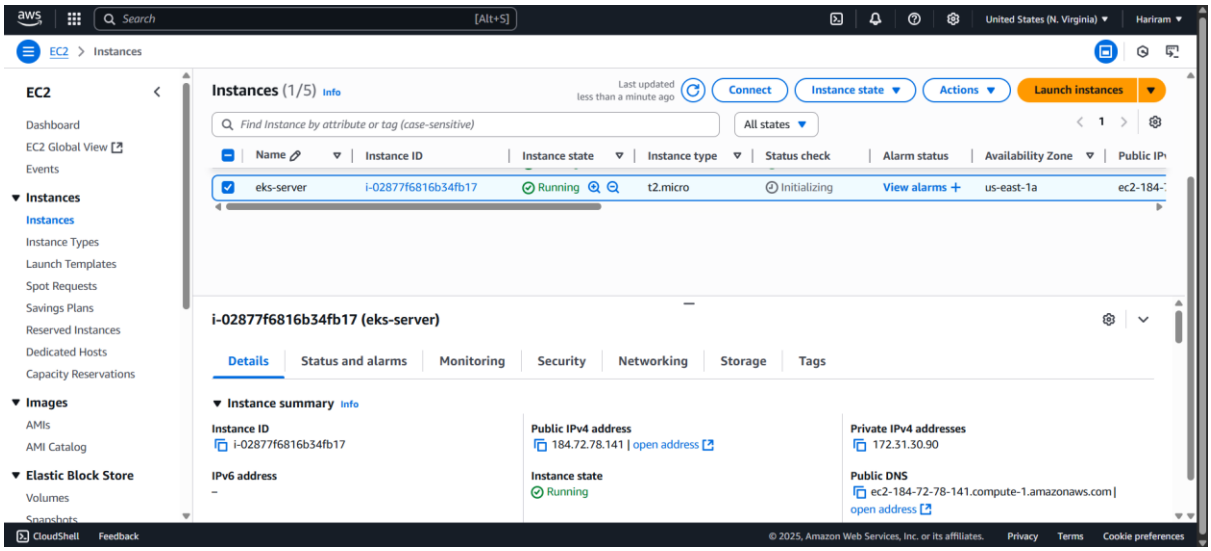
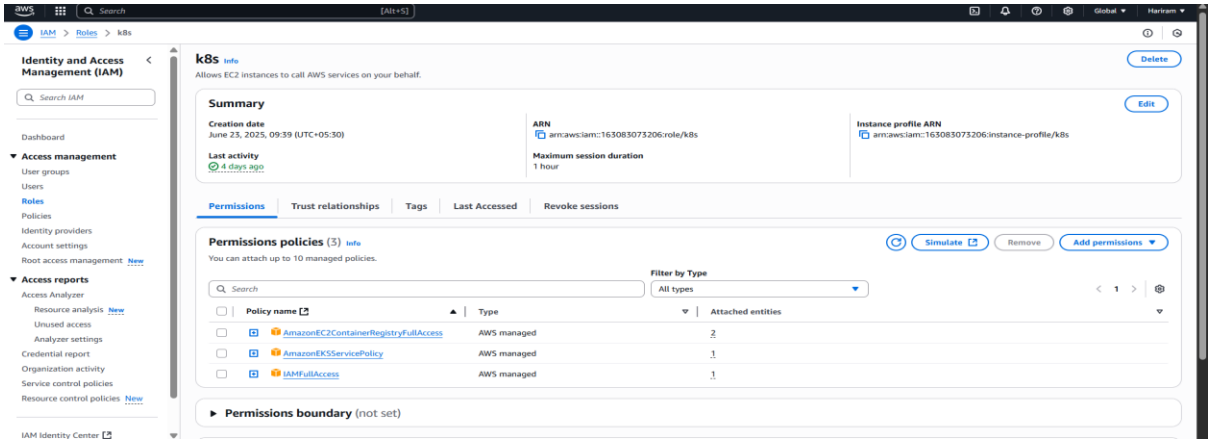


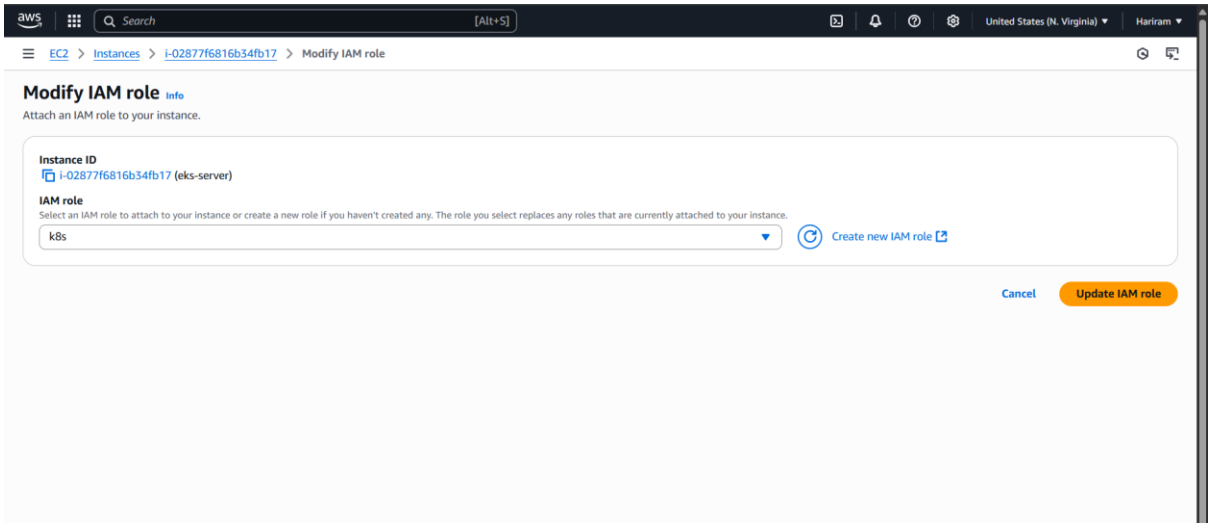
Image is created in ECR.



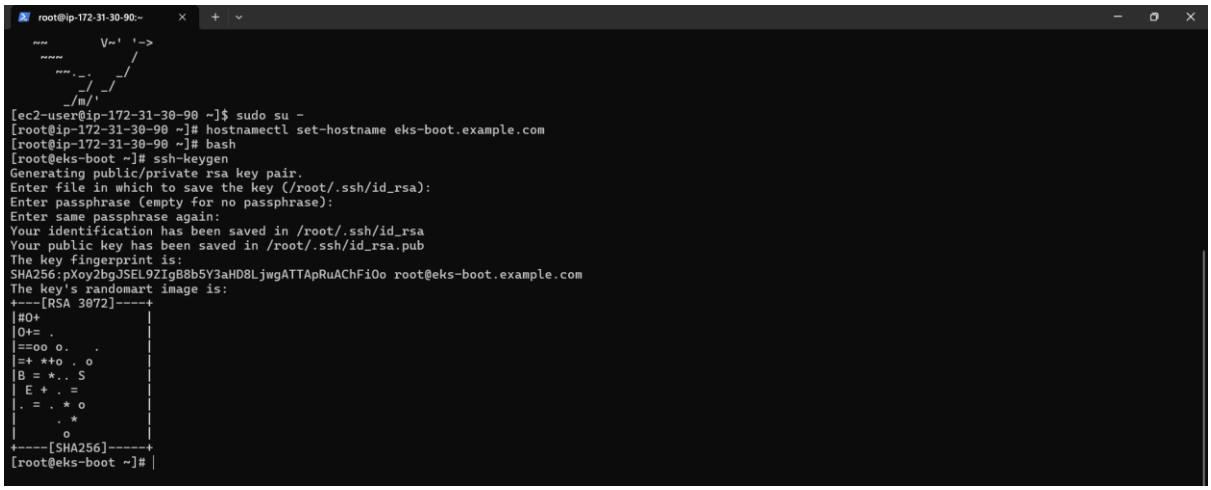
EKS server is created with amazon linux and existing security group k8s-sg with all k8s necessary ports allowed.



Roles is created with necessary permissions.



Roles is added to the eks-server.



EKS-server is started. Hostname is changed. Ssh key is generated.

Eks-server pub key is added to Jenkins.

```
root@ip-172-31-30-90:~/.ssh
no-port-forwarding,no-agent-forwarding,no-X11-forwarding,command="echo 'Please login as the user \"ec2-user\" rather than the user \"root\".';echo;sleep 10;
exit 142" ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQDowCDZw93pIkZ5Zyr4FMsOjcgHTQnJdrf8gRXklMU5mZUBlAqJABcR5lsyzodN+N1iAX9Yee4zd3f+0meqX45ZxTXzDRNKGT50m7fbdB5Z4d
w/IUJsRcSOICVPvJQwKqB6Fcrm1Maa3yguwvt2pBWJcqgfvdOr1yisqKOGwztIUoHUDWCrlag8Lh+gXBLZx5FRF4mpglg67WRNP22dzE+hD5TiBe2DKQvSjU38FiiDcFRVm5X+EKa0VnqJBxVjqUspolzK/z
3W25cVLqh0/NrBQrdn+0J8YNPtYzo2Np5ghhKV3ie4z0tQcNY8Aa5ZWjFlptqmcIvg9Nzvdm25QDE/ aws_Ltimindtree

ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABgQCY5oIJGoTByF5rADOTAHB44WmJmloJctqnyUx2ua6oVcQRxJkVwXkhBbgNt5oVZnfgwR8sXSGsll5yBOOhdLAhdUcTLaWnV5pabz35wakqaT96hEYOLVFS
gjbPVrMcfoW77YZVN79U5zTs4+IxorA/EUB+0T81CeP8rbFnCdLNUGbxQBk1x7i7qAPUzTbU/O2XXhKFhfQ3eRU+ZApipOJzDLTuf6pqaamfWGSpFPL1BY4334qthfZnbA846stRL5zd5BDw5B7USP27MA5q
h/nJkmSfv8pPHQIOWAXPDKFG2fPUTCgRiJqMz6tuzCBBFJlbfCgfBAn4izGd4ygVn0qXS9axUxpnJbYpoI+y08cNHpvfiuXXLIjbTkf+g7aHODGTeVhQ4aOjWuQJ3Flsbom/yisOChncRFM7hlb1UPnUZtRL
+HJRHczy9cwB4V5vf5n+S7PNEUi8Bj4YneqNgzf5WYfgiz9JGiZfe9Tn/hkkmB6n3LRSJPJ5VDqkWywNWeU= root@jenkins.example.com
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-- INSERT --                                                                                                                                  3,578         All
```

Jenkins pub key is added to eks-server.

```
root@ip-172-31-30-90:~
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin yes
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2
# but this is overridden so installations will only check .ssh/authorized_keys
AuthorizedKeysFile      .ssh/authorized_keys

#AuthorizedPrincipalsFile none


# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# Explicitly disable PasswordAuthentication. By presetting it, we
# avoid the cloud-init set_passwords module modifying sshd_config and
# restarting sshd in the default instance launch configuration.
PasswordAuthentication yes
PermitEmptyPasswords no

-- INSERT --                                                                                                                                  33,10         29%
```

```
root@ip-172-31-30-90:~
[root@eks-boot ~]# vim /etc/ssh/sshd_config
[root@eks-boot ~]# systemctl restart sshd
[root@eks-boot ~]# systemctl enable sshd
```

Eks-server sshd config is edited.

```
root@ip-172-31-30-90:~
[root@eks-boot ~]# yum update -y
Amazon Linux 2023 Kernel Livepatch repository                                                    137 kB/s |  17 kB     00:00
Dependencies resolved.
Nothing to do.
Complete!
[root@eks-boot ~]# curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscliv2.zip"
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 63.1M  100 63.1M    0     0  68.1M      0 --:--:-- --:--:-- --:--:-- 68.0M
[root@eks-boot ~]# unzip awscliv2.zip
Archive:  awscliv2.zip
   creating: aws/
   creating: aws/dist/
  inflating: aws/install
  inflating: aws/README.md
  inflating: aws/THIRD_PARTY_LICENSES
   creating: aws/dist/awscli/
   creating: aws/dist/docutils/
   creating: aws/dist/lib-dynload/
  inflating: aws/dist/aws
  inflating: aws/dist/aws_completer
  inflating: aws/dist/libpython3.13.so.1.0
  inflating: aws/dist/_awscrt.abi3.so
  inflating: aws/dist/_ruamel_yaml.cpython-313-x86_64-linux-gnu.so
  inflating: aws/dist/libz.so.1
  inflating: aws/dist/liblzma.so.5
  inflating: aws/dist/libbz2.so.1
  inflating: aws/dist/libffi.so.6
  inflating: aws/dist/libuuid.so.1
  inflating: aws/dist/libreadline.so.6
  inflating: aws/dist/libtinfo.so.5
  inflating: aws/dist/libsqlite3.so.0
  inflating: aws/dist/base_library.zip
  inflating: aws/dist/lib-dynload/_datetime.cpython-313-x86_64-linux-gnu.so
  inflating: aws/dist/lib-dynload/unicodedata.cpython-313-x86_64-linux-gnu.so
  inflating: aws/dist/lib-dynload/_csv.cpython-313-x86_64-linux-gnu.so
  inflating: aws/dist/lib-dynload/_statistics.cpython-313-x86_64-linux-gnu.so
  inflating: aws/dist/lib-dynload/_contextvars.cpython-313-x86_64-linux-gnu.so
  inflating: aws/dist/lib-dynload/_decimal.cpython-313-x86_64-linux-gnu.so
  inflating: aws/dist/lib-dynload/_pickle.cpython-313-x86_64-linux-gnu.so
  inflating: aws/dist/lib-dynload/_hashlib.cpython-313-x86_64-linux-gnu.so
```

Amazon Cli package is installed.

```
root@ip-172-31-30-90:~
  inflating: aws/dist/awscli/botocore/data/auditmanager/2017-07-25/endpoint-rule-set-1.json
  inflating: aws/dist/awscli/botocore/data/auditmanager/2017-07-25/paginators-1.json
  inflating: aws/dist/awscli/botocore/data/auditmanager/2017-07-25/service-2.json
   creating: aws/dist/awscli/botocore/data/lightsail/2016-11-28/
  inflating: aws/dist/awscli/botocore/data/lightsail/2016-11-28/paginators-1.json
  inflating: aws/dist/awscli/botocore/data/lightsail/2016-11-28/endpoint-rule-set-1.json
  inflating: aws/dist/awscli/botocore/data/lightsail/2016-11-28/service-2.json
  inflating: aws/dist/awscli/botocore/data/lightsail/2016-11-28/completions-1.json
   creating: aws/dist/awscli/botocore/.changes/next-release/
  inflating: aws/dist/awscli/botocore/.changes/next-release/api-change-connect-59117.json
  inflating: aws/dist/awscli/topics/topic-tags.json
  inflating: aws/dist/awscli/topics/config-vars.rst
  inflating: aws/dist/awscli/topics/s3-config.rst
  inflating: aws/dist/awscli/topics/ddb-expressions.rst
  inflating: aws/dist/awscli/topics/return-codes.rst
  inflating: aws/dist/awscli/topics/s3-faq.rst
  inflating: aws/dist/awscli/data/cli.json
  inflating: aws/dist/awscli/data/metadata.json
  inflating: aws/dist/awscli/data/ac.index
   creating: aws/dist/awscli/customizations/sso/
   creating: aws/dist/awscli/customizations/Wizard/
   creating: aws/dist/awscli/customizations/wizard/wizards/
   creating: aws/dist/awscli/customizations/wizard/wizards/configure/
   creating: aws/dist/awscli/customizations/wizard/wizards/dynamodb/
   creating: aws/dist/awscli/customizations/wizard/wizards/events/
   creating: aws/dist/awscli/customizations/wizard/wizards/iam/
   creating: aws/dist/awscli/customizations/wizard/wizards/lambda/
  inflating: aws/dist/awscli/customizations/wizard/wizards/dynamodb/new-table.yml
  inflating: aws/dist/awscli/customizations/wizard/wizards/configure/_main.yml
  inflating: aws/dist/awscli/customizations/wizard/wizards/iam/new-role.yml
  inflating: aws/dist/awscli/customizations/wizard/wizards/lambda/new-function.yml
  inflating: aws/dist/awscli/customizations/wizard/wizards/events/new-rule.yml
  inflating: aws/dist/awscli/customizations/sso/index.html
[root@eks-boot ~]#  sudo ./aws/install
You can now run: /usr/local/bin/aws --version
[root@eks-boot ~]#  aws configure
AWS Access Key ID [None]: AKIASL6ECG23OZQMXIHS
AWS Secret Access Key [None]: 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
Default region name [None]: us-east-1
Default output format [None]: table
[root@eks-boot ~]#
```

Aws configuration credentials are given from the IAM user.

```
root@ip-172-31-30-90:~
  inflating: aws/dist/awscli/topics/ddb-expressions.rst
  inflating: aws/dist/awscli/topics/return-codes.rst
  inflating: aws/dist/awscli/topics/s3-faq.rst
  inflating: aws/dist/awscli/data/cli.json
  inflating: aws/dist/awscli/data/metadata.json
  inflating: aws/dist/awscli/data/ac.index
   creating: aws/dist/awscli/customizations/sso/
   creating: aws/dist/awscli/customizations/Wizard/
   creating: aws/dist/awscli/customizations/wizard/wizards/
   creating: aws/dist/awscli/customizations/wizard/wizards/configure/
   creating: aws/dist/awscli/customizations/wizard/wizards/dynamodb/
   creating: aws/dist/awscli/customizations/wizard/wizards/events/
   creating: aws/dist/awscli/customizations/wizard/wizards/iam/
   creating: aws/dist/awscli/customizations/wizard/wizards/lambda/
  inflating: aws/dist/awscli/customizations/wizard/wizards/dynamodb/new-table.yml
  inflating: aws/dist/awscli/customizations/wizard/wizards/configure/_main.yml
  inflating: aws/dist/awscli/customizations/wizard/wizards/iam/new-role.yml
  inflating: aws/dist/awscli/customizations/wizard/wizards/lambda/new-function.yml
  inflating: aws/dist/awscli/customizations/wizard/wizards/events/new-rule.yml
  inflating: aws/dist/awscli/customizations/sso/index.html
[root@eks-boot ~]#  sudo ./aws/install
You can now run: /usr/local/bin/aws --version
[root@eks-boot ~]#  aws configure
AWS Access Key ID [None]: AKIASL6ECG23OZQMXIHS
AWS Secret Access Key [None]: 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
Default region name [None]: us-east-1
Default output format [None]: table
[root@eks-boot ~]# curl --silent --location "https://github.com/weaveworks/eksctl/releases/latest/download/eksctl_$(uname -s)_amd64.tar.gz" | tar xz -C /tmp
[root@eks-boot ~]# sudo mv /tmp/eksctl /usr/local/bin
[root@eks-boot ~]# eksctl version
0.210.0
[root@eks-boot ~]# curl -LO https://storage.googleapis.com/kubernetes-release/release/$(curl -s https://storage.googleapis.com/kubernetes-release/release/st
able.txt)/bin/linux/amd64/kubectl
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 53.7M  100 53.7M    0     0  85.7M      0 --:--:-- --:--:-- --:--:-- 85.8M
[root@eks-boot ~]# sudo install -o root -g root -m 0755 kubectl /usr/local/bin/kubectl
[root@eks-boot ~]#  kubectl version --client
Client Version: v1.31.0
Kustomize Version: v5.4.2
[root@eks-boot ~]#
```

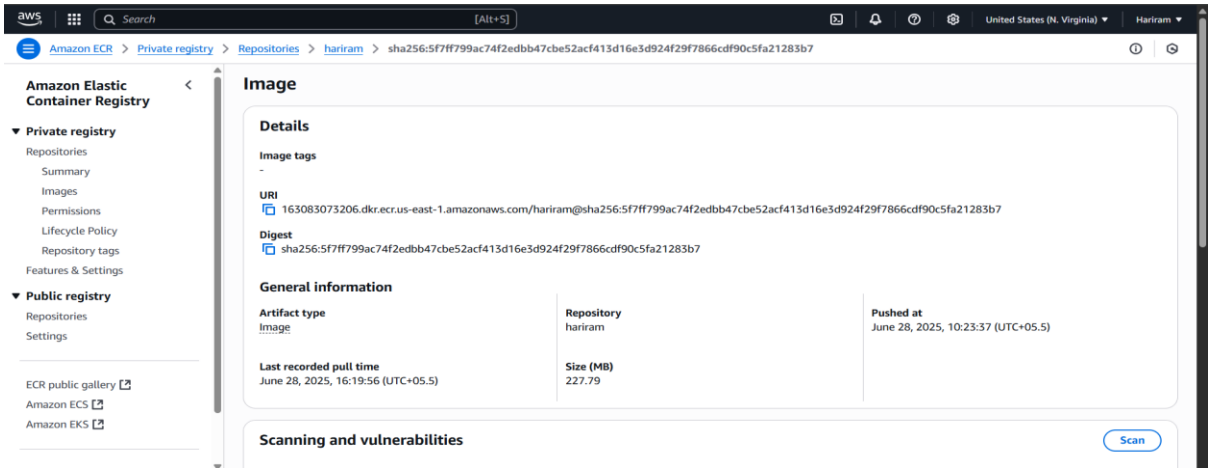
Kubernetes package is installed.

```
root@ip-172-31-30-90:/opt
[root@eks-boot ~]# eksctl create cluster --name milestone-2 --region us-east-1 --version 1.32 --vpc-public-subnets subnet-00608da34c56e569d,subnet-0541b7ff15bd92fd9 --without-nodegroup
2025-06-28 10:09:31 [ℹ]  eksctl version 0.210.0
2025-06-28 10:09:31 [ℹ]  using region us-east-1
2025-06-28 10:09:32 [✓]  using existing VPC (vpc-0c1e9f84aa49d9d9e) and subnets (private:map[] public:map[us-east-1a:{subnet-00608da34c56e569d us-east-1a 172.31.16.0/20 0 } us-east-1b:{sub
net-0541b7ff15bd92fd9 us-east-1b 172.31.32.0/20 0 }])
2025-06-28 10:09:32 [!]  custom VPC/subnets will be used; if resulting cluster doesn't function as expected, make sure to review the configuration of VPC/subnets
2025-06-28 10:09:32 [ℹ]  using Kubernetes version 1.32
2025-06-28 10:09:32 [ℹ]  creating EKS cluster "milestone-2" in "us-east-1" region with
2025-06-28 10:09:32 [ℹ]  if you encounter any issues, check CloudFormation console or try 'eksctl utils describe-stacks --region=us-east-1 --cluster=milestone-2'
2025-06-28 10:09:32 [ℹ]  Kubernetes API endpoint access will use default of {publicAccess=true, privateAccess=false} for cluster "milestone-2" in "us-east-1"
2025-06-28 10:09:32 [ℹ]  CloudWatch logging will not be enabled for cluster "milestone-2" in "us-east-1"
2025-06-28 10:09:32 [ℹ]  you can enable it with 'eksctl utils update-cluster-logging --enable-types={SPECIFY-YOUR-LOG-TYPES-HERE (e.g. all)} --region=us-east-1 --cluster=milestone-2'
2025-06-28 10:09:32 [ℹ]  default addons coredns, metrics-server, vpc-cni, kube-proxy were not specified, will install them as EKS addons
2025-06-28 10:09:32 [ℹ]
2 sequential tasks: { create cluster control plane "milestone-2",
    2 sequential sub-tasks: {
        1 task: { create addons },
        wait for control plane to become ready,
    }
}
2025-06-28 10:09:32 [ℹ]  building cluster stack "eksctl-milestone-2-cluster"
2025-06-28 10:09:32 [ℹ]  deploying stack "eksctl-milestone-2-cluster"
2025-06-28 10:10:02 [ℹ]  waiting for CloudFormation stack "eksctl-milestone-2-cluster"
2025-06-28 10:10:32 [ℹ]  waiting for CloudFormation stack "eksctl-milestone-2-cluster"
2025-06-28 10:11:32 [ℹ]  waiting for CloudFormation stack "eksctl-milestone-2-cluster"
2025-06-28 10:12:32 [ℹ]  waiting for CloudFormation stack "eksctl-milestone-2-cluster"
2025-06-28 10:13:32 [ℹ]  waiting for CloudFormation stack "eksctl-milestone-2-cluster"
2025-06-28 10:14:33 [ℹ]  waiting for CloudFormation stack "eksctl-milestone-2-cluster"
2025-06-28 10:15:33 [ℹ]  waiting for CloudFormation stack "eksctl-milestone-2-cluster"
2025-06-28 10:16:33 [ℹ]  waiting for CloudFormation stack "eksctl-milestone-2-cluster"
2025-06-28 10:16:33 [ℹ]  creating addon: coredns
2025-06-28 10:16:34 [ℹ]  successfully created addon: coredns
2025-06-28 10:16:34 [ℹ]  creating addon: metrics-server
2025-06-28 10:16:35 [ℹ]  successfully created addon: metrics-server
2025-06-28 10:16:35 [!]  recommended policies were found for "vpc-cni" addon, but since OIDC is disabled on the cluster, eksctl cannot configure the requested permissions; the recommended
way to provide IAM permissions for "vpc-cni" addon is via pod identity associations; after addon creation is completed, add all recommended policies to the config file, under `addon.PodIde
ntityAssociations`, and run `eksctl update addon`
2025-06-28 10:16:35 [ℹ]  creating addon: vpc-cni
2025-06-28 10:16:35 [ℹ]  successfully created addon: vpc-cni
2025-06-28 10:16:36 [ℹ]  creating addon: kube-proxy
2025-06-28 10:16:36 [ℹ]  successfully created addon: kube-proxy
2025-06-28 10:18:36 [ℹ]  waiting for the control plane to become ready
2025-06-28 10:18:37 [✓]  saved kubeconfig as "/root/.kube/config"
2025-06-28 10:18:37 [ℹ]  no tasks
2025-06-28 10:18:37 [✓]  all EKS cluster resources for "milestone-2" have been created
2025-06-28 10:18:38 [ℹ]  kubectl command should work with "/root/.kube/config", try 'kubectl get nodes'
2025-06-28 10:18:38 [✓]  EKS cluster "milestone-2" in "us-east-1" region is ready
```

Cluster is created.

```
root@ip-172-31-30-90:/opt
2025-06-28 10:09:32 [ℹ]  deploying stack "eksctl-milestone-2-cluster"
2025-06-28 10:10:02 [ℹ]  waiting for CloudFormation stack "eksctl-milestone-2-cluster"
2025-06-28 10:10:32 [ℹ]  waiting for CloudFormation stack "eksctl-milestone-2-cluster"
2025-06-28 10:11:32 [ℹ]  waiting for CloudFormation stack "eksctl-milestone-2-cluster"
2025-06-28 10:12:32 [ℹ]  waiting for CloudFormation stack "eksctl-milestone-2-cluster"
2025-06-28 10:13:32 [ℹ]  waiting for CloudFormation stack "eksctl-milestone-2-cluster"
2025-06-28 10:14:33 [ℹ]  waiting for CloudFormation stack "eksctl-milestone-2-cluster"
2025-06-28 10:15:33 [ℹ]  waiting for CloudFormation stack "eksctl-milestone-2-cluster"
2025-06-28 10:16:33 [ℹ]  waiting for CloudFormation stack "eksctl-milestone-2-cluster"
2025-06-28 10:16:33 [ℹ]  creating addon: coredns
2025-06-28 10:16:34 [ℹ]  successfully created addon: coredns
2025-06-28 10:16:34 [ℹ]  creating addon: metrics-server
2025-06-28 10:16:35 [ℹ]  successfully created addon: metrics-server
2025-06-28 10:16:35 [!]  recommended policies were found for "vpc-cni" addon, but since OIDC is disabled on the cluster, eksctl cannot configure
 the requested permissions; the recommended way to provide IAM permissions for "vpc-cni" addon is via pod identity associations; after addon cre
ation is completed, add all recommended policies to the config file, under `addon.PodIdentityAssociations`, and run `eksctl update addon`
2025-06-28 10:16:35 [ℹ]  creating addon: vpc-cni
2025-06-28 10:16:35 [ℹ]  successfully created addon: vpc-cni
2025-06-28 10:16:36 [ℹ]  creating addon: kube-proxy
2025-06-28 10:16:36 [ℹ]  successfully created addon: kube-proxy
2025-06-28 10:18:36 [ℹ]  waiting for the control plane to become ready
2025-06-28 10:18:37 [✓]  saved kubeconfig as "/root/.kube/config"
2025-06-28 10:18:37 [ℹ]  no tasks
2025-06-28 10:18:37 [✓]  all EKS cluster resources for "milestone-2" have been created
2025-06-28 10:18:38 [ℹ]  kubectl command should work with "/root/.kube/config", try 'kubectl get nodes'
2025-06-28 10:18:38 [✓]  EKS cluster "milestone-2" in "us-east-1" region is ready
[root@eks-boot ~]# kubectl get pods
No resources found in default namespace.
[root@eks-boot ~]# cd /opt/
[root@eks-boot opt]# vim regapp-deploy.yml
"regapp-deploy.yml" [New] 29L, 530B written
[root@eks-boot opt]# vim regapp-service.yml
[root@eks-boot opt]# ll
total 8
drwxr-xr-x. 4 root root  33 Jun 20 06:56 aws
-rw-r--r--. 1 root root 530 Jun 28 10:20 regapp-deploy.yml
-rw-r--r--. 1 root root 196 Jun 28 10:20 regapp-service.yml
[root@eks-boot opt]#
```

Deployment and service yaml file is added in /opt.



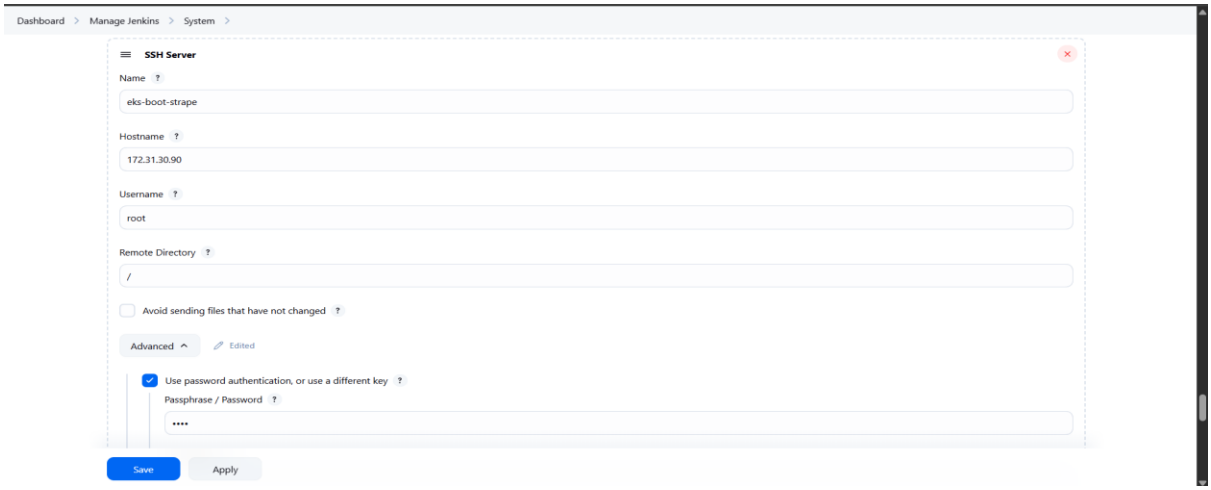
Copy the URI from ECR.

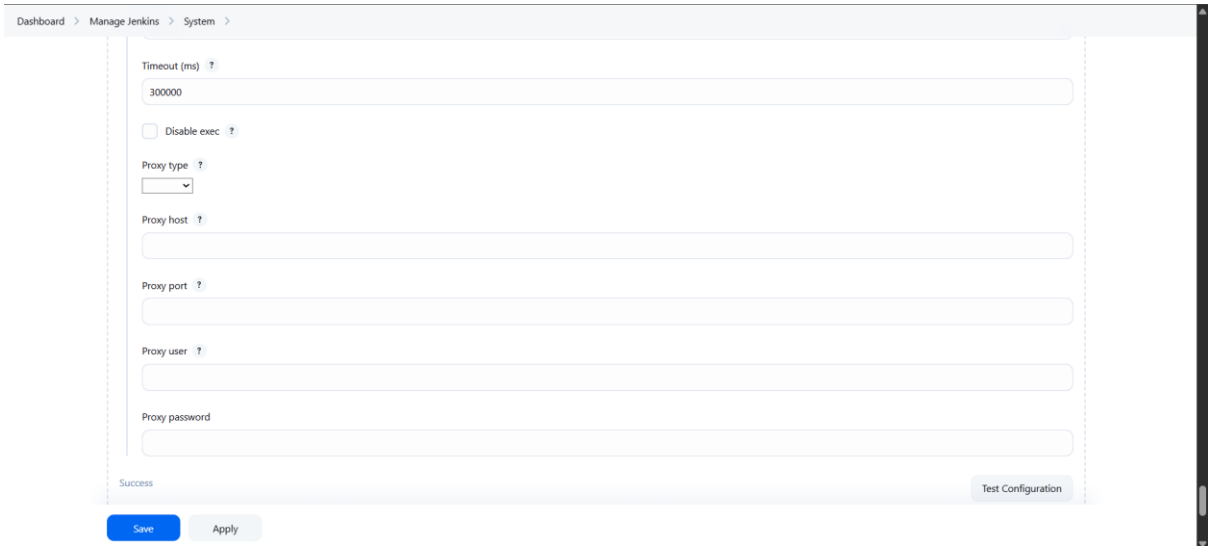
```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: regapp-deployment
  labels:
    app: regapp

spec:
  replicas: 2
  selector:
    matchLabels:
      app: regapp

  template:
    metadata:
      labels:
        app: regapp
    spec:
      containers:
      - name: regapp
        image: 163083073206.dkr.ecr.us-east-1.amazonaws.com/hariram:latest
        imagePullPolicy: Always
        ports:
        - containerPort: 8080
  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxSurge: 1
      maxUnavailable: 1
~
~
~
~
~
~
"regapp-deploy.yml" 29L, 530B                                    21,74         All
```

This is the deployment yaml file. Paste the URI in image path.

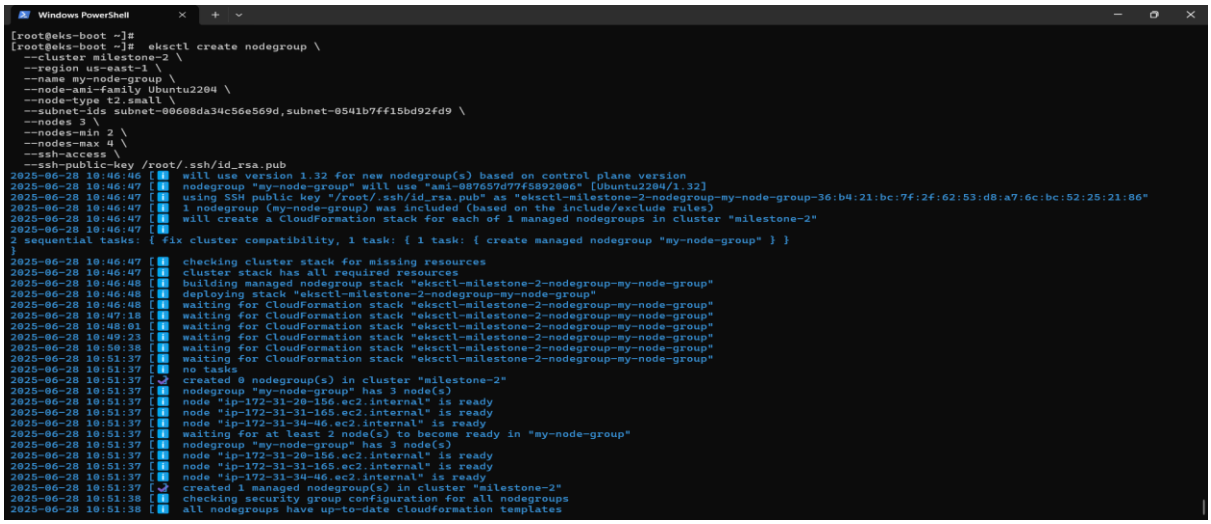
```yaml
apiVersion: v1
kind: Service
metadata:
  name: regapp-service
  labels:
    app: regapp
spec:
  selector:
    app: regapp

  ports:
    - port: 8080
      targetPort: 8080

  type: LoadBalancer
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
"regapp-service.yml" 15L, 196B                                   15,20         All
```

This is the service yaml file.



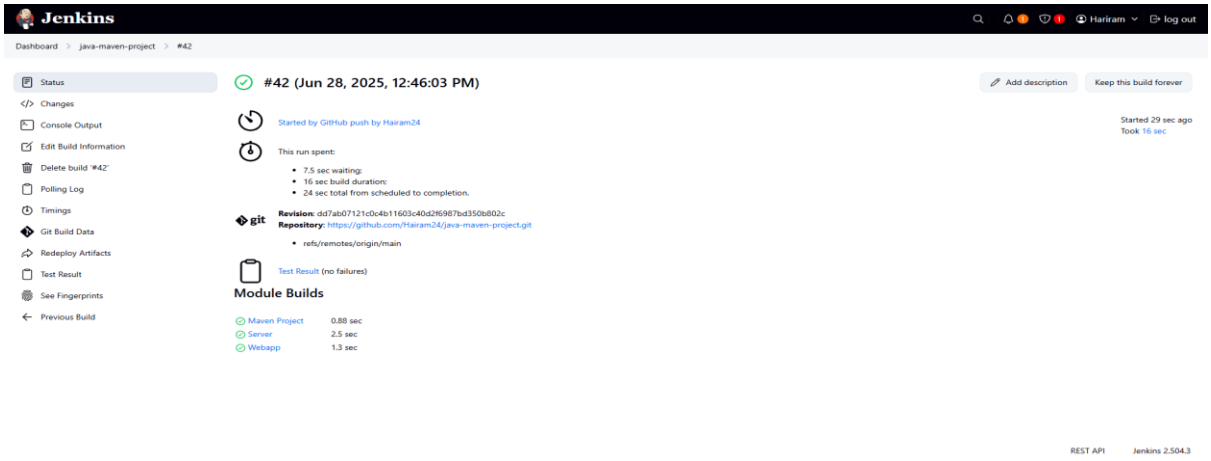


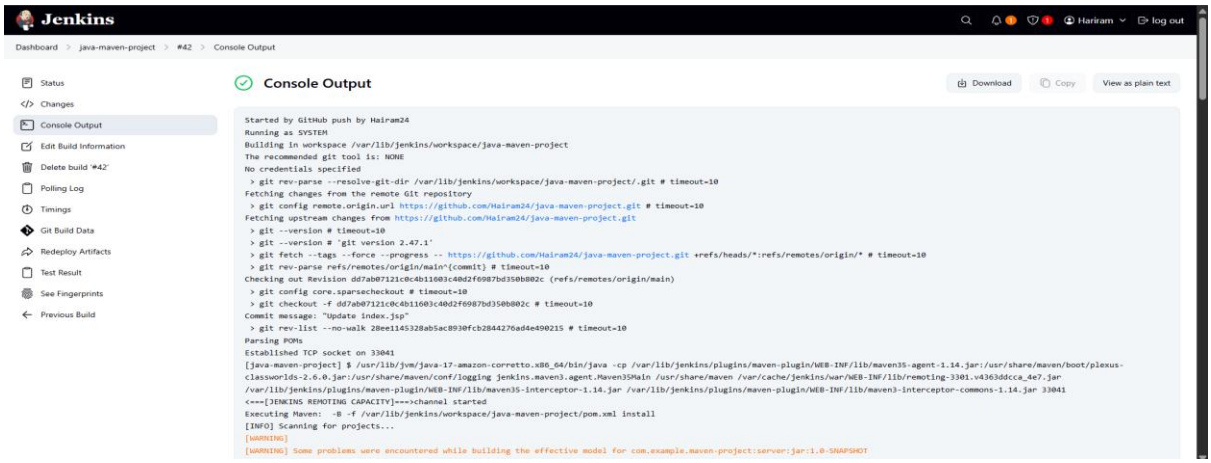
Eks-server ssh is added to Jenkins system.



Node group is created.

Post build eks-server is added with exec commands.

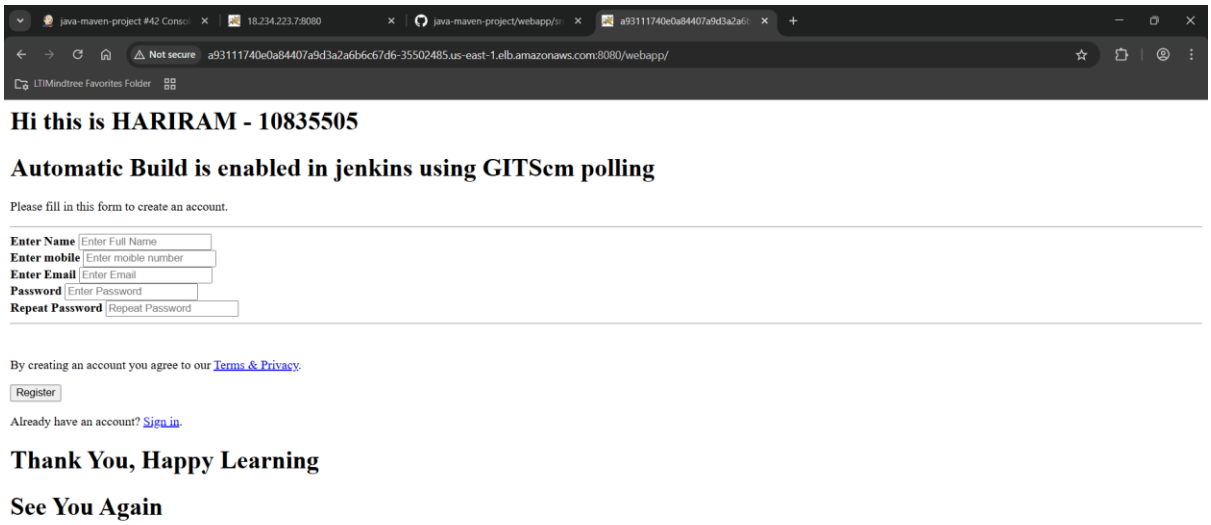




Build successfully.

```
[root@eks-boot ~]# kubectl get pod
NAME                                READY   STATUS    RESTARTS   AGE
regapp-deployment-5fc55cbd8d-mhmkl   1/1     Running   0          38m
regapp-deployment-5fc55cbd8d-ssxtk   1/1     Running   0          38m
[root@eks-boot ~]# kubectl get svc
NAME             TYPE           CLUSTER-IP       EXTERNAL-IP                                                              PORT(S)          AGE
kubernetes       ClusterIP      10.100.0.1       <none>                                                                   443/TCP          51m
regapp-service   LoadBalancer   10.100.125.236   a93111740e0a84407a9d3a2a6b6c67d6-35502485.us-east-1.elb.amazonaws.com   8080:31288/TCP   38m
[root@eks-boot ~]#
```

Two pods are running after the build successful.



Final output is displayed in server ip:8080/webapp/.

SUBMITTED BY –

HARIRAM S

PS.NO: 10835505